YEAR 12 MATHEMATICS EXTENSION 1

Year 12 Integration Part 1 Integration Basics

Lesson and Homework Pack

Revision	Indices, derivatives, sigma notation, areas	Year 11 · 3G, 7A–8G · Year 12 · 1E
Lesson 4J	Primitive functions	Past papers: timed practice
Lesson 5A	Areas and the definite integral	Past papers: 1 mark · 5 min
Lesson 5B	The fundamental theorem of calculus	Past papers: 1 mark · 5 min
Lesson 5C	The definite integral and its properties	Past papers: 3 marks · 5 min
Lesson 5D	Proving the fundamental theorem	Past papers: timed practice
Lesson 5E	The indefinite integral	Past papers: timed practice
Lesson 5F	Finding areas by integration	Past papers: 1 mark · 5 min
Lesson 5G	Areas of compound regions	Past papers: timed practice
Lesson 5H	The trapezoidal rule	Past papers: 1 mark · 5 min
Lesson 5I	The reverse chain rule	Past papers: 2 marks · 5 min
Review	Chapter 5 review exercise	Mixed practice

This is the first of three integration packs. Part 2 covers exponential, logarithmic and trigonometric integrals, and Part 3 covers the Extension 1 methods and volumes of rotation. Each lesson runs in the same order: **concept** → **tutor demo** → **your attempt** → **worked example** to keep for homework. Then do the **textbook homework** and finish with the **past-paper set** under exam conditions. Answers and full past-paper solutions are in the separate Homework Answers booklet.

REVISION

Indices, derivatives, sigma notation and areas

A short review of the Year 11 work on indices (7A–7B), differentiating powers of x (8C, 8E, 8F), the chain rule (8G) and semicircles (3G), and of sigma notation (Year 12, 1E) and the area formulae. Every lesson in this pack uses the power rule and the index laws.

SUMMARY · keep these pages open while you work

INDEX LAWS, ZERO AND NEGATIVE INDICES (YEAR 11, 7A)

In the power a^n, the number a is the **base** and n is the **index** (or exponent). For positive integers n, $a^n = a \times a \times \cdots \times a$ with n factors.

Combining powers with the same base

- Multiply: add the indices, $a^m \times a^n = a^{m+n}$.
- Divide: subtract the indices, $a^m \div a^n = a^{m-n}$.
- Power of a power: multiply, $(a^m)^n = a^{mn}$.

Powers of products and quotients

- $(ab)^n = a^n \times b^n$
- $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$

Zero and negative indices ($a \neq 0$) are defined so that the index laws still work. We know $a^3 \div a^3 = 1$, and the laws give $a^{3-3} = a^0$, so we define $a^0 = 1$. We know $a^2 \div a^3 = \frac{1}{a}$, and the laws give $a^{2-3} = a^{-1}$, so we define $a^{-1} = \frac{1}{a}$. In general,

$$a^0 = 1 \qquad a^{-1} = \frac{1}{a} \qquad a^{-2} = \frac{1}{a^2} \qquad a^{-n} = \frac{1}{a^n}.$$

- The negative sign in the index says, 'Take the reciprocal'. Always do this first. The reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$, so $\left(\frac{2}{3}\right)^{-3} = \left(\frac{3}{2}\right)^3 = \frac{27}{8}$.
- A constant factor stays where it is: $\frac{5}{x^2} = 5 \times \frac{1}{x^2} = 5x^{-2}$ and $\frac{1}{3x^5} = \frac{1}{3} \times \frac{1}{x^5} = \frac{1}{3}x^{-5}$.

WORKED EXAMPLE **Year 11 Example 5 (7A)**

Use the index laws, extended to negative indices, to simplify these expressions, giving answers in fraction form without negative indices.

a $x^4 \times x^{-10}$ b $12a^2b^4 \div 6a^3b^4$ c $(2^n)^{-3}$

SOLUTION

a
$$\begin{aligned} x^4 \times x^{-10} &= x^{4-10} \\ &= x^{-6} \\ &= \frac{1}{x^6} \end{aligned}$$

b
$$\begin{aligned} 12a^2b^4 \div 6a^3b^4 &= \tfrac{12}{6}a^{2-3}b^{4-4} \\ &= 2a^{-1}b^0 \\ &= 2a^{-1} \\ &= \frac{2}{a} \end{aligned}$$

c
$$\begin{aligned} (2^n)^{-3} &= 2^{n\times(-3)} \\ &= 2^{-3n} \\ &= \frac{1}{2^{3n}} \end{aligned}$$

STEPS THE BOOK SKIPS: YEAR 11 EXAMPLE 5 In part **b**, $b^0 = 1$, which is why the factor b^0 disappears in the next line.

FRACTIONAL INDICES (YEAR 11, 7B)

Let $a \geq 0$. Because $\left(\sqrt{a}\right)^2 = a$ and the index laws give $\left(a^{\frac{1}{2}}\right)^2 = a^1 = a$, we define $a^{\frac{1}{2}} = \sqrt{a}$. In the same way $a^{\frac{1}{3}} = \sqrt[3]{a}$, and in general

$$a^{\frac{1}{n}} = \sqrt[n]{a} \qquad a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m} \qquad a^{-\frac{m}{n}} = \frac{1}{a^{\frac{m}{n}}},$$

where the non-negative root is always taken (for $a^{\frac{m}{n}}$ and $a^{-\frac{m}{n}}$, take $a > 0$).

Deal with a complicated index in this order: (1) if the index is negative, take the reciprocal; (2) if the index has a denominator, take that root; (3) finally, take the power given by the numerator. For example, $\left(\frac{8}{27}\right)^{-\frac{4}{3}} = \left(\frac{27}{8}\right)^{\frac{4}{3}} = \left(\frac{3}{2}\right)^4 = \frac{81}{16}$.

Conversions used all the time in calculus (here $x > 0$):

$$\sqrt{x} = x^{\frac{1}{2}} \qquad \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}} \qquad \sqrt[3]{x^2} = x^{\frac{2}{3}}$$

$$x\sqrt{x} = x^1 \times x^{\frac{1}{2}} = x^{\frac{3}{2}} \qquad \frac{1}{x\sqrt{x}} = x^{-\frac{3}{2}}$$

- The textbook often writes these indices as mixed numbers: $x^{1\frac{1}{2}} = x^{\frac{3}{2}}$ and $x^{-1\frac{1}{2}} = x^{-\frac{3}{2}}$.
- $\sqrt{9} = 9^{\frac{1}{2}} = 3$ (not -3), and $\sqrt{-9}$ and $(-9)^{\frac{1}{2}}$ are undefined. So $\sqrt{x}$ and $x^{\frac{1}{2}}$ are defined only for $x \geq 0$.

WORKED EXAMPLE **Year 11 Example 10 (7B)**

Write each expression using a fractional index.

a $\sqrt{x^3}$ b $\frac{1}{\sqrt{x^3}}$ c $\sqrt[3]{x^2}$ d $\frac{1}{\sqrt[3]{x^2}}$

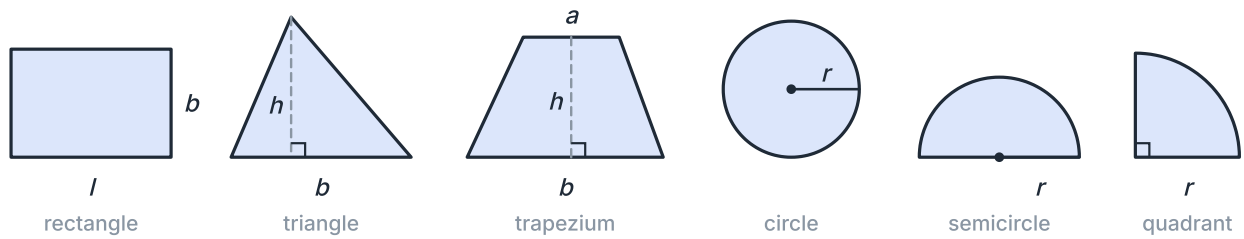
SOLUTION

a $\sqrt{x^3} = x^{\frac{3}{2}}$ b $\frac{1}{\sqrt{x^3}} = x^{-\frac{3}{2}}$ c $\sqrt[3]{x^2} = x^{\frac{2}{3}}$ d $\frac{1}{\sqrt[3]{x^2}} = x^{-\frac{2}{3}}$

STEPS THE BOOK SKIPS: YEAR 11 EXAMPLE 10 The root becomes the denominator of the index, the power inside becomes the numerator, and a power in the denominator of a fraction makes the index negative.

AREA FORMULAE

All area formulae are based on two principles: the area of a rectangle is length × breadth, and when a region is dissected (cut up and rearranged), its area is unchanged.



- Rectangle: $A = lb$
- Triangle: $A = \frac{1}{2}bh = \frac{1}{2} \times \text{base} \times \text{height}$
- Trapezium: $A = \frac{1}{2}(a+b)h$, the average of the parallel sides × the width
- Circle: $A = \pi r^2$
- Semicircle: $A = \frac{1}{2}\pi r^2$
- Quadrant (quarter circle): $A = \frac{1}{4}\pi r^2$

The height h of a triangle and the width h of a trapezium are **perpendicular** distances. For a trapezium, h is the distance between the parallel sides.

OUR OWN EXAMPLE: USING THE AREA FORMULAE

Question. Find the area of:

a a trapezium with parallel sides 3 cm and 7 cm, at a perpendicular distance of 4 cm apart,

b a quadrant of radius 6 cm,

c a triangle with base 5 cm and perpendicular height 9 cm.

Solution.

a $A = \frac{1}{2}(a+b)h$
$= \frac{1}{2} \times (3+7) \times 4$
$= 20 \text{ cm}^2$

b $A = \frac{1}{4}\pi r^2$
$= \frac{1}{4} \times \pi \times 6^2$
$= 9\pi \text{ cm}^2$

c $A = \frac{1}{2}bh$
$= \frac{1}{2} \times 5 \times 9$
$= 22\frac{1}{2} \text{ cm}^2$

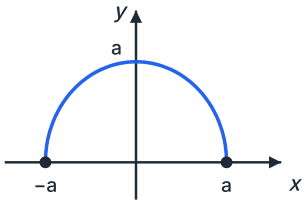
Leave π in an exact answer, as in part **b**, unless the question asks for a decimal.

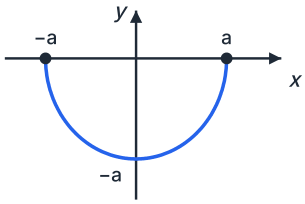
CIRCLES AND SEMICIRCLES (YEAR 11, 3G), AND THE GRAPH OF $y = |x|$ (4D)

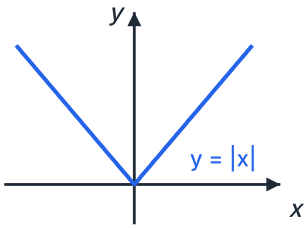
By Pythagoras' theorem, the circle with centre the origin and radius a is $x^2 + y^2 = a^2$. It is not a function. Solving for y gives two functions:

- $y = \sqrt{a^2 - x^2}$ is the **upper semicircle**,
- $y = -\sqrt{a^2 - x^2}$ is the **lower semicircle**.

The graph of $y = |x|$ is two half-lines meeting at the origin, $y = x$ for $x \geq 0$ and $y = -x$ for $x < 0$.







SUMS OF TERMS AND SIGMA NOTATION (YEAR 12, 1E)

For a sequence $T_1,\ T_2,\ T_3,\ \ldots$, the **sum of the first n terms** (the **nth partial sum**) is $S_n = T_1 + T_2 + T_3 + \cdots + T_n$. The terms can be recovered from the partial sums: $T_1 = S_1$ and $T_n = S_n - S_{n-1}$ for $n \geq 2$.

Sigma notation is a concise notation for these sums. The symbol Σ is the Greek capital letter sigma and stands for 'sum':

$$\sum_{n=5}^{20} T_n = T_5 + T_6 + T_7 + T_8 + \cdots + T_{20},$$

and any two integers can be used in place of 5 and 20. For example,

$$\sum_{n=2}^{5} n^2 = 2^2 + 3^2 + 4^2 + 5^2 = 4 + 9 + 16 + 25 = 54.$$

- This says 'evaluate the function n^2 for all the integers from $n = 2$ to $n = 5$, then add up the resulting values'. There are $5 - 2 + 1 = 4$ terms.
- The letter under the sigma is only a counter. Other letters such as k, r and ℓ are also used.
- A **series** is the sequence of partial sums: $S_1 = T_1$, $S_2 = T_1 + T_2$, $S_3 = T_1 + T_2 + T_3$, and so on.
- In Lesson 5A, the area under a curve is approximated by adding up the areas $f(x)\,\delta x$ of thin strips, written as $\sum_{x=a}^{b} f(x)\,\delta x$.

WORKED EXAMPLE **Year 12 Example 21 (1E)**

Evaluate these sums.

a $\sum_{n=4}^{7}(5n+1)$

b $\sum_{n=1}^{5}(-2)^n$

SOLUTION

a $\sum_{n=4}^{7}(5n+1) = 21+26+31+36$

$= 114$

b $\sum_{n=1}^{5}(-2)^n = -2+4-8+16-32$

$= -22$

STEPS THE BOOK SKIPS: YEAR 12 EXAMPLE 21 The terms in **a** are $5\times4+1=21$, $5\times5+1=26$, $5\times6+1=31$ and $5\times7+1=36$. In **b**, odd powers of -2 are negative and even powers are positive.

DIFFERENTIATING POWERS OF x (YEAR 11, 8C, 8E, 8F)

The derivative of any power of x. Let $f(x)=x^n$, where n is any real number. Then $f'(x)=nx^{n-1}$. This rule is usually memorised as: 'Take the index as a factor, and reduce the index by 1.'

- It applies unchanged when n is negative or a fraction. For example, if $y=x^{-4}$, then $\frac{dy}{dx}=-4x^{-5}$, and if $y=x^{\frac{5}{2}}$, then $\frac{dy}{dx}=\frac{5}{2}x^{\frac{3}{2}}$.
- The derivative of a constant is zero.

Linear combinations. Sums and multiples are differentiated one term at a time. If $f(x)=g(x)+k(x)$, then $f'(x)=g'(x)+k'(x)$. If $f(x)=ag(x)$, then $f'(x)=ag'(x)$.

Get the function into powers of x first.

- Expand a product of brackets first: $x^3(x-10)=x^4-10x^3$.
- Convert reciprocals of powers to negative indices, and surds to fractional indices.
- If the denominator of a fraction is a single term, divide it into each term of the numerator.
- Then apply the rule. Convert the answer back to fraction or surd form when the question asks.

WORKED EXAMPLE **Year 11 Example 17 (8E)**

Differentiate each function, giving your answer without negative indices.

a $f(x) = \dfrac{1}{x}$ b $f(x) = \dfrac{5}{x^2}$ c $f(x) = \dfrac{1}{3x^5}$

SOLUTION

In each example, first convert the expression to negative indices, then apply the rule, then convert back to fraction notation.

a
$$\begin{aligned} f(x) &= \frac{1}{x} \\ &= x^{-1} \\ f'(x) &= -x^{-2} \\ &= -\frac{1}{x^2} \end{aligned}$$

b
$$\begin{aligned} f(x) &= \frac{5}{x^2} \\ &= 5x^{-2} \\ f'(x) &= -10x^{-3} \\ &= -\frac{10}{x^3} \end{aligned}$$

c
$$\begin{aligned} f(x) &= \frac{1}{3x^5} \\ &= \tfrac{1}{3}x^{-5} \\ f'(x) &= -\tfrac{5}{3}x^{-6} \\ &= -\frac{5}{3x^6} \end{aligned}$$

STEPS THE BOOK SKIPS: YEAR 11 EXAMPLE 17 The missing middle steps: in **b**, $5 \times (-2)x^{-2-1} = -10x^{-3}$; in **c**, $\frac{1}{3} \times (-5)x^{-5-1} = -\frac{5}{3}x^{-6}$.

WORKED EXAMPLE **Year 11 Example 21 (8F)**

Write each function using a fractional index, then differentiate it, then convert the answer back to surd form.

a $y = 12\sqrt{x}$ b $y = \sqrt{16x}$ c $y = \dfrac{3}{\sqrt{x}}$

SOLUTION

a
$$\begin{aligned} y &= 12\sqrt{x} \\ &= 12x^{\frac{1}{2}} \\ \frac{dy}{dx} &= 12 \times \tfrac{1}{2} \times x^{-\frac{1}{2}} \\ &= \frac{6}{\sqrt{x}} \end{aligned}$$

b
$$\begin{aligned} y &= \sqrt{16x} \\ &= 4x^{\frac{1}{2}} \\ \frac{dy}{dx} &= 4 \times \tfrac{1}{2} \times x^{-\frac{1}{2}} \\ &= \frac{2}{\sqrt{x}} \end{aligned}$$

c
$$\begin{aligned} y &= \frac{3}{\sqrt{x}} \\ &= 3x^{-\frac{1}{2}} \\ \frac{dy}{dx} &= -\tfrac{1}{2} \times 3 \times x^{-1\frac{1}{2}} \\ &= -\frac{3}{2x\sqrt{x}} \end{aligned}$$

STEPS THE BOOK SKIPS: YEAR 11 EXAMPLE 21 In **b**, $\sqrt{16x} = \sqrt{16} \times \sqrt{x} = 4x^{\frac{1}{2}}$. In **c**, $x^{-1\frac{1}{2}} = x^{-\frac{3}{2}} = \dfrac{1}{x^{\frac{3}{2}}} = \dfrac{1}{x\sqrt{x}}$, so $-\frac{3}{2}x^{-1\frac{1}{2}} = -\dfrac{3}{2x\sqrt{x}}$.

WORKED EXAMPLE **Year 11 Example 23 (8F)**

Divide through by the denominator, then differentiate.

a $y = \dfrac{10x - 6}{\sqrt{x}}$

b $y = \dfrac{3x + 4\sqrt{x}}{x}$

SOLUTION

a $y = \dfrac{10x^1}{x^{\frac{1}{2}}} - \dfrac{6}{x^{\frac{1}{2}}}$

$= 10x^{\frac{1}{2}} - 6x^{-\frac{1}{2}}$

$\dfrac{dy}{dx} = 5x^{-\frac{1}{2}} + 3x^{-1\frac{1}{2}}$

b $y = \dfrac{3x}{x} + \dfrac{4x^{\frac{1}{2}}}{x^1}$

$= 3 + 4x^{-\frac{1}{2}}$

$\dfrac{dy}{dx} = -2x^{-1\frac{1}{2}}$

STEPS THE BOOK SKIPS: YEAR 11 EXAMPLE 23 The index steps: in **a**, $x^1 \div x^{\frac{1}{2}} = x^{1-\frac{1}{2}} = x^{\frac{1}{2}}$, then $10 \times \frac{1}{2} = 5$ and $-6 \times (-\frac{1}{2}) = 3$. In **b**, $x^{\frac{1}{2}} \div x^1 = x^{-\frac{1}{2}}$, the constant 3 has derivative 0, and $4 \times (-\frac{1}{2}) = -2$.

THE CHAIN RULE (YEAR 11, 8G)

A **composite function** is a chain of two functions. For example, $y = \sqrt{25 - x^2}$ is 'square and subtract from 25', followed by 'take the positive square root': let $u = 25 - x^2$, then $y = \sqrt{u}$.

The chain rule. Suppose that y is a function of u, where u is a function of x. Then

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}.$$

- Recommended setting-out: work on the right first. Write u as a function of x and y as a function of u, find $\dfrac{du}{dx}$ and $\dfrac{dy}{du}$, then multiply them on the left.
- **Powers of a linear function.** With $u = ax + b$ and $y = u^n$, $\dfrac{du}{dx} = a$ and $\dfrac{dy}{du} = nu^{n-1}$, so $\dfrac{d}{dx}(ax + b)^n = an(ax + b)^{n-1}$.
- **A shorter setting-out** uses the standard form $\dfrac{d}{dx}\big(f(x)\big)^n = n\big(f(x)\big)^{n-1} f'(x)$. It is safer to write $f(x)$ or u on the right and keep at least one middle step.
- Lesson 5I reverses this process, so the full setting-out must be clearly understood.

WORKED EXAMPLE **Year 11 Example 24 (8G)**

Use the chain rule to differentiate each function.

a $(x^2+1)^6$ b $7(3x+4)^5$

SOLUTION

a Let $y=(x^2+1)^6$.

Then $\dfrac{dy}{dx}=\dfrac{dy}{du}\times\dfrac{du}{dx}$

$=6(x^2+1)^5\times 2x$

$=12x(x^2+1)^5$.

Let $u=x^2+1$.

Then $y=u^6$.

Hence $\dfrac{du}{dx}=2x$

and $\dfrac{dy}{du}=6u^5$.

b Let $y=7(3x+4)^5$.

Then $\dfrac{dy}{dx}=\dfrac{dy}{du}\times\dfrac{du}{dx}$

$=35(3x+4)^4\times 3$

$=105(3x+4)^4$.

Let $u=3x+4$.

Then $y=7u^5$.

Hence $\dfrac{du}{dx}=3$

and $\dfrac{dy}{du}=35u^4$.

STEPS THE BOOK SKIPS: YEAR 11 EXAMPLE 24 In each line $\dfrac{dy}{du}$ is written in terms of x by putting u back: in **a**, $6u^5=6(x^2+1)^5$; in **b**, $35u^4=35(3x+4)^4$.

WORKED EXAMPLE **Year 11 Example 25 (8G)**

Use the formula $\dfrac{d}{dx}(ax+b)^n=an(ax+b)^{n-1}$ to differentiate each function. Convert the answers to parts **b** and **c** back to fraction or surd form.

a $y=(4x-1)^7$ b $y=\dfrac{1}{3x-8}$ c $y=\sqrt{5x+3}$

SOLUTION

a $y=(4x-1)^7$

$\dfrac{dy}{dx}=4\times 7\times(4x-1)^6$ (here $a=4,\ b=-1$ and $n=7$)

$=28(4x-1)^6$

b $y=(3x-8)^{-1}$ (convert to index form)

$\dfrac{dy}{dx}=3\times(-1)\times(3x-8)^{-2}$ (here $a=3,\ b=-8$ and $n=-1$)

$=-\dfrac{3}{(3x-8)^2}$

c $y = (5x+3)^{\frac{1}{2}}$ (convert to index form)

$$\frac{dy}{dx} = 5 \times \tfrac{1}{2} \times (5x+3)^{-\frac{1}{2}} \quad \text{(here } a=5,\ b=3 \text{ and } n=\tfrac{1}{2}\text{)}$$

$$= \frac{5}{2\sqrt{5x+3}}$$

WARM-UP · answers are in the answers booklet

Each question is from the textbook exercise named, with some parts left out. Do not use a calculator.

1 Year 11 Exercise 7B Q13

Rewrite these expressions, using fractional and negative indices.

a $\dfrac{1}{\sqrt{x}}$ b $\dfrac{12}{\sqrt{x}}$ c $-\dfrac{5}{\sqrt{x}}$ d $\dfrac{15}{\sqrt[3]{x}}$

e $-\dfrac{4}{\sqrt[3]{x^2}}$ f $x\sqrt{x}$ g $\dfrac{5}{x\sqrt{x}}$ h $8x^2\sqrt{x}$

2 Year 11 Exercise 8E Q5

For each function, first divide through by the denominator. Then find the derivative. Leave your answer with negative indices where appropriate.

a $y = \dfrac{x^3+x^2}{x}$ b $y = \dfrac{x^3+x^2}{x^5}$ c $y = \dfrac{4x-5x^4}{x^2}$ d $y = \dfrac{6x^2+6x^6}{2x^6}$

3 Year 11 Exercise 8G Q4

Use the standard form $\frac{d}{dx}(ax+b)^n = an(ax+b)^{n-1}$ to differentiate these functions.

a $y = (5x-7)^5$ b $y = 9(5x+3)^4$ c $y = (4-3x)^7$

d $y = (4x-5)^{-7}$ e $y = 2(3x+7)^{-5}$ f $y = 3(5-7x)^{-4}$

4 Year 11 Exercise 8G Q8

Differentiate these functions.

a $y = (5-x^2)^3$ b $y = (3x^2+7)^7$

c $y = (3x^3-7)^5$ d $y = (x^3-x^2)^{-5}$

5 Year 12 Exercise 1E Q4

Rewrite each partial sum without sigma notation, then evaluate it.

a $\sum_{n=1}^{6} 2n$ b $\sum_{n=1}^{6}(3n+2)$ c $\sum_{n=5}^{8} n^2$ d $\sum_{n=0}^{5} 2^n$

e $\sum_{n=2}^{4} 3^n$ f $\sum_{n=5}^{105} 4$ g $\sum_{n=0}^{4}(-1)^n(n+5)$

LESSON 4J

Primitive functions

This lesson reverses the process of differentiation and asks, 'What can we say about a function if we know its derivative?' Any two functions with the same derivative differ only by a constant, so the answer is a whole family of functions, the *primitive* $F(x) + C$. The rule 'increase the index by 1 and divide by the new index' finds primitives of powers of x, and of powers of a linear function $ax + b$. An initial condition then picks out one member of the family. The results of this lesson are needed when integration is introduced in Lesson 5A.

CONCEPT 1 · functions with the same derivative

REVERSING DIFFERENTIATION

This section reverses the process of differentiation and asks, 'What can we say about a function if we know its derivative?' The results of this section will be needed when integration is introduced in Chapter 5.

This topic was briefly mentioned in Section 9D of the Year 11 book, but without any terminology or formulae, and we begin the discussion again here.

FUNCTIONS WITH THE SAME DERIVATIVE

Many different functions may all have the same derivative. For example, all these functions have the same derivative $2x$:

$$x^2, \qquad x^2 + 3, \qquad x^2 - 2, \qquad x^2 + 4\tfrac{1}{2}, \qquad x^2 - \pi.$$

These functions are all the same apart from a constant term. This is true generally: *any two functions with the same derivative differ only by a constant.*

WHY THE CONSTANT TERM MAKES NO DIFFERENCE

The derivative of a constant is zero, so the constant term adds nothing to the derivative.

For example, $\dfrac{d}{dx}\left(x^2 + 3\right) = 2x + 0 = 2x$ and $\dfrac{d}{dx}\left(x^2 - \pi\right) = 2x - 0 = 2x$.

14 FUNCTIONS WITH THE SAME DERIVATIVE

A If a function $f(x)$ has derivative zero in an interval $a < x < b$, then $f(x)$ is a constant function in $a < x < b$.

B If $f'(x) = g'(x)$ for all x in an interval $a < x < b$, then $f(x)$ and $g(x)$ differ by a constant in $a < x < b$.

PROOF

A Because the derivative is zero, the gradient of the curve is zero throughout the interval. The curve is therefore a horizontal straight line, and $f(x)$ is a constant function.

B Take the difference between $f(x)$ and $g(x)$ and apply part **A**.

$$\begin{aligned} \text{Let} \quad & h(x) = f(x) - g(x). \\ \text{Then} \quad & h'(x) = f'(x) - g'(x) \\ & = 0, \qquad \text{for all } x \text{ in the interval } a < x < b. \\ \text{Hence by part A,} \quad & h(x) = C, \qquad \text{where } C \text{ is a constant,} \\ \text{so} \quad & f(x) - g(x) = C, \qquad \text{as required.} \end{aligned}$$

READING THE PROOF

The line $h'(x) = f'(x) - g'(x)$ holds because the derivative of a difference is the difference of the derivatives. The next line is zero because $f'(x) = g'(x)$ for all x in the interval.

Notice that both parts of Box 14 are stated for an **interval** $a < x < b$, with no gaps in it.

TUTOR DEMO **Two functions with the same derivative**

Let $f(x) = (x+1)^2$ and $g(x) = x^2 + 2x$.
a Find $f'(x)$ and $g'(x)$.
b Find $f(x) - g(x)$, and explain how your answer agrees with Box 14B.

YOUR TURN **Same method**

Let $f(x) = (x-3)^2$ and $g(x) = x^2 - 6x + 2$.
a Find $f'(x)$ and $g'(x)$.
b Find $f(x) - g(x)$, and explain how your answer agrees with Box 14B.

OUR OWN EXAMPLE: THE SAME DERIVATIVE $2x$

Question. Show that the functions x^2, x^2+3, x^2-2, $x^2+4\frac{1}{2}$ and $x^2-\pi$ all have the same derivative $2x$.

Solution. Differentiate each function. The derivative of x^2 is $2x$, and the derivative of each constant term is zero.

$$\frac{d}{dx}\left(x^2\right)=2x \qquad \frac{d}{dx}\left(x^2+3\right)=2x+0=2x$$

$$\frac{d}{dx}\left(x^2-2\right)=2x-0=2x \qquad \frac{d}{dx}\left(x^2+4\tfrac{1}{2}\right)=2x+0=2x$$

$$\frac{d}{dx}\left(x^2-\pi\right)=2x-0=2x$$

The five functions are the same apart from their constant terms, and any two of them differ by a constant. For example, $\left(x^2+3\right)-\left(x^2-2\right)=5$.

CONCEPT 2 · families of curves, initial conditions and primitives

THE FAMILY OF CURVES WITH THE SAME DERIVATIVE

Continuing with the very first example, the various functions whose derivatives are $2x$ are all of the form

$$f(x)=x^2+C, \qquad \textbf{where } C \textbf{ is a constant.}$$

By taking different values of the constant C, these functions form an infinite family of curves, each consisting of the parabola $y=x^2$ translated upwards or downwards.

INITIAL OR BOUNDARY CONDITIONS

If we know also that the graph of the function passes through a particular point, say $(2,7)$, then we can evaluate the constant C by substituting the point into $f(x)=x^2+C$:

$$7=4+C.$$

Thus $C=3$ and hence $f(x)=x^2+3$. In place of the infinite family of functions, there is now a single function.

Such an extra condition is called an *initial condition* or *boundary condition*.

PRIMITIVES

We need a suitable name for the result of this reverse process. The words *primitive* and *anti-derivative* are both used.

15 A PRIMITIVE OF A FUNCTION

- A function $F(x)$ is called *a primitive* or *an anti-derivative* of $f(x)$ if the derivative of $F(x)$ is $f(x)$,
  $F'(x) = f(x)$.
- If $F(x)$ is any primitive of $f(x)$, then *the general primitive* of $f(x)$ is
  $F(x) + C, \qquad \textbf{where } C \textbf{ is a constant.}$

A PRIMITIVE AND THE PRIMITIVE

The general primitive of $f(x)$ is also called just *the primitive* of $f(x)$. For example, each of these functions is *a primitive* of $x^2 + 1$,

$\frac{1}{3}x^3 + x, \qquad \frac{1}{3}x^3 + x + 7, \qquad \frac{1}{3}x^3 + x - 13, \qquad \frac{1}{3}x^3 + x + 4\pi,$

whereas *the primitive* of $x^2 + 1$ is $\frac{1}{3}x^3 + x + C$, where C is a constant.

CHECKING A PRIMITIVE

Each function in the list is a primitive of $x^2 + 1$ because the derivative of each one is $x^2 + 1$. For example,

$$\frac{d}{dx}\left(\tfrac{1}{3}x^3 + x + 7\right) = \tfrac{1}{3} \times 3x^2 + 1 + 0 = x^2 + 1.$$

TUTOR DEMO **The family with derivative $3x^2$**

a Write down four different functions whose derivative is $3x^2$, and check each one by differentiating.
b Write down the primitive of $3x^2$.
c Find the member of this family whose graph passes through the point $(2, 5)$.

YOUR TURN **Same method**

a Write down four different functions whose derivative is $4x^3$, and check each one by differentiating.
b Write down the primitive of $4x^3$.
c Find the member of this family whose graph passes through the point $(1, -2)$.

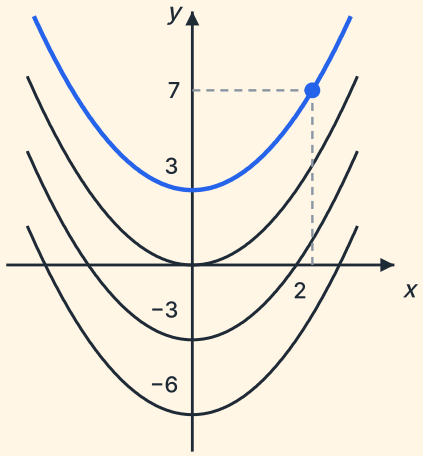
OUR OWN EXAMPLE: THE FAMILY WITH DERIVATIVE $2x$

Question.
a Find all the functions whose derivative is $2x$, and describe their graphs.
b Find the function whose derivative is $2x$ and whose graph passes through the point $(2, 7)$.

Solution. a The derivative of x^2 is $2x$, so x^2 is a primitive of $2x$. By Box 14B, every function whose derivative is $2x$ differs from x^2 by a constant. So the functions are all of the form

$$f(x) = x^2 + C, \qquad \textbf{where } C \textbf{ is a constant.}$$

They form an infinite family of curves. Each curve is the parabola $y = x^2$ translated upwards or downwards. The diagram shows the curves with $C = 3, 0, -3$ and -6.



b Substitute the point $(2, 7)$ into $f(x) = x^2 + C$:

$7 = 2^2 + C$
$7 = 4 + C,$
so $C = 3$, and hence $f(x) = x^2 + 3$. This is the blue curve in the diagram.

CONCEPT 3 · a rule for finding primitives

A RULE FOR FINDING PRIMITIVES

We have seen that a primitive of x is $\frac{1}{2}x^2$, and a primitive of x^2 is $\frac{1}{3}x^3$. Reversing the formula

$$\frac{d}{dx}\left(x^{n+1}\right) = (n+1)x^n$$

gives the general rule:

16 FINDING PRIMITIVES

$$\text{If} \quad \frac{dy}{dx} = x^n, \quad \text{where } n \neq -1,$$

$$\text{then} \quad y = \frac{x^{n+1}}{n+1} + C, \quad \text{for some constant } C.$$

'Increase the index by 1 and divide by the new index.'

WHY THE RULE WORKS

Divide both sides of the formula above by $n+1$. This is allowed only when $n+1 \neq 0$, that is, when $n \neq -1$:

$$\frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = \frac{(n+1)x^n}{n+1} = x^n.$$

So $\frac{x^{n+1}}{n+1}$ is a primitive of x^n, and by Box 15 the general primitive is $\frac{x^{n+1}}{n+1} + C$.

SUMS, MULTIPLES AND CONSTANTS

Derivatives of sums and multiples are found one term at a time (revision, Year 11, 8C), so primitives are found one term at a time too:

- A constant factor stays where it is. For example, a primitive of $5x^3$ is $5 \times \frac{x^4}{4} = \frac{5}{4}x^4$.
- A constant term k is kx^0, so a primitive of it is $k \times \frac{x^1}{1} = kx$. For example, a primitive of 7 is $7x$.
- Only one constant C is needed at the end, because the sum of several constants is a constant.
- Expand a product of brackets first, so that each term is a multiple of a power of x.

CHECK A PRIMITIVE BY DIFFERENTIATING IT

Differentiate your answer. You should get back the function you started with. For example, $\frac{d}{dx}\left(\frac{5}{4}x^4 + 7x + C\right) = 5x^3 + 7$.

TUTOR DEMO **Primitives of polynomials**

Find the primitives (or anti-derivatives) of:

a $x^4 + x^3 + x^2 + 3$ b $4x^5 - 2$ c $(2x + 1)(x - 3)$

YOUR TURN **Same method**

Find the primitives (or anti-derivatives) of:

a $x^5 + x^2 + x + 4$ b $3x^4 + 5$ c $(x + 3)(x - 1)$

WORKED EXAMPLE 26 **From the text: keep for homework**

Find the primitives (or anti-derivatives) of:

a $x^3 + x^2 + x + 1$ b $5x^3 + 7$

SOLUTION

a Let $\frac{dy}{dx} = x^3 + x^2 + x + 1$.
Then $y = \frac{1}{4}x^4 + \frac{1}{3}x^3 + \frac{1}{2}x^2 + x + C$, for some constant C.

b Let $f'(x) = 5x^3 + 7$.
Then $f(x) = \frac{5}{4}x^4 + 7x + C$, for some constant C.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 26

a Increase each index by 1 and divide by the new index:
$y = \frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2}{2} + \frac{x^1}{1} + C = \frac{1}{4}x^4 + \frac{1}{3}x^3 + \frac{1}{2}x^2 + x + C$. The term 1 is x^0,

and its primitive is $\frac{x^1}{1} = x$.

b The 5 is a constant factor, and the primitive of 7 is $7x$:

$$f(x) = 5 \times \frac{x^4}{4} + 7x + C = \tfrac{5}{4}x^4 + 7x + C.$$

CONCEPT 4 · negative and fractional indices

REWRITE WITH INDICES FIRST

The rule in Box 16 works for every index n except $n = -1$, including negative and fractional indices. So first rewrite reciprocals of powers with negative indices, and surds with fractional indices (revision, Year 11, 7A–7B). Then apply the rule.

- **Dividing by a negative index.** For example, $\frac{x^{-1}}{-1} = -x^{-1} = -\frac{1}{x}$.
- **Dividing by a fractional index.** Dividing by $\frac{3}{2}$ is the same as multiplying by $\frac{2}{3}$, so $\frac{x^{\frac{3}{2}}}{\frac{3}{2}} = \frac{2}{3}x^{\frac{3}{2}}$.
- Convert the answer back to fraction form (without negative indices) when the question asks.

TUTOR DEMO **Negative and fractional indices**

Rewrite each function with negative or fractional indices, and find the primitive. Write the answer to part **a** as a fraction without a negative index.

a $\frac{4}{x^3}$

b $\sqrt[3]{x^2}$

YOUR TURN **Same method**

Rewrite each function with negative or fractional indices, and find the primitive. Write the answer to part **a** as a fraction without a negative index.

a $\dfrac{6}{x^4}$ b $x\sqrt{x}$

WORKED EXAMPLE 27 **From the text: keep for homework**

Rewrite each function with negative or fractional indices, and find the primitive.

a $\dfrac{1}{x^2}$ b $\sqrt{x}$

SOLUTION

a Let $f'(x) = \dfrac{1}{x^2}$
$= x^{-2}.$

Then
$f(x) = -x^{-1} + C,$ **where C is a constant,**
$= -\dfrac{1}{x} + C.$

b Let $\dfrac{dy}{dx} = \sqrt{x}$
$= x^{\frac{1}{2}}.$

Then $y = \frac{2}{3}x^{\frac{3}{2}} + C,$ where C is a constant.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 27

a The index -2 increases by 1 to -1, and we divide by -1:

$$f(x) = \frac{x^{-1}}{-1} + C = -x^{-1} + C.$$

b The index $\frac{1}{2}$ increases by 1 to $\frac{3}{2}$, and dividing by $\frac{3}{2}$ is multiplying by $\frac{2}{3}$:

$$y = \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + C = \tfrac{2}{3}x^{\frac{3}{2}} + C.$$

CONCEPT 5 · the linear extension

LINEAR EXTENSION

Reversing the formula

$$\frac{d}{dx}(ax+b)^{n+1} = a(n+1)(ax+b)^n$$

gives:

17 EXTENSION TO POWERS OF LINEAR FUNCTIONS

$$\text{If} \quad \frac{dy}{dx} = (ax+b)^n, \quad \text{where } n \neq -1,$$

$$\text{then} \quad y = \frac{(ax+b)^{n+1}}{a(n+1)} + C, \quad \text{for some constant } C.$$

'Increase the index by 1, then divide by the new index and by the coefficient of x.'

WHY THE RULE WORKS

The formula above comes from the chain rule (revision, Year 11, 8G), with $u = ax + b$ and $\frac{du}{dx} = a$. Divide both sides by $a(n+1)$. This is allowed when $a \neq 0$ and $n \neq -1$:

$$\frac{d}{dx}\left(\frac{(ax+b)^{n+1}}{a(n+1)}\right) = \frac{a(n+1)(ax+b)^n}{a(n+1)} = (ax+b)^n.$$

WATCH THE COEFFICIENT OF x

- In $(1-3x)^6$ the coefficient of x is $a = -3$, so we divide by -3. In $(x+1)^4$ the coefficient is $a = 1$.
- Box 17 is only for a power of a **linear** function $ax + b$. For a power of something like $x^2 + 1$, expand the brackets first for now (Lesson 5I deals with some of these).

TUTOR DEMO **Powers of linear functions**

Find the primitives of:

a $(2x+5)^3$ b $(3-2x)^5$ c $\dfrac{1}{(x+3)^3}$ d $\sqrt{4x+1}$

YOUR TURN **Same method**

Find the primitives of:

a $(4x+3)^5$ b $(2-5x)^4$ c $\dfrac{1}{(x-4)^3}$ d $\sqrt{x+3}$

WORKED EXAMPLE 28 **From the text: keep for homework**

Find the primitives of:

a $(3x+1)^4$ b $(1-3x)^6$ c $\dfrac{1}{(x+1)^2}$ d $\sqrt{x+1}$

SOLUTION

a Let $\dfrac{dy}{dx} = (3x+1)^4$.

Then $y = \dfrac{(3x+1)^5}{5\times 3} + C$

where C is a constant,

$= \dfrac{(3x+1)^5}{15} + C.$

b Let $\dfrac{dy}{dx} = (1-3x)^6$.

Then $y = \dfrac{(1-3x)^7}{7\times(-3)} + C,$

where C is a constant,

$y = -\dfrac{(1-3x)^7}{21} + C.$

c Let $\dfrac{dy}{dx} = \dfrac{1}{(x+1)^2}$

$= (x+1)^{-2}.$

Then $y = \dfrac{(x+1)^{-1}}{-1} + C,$

where C is a constant,

$= -\dfrac{1}{x+1} + C.$

d Let $\dfrac{dy}{dx} = \sqrt{x+1}$

$= (x+1)^{\frac{1}{2}}.$

Then $y = \dfrac{(x+1)^{\frac{3}{2}}}{\frac{3}{2}} + C,$

where C is a constant,

$y = \frac{2}{3}(x+1)^{\frac{3}{2}} + C.$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 28

In Box 17, a is the coefficient of x and n is the index. Divide by the new index $n+1$ and by a.

a Here $a = 3$ and $n = 4$, so $y = \dfrac{(3x+1)^5}{5\times 3} + C = \dfrac{(3x+1)^5}{15} + C.$

b Here $a = -3$ and $n = 6$, so $y = \dfrac{(1-3x)^7}{7\times(-3)} + C = -\dfrac{(1-3x)^7}{21} + C.$

c Here $a = 1$ and $n = -2$, so $y = \dfrac{(x+1)^{-1}}{-1} + C = -\dfrac{1}{x+1} + C.$

d Here $a = 1$ and $n = \frac{1}{2}$, so $y = \dfrac{(x+1)^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{2}{3}(x+1)^{\frac{3}{2}} + C.$

CONCEPT 6 · finding the primitive, given an initial condition

USING AN INITIAL CONDITION

If the derivative of a function is known, plus an initial condition (or boundary condition), then substitute the condition into the general primitive to find the constant, and hence the original function.

18 FINDING A FUNCTION, GIVEN ITS DERIVATIVE AND AN INITIAL CONDITION

- First find the primitive, taking care to include the constant of integration.
- Then substitute the known value of the function to work out the constant.

TUTOR DEMO **One initial condition**

a Given that $\frac{dy}{dx} = 12x^2 - 2$, and $y = 5$ when $x = 1$, find y as a function of x.

b Given that $\frac{dy}{dx} = (2x - 1)^3$, and $y = 2$ when $x = 1$, find y as a function of x.

YOUR TURN **Same method**

a Given that $\frac{dy}{dx} = 6x^2 + 2x$, and $y = 7$ when $x = 1$, find y as a function of x.

b Given that $\frac{dy}{dx} = (3x + 1)^2$, and $y = 1$ when $x = 0$, find y as a function of x.

WORKED EXAMPLE 29 **From the text: keep for homework**

Given that $\frac{dy}{dx} = 6x^2 + 1$, and $y = 12$ when $x = 2$, find y as a function of x.

SOLUTION

We know that $\frac{dy}{dx} = 6x^2 + 1$,

So $y = 2x^3 + x + C$, for some constant C.

Substituting $x = 2$ and $y = 12$, $12 = 16 + 2 + C$,

so $C = -6$, and hence $y = 2x^3 + x - 6$.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 29

The primitive: $y = 6 \times \frac{x^3}{3} + x + C = 2x^3 + x + C$.

Substituting the condition into this primitive, $12 = 2 \times 2^3 + 2 + C$, that is, $12 = 16 + 2 + C$, so $C = -6$.

CONCEPT 7 · starting from the second derivative

TWO PRIMITIVES, TWO CONSTANTS

If the second derivative $f''(x)$ is known, take the primitive twice. The first primitive gives $f'(x)$ with a constant C. The second primitive gives $f(x)$ with a second constant D. The primitive of the constant C is Cx, so $f(x)$ has the form $(\cdots) + Cx + D$.

Two conditions are then needed to find the two constants:

- two values of the function, as in Worked example 30, or
- one value of $f'(x)$ and one value of $f(x)$. Then use the condition on $f'(x)$ to find C before taking the second primitive.

TUTOR DEMO **Two constants**

a Given that $f''(x) = 12(x-2)^2$, and $f(2) = f(0) = 0$, find $f(3)$.
b Given that $y'' = 6x - 2$, and that when $x = 1$, $y' = 4$ and $y = 3$, find y as a function of x.

YOUR TURN **Same method**

a Given that $f''(x) = 6(x+1)$, and $f(0) = 0$ and $f(1) = 6$, find $f(2)$.
b Given that $y'' = 12x + 2$, and that when $x = 0$, $y' = 1$ and $y = -2$, find y as a function of x.

WORKED EXAMPLE 30 **From the text: keep for homework**

Given that $f''(x) = 12(x-1)^2$, and $f(0) = f(1) = 0$, find $f(4)$.

SOLUTION

We know that $f''(x) = 12(x-1)^2$.

Taking the primitive, $f'(x) = 4(x-1)^3 + C$, for some constant C,

and taking the primitive again, $f(x) = (x-1)^4 + Cx + D$, for some constant D.

Because $f(0) = 0$, $0 = 1 + 0 + D$

$D = -1.$

Because $f(1) = 0$, $0 = 0 + C - 1.$

Hence $C = 1$, so $f(4) = 81 + 4 - 1$

$= 84.$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 30

The first primitive: $f'(x) = 12 \times \dfrac{(x-1)^3}{3 \times 1} + C = 4(x-1)^3 + C.$

The second primitive: $f(x) = 4 \times \dfrac{(x-1)^4}{4 \times 1} + Cx + D = (x-1)^4 + Cx + D.$

Because $f(0) = 0$, $0 = (0-1)^4 + C \times 0 + D$. Because $f(1) = 0$,
$0 = (1-1)^4 + C \times 1 + D$.

With $C = 1$ and $D = -1$, $f(x) = (x-1)^4 + x - 1$, and
$f(4) = (4-1)^4 + 4 - 1 = 81 + 4 - 1 = 84.$

TEXTBOOK HOMEWORK · Exercise 4J

FOUNDATION

1 Find the primitive of each function.

a x^6 b x^3 c x^{10} d $3x$

e 5 f $5x^9$ g $21x^6$ h 0

2 Find the primitive of each function.

a $x^2 + x^4$ b $4x^3 - 5x^4$ c $2x^2 + 5x^7$

d $x^2 - x + 1$ e $3 - 4x + 16x^7$ f $3x^2 - 4x^3 - 5x^4$

3 Find the primitive of each function, after first expanding the product.

a $x(x - 3)$ b $(x + 1)(x - 2)$ c $(3x - 1)(x + 4)$

d $x^2(5x^3 - 4x)$ e $2x^3(4x^4 + 1)$ f $(x - 3)(1 + x^2)$

4 Find y as a function of x if:

a $y' = 2x + 3$ and: i $y = 3$ when $x = 0$, ii $y = 8$ when $x = 1$.

b $y' = 9x^2 + 4$ and: i $y = 1$ when $x = 0$, ii $y = 5$ when $x = 1$.

c $y' = 3x^2 - 4x + 7$ and: i $y = 0$ when $x = 0$, ii $y = -1$ when $x = 1$.

DEVELOPMENT

5 Write each function using a negative power of x. Then find the primitive function, writing it as a fraction without a negative index.

a $\dfrac{1}{x^2}$ b $\dfrac{1}{x^3}$ c $-\dfrac{2}{x^3}$ d $-\dfrac{3}{x^4}$ e $\dfrac{1}{x^2} - \dfrac{1}{x^3}$

6 Write each function using a fractional index, and hence find the primitive.

a $\sqrt{x}$ b $\dfrac{1}{\sqrt{x}}$ c $\sqrt[3]{x}$ d $\dfrac{2}{\sqrt{x}}$ e $\sqrt[5]{x^3}$

7 Find y as a function of x if $\dfrac{dy}{dx} = \sqrt{x}$ and:

a $y = 1$ when $x = 0$,

b $y = 2$ when $x = 9$.

8 Find each family of curves whose gradient function is given below. Then sketch the family, and find the member of the family passing through $A(1, 2)$.

a $\dfrac{dy}{dx} = -4x$

b $\dfrac{dy}{dx} = 3$

c $\dfrac{dy}{dx} = 3x^2$

d $\dfrac{dy}{dx} = -\dfrac{1}{x^2}$

9 Recall that if $\dfrac{dy}{dx} = (ax+b)^n$, then $y = \dfrac{(ax+b)^{n+1}}{a(n+1)} + C$, for some constant C. Use this result to find the primitive of each function.

a $(x+1)^3$ b $(x-2)^5$ c $(x+5)^2$ d $(2x+3)^4$ e $(3x-4)^6$

f $(5x-1)^3$ g $(1-x)^3$ h $(1-7x)^3$ i $\dfrac{1}{(x-2)^4}$ j $\dfrac{1}{(1-x)^{10}}$

10 Find the primitive of each function. Use the rule given in the previous question and the fact that $\sqrt{u} = u^{\frac{1}{2}}$.

a $\sqrt{x+1}$ b $\sqrt{x-5}$ c $\sqrt{1-x}$ d $\sqrt{2x-7}$ e $\sqrt{3x-4}$

11 a Find y if $y' = (x-1)^4$, given that $y = 0$ when $x = 1$.

b Find y if $y' = (2x+1)^3$, given that $y = -1$ when $x = 0$.

c Find y if $y' = \sqrt{2x+1}$, given that $y = \frac{1}{3}$ when $x = 0$.

12 a Find the equation of the curve through the origin whose gradient is $\frac{dy}{dx} = 3x^4 - x^3 + 1$.

b Find the curve passing through $(2, 6)$ with gradient function $\frac{dy}{dx} = 2 + 3x^2 - x^3$.

c Find the curve through the point $\left(\frac{1}{5}, 1\right)$ with gradient function $y' = (2 - 5x)^3$.

13 Find y if $\frac{dy}{dt} = 8t^3 - 6t^2 + 5$, and $y = 4$ when $t = 0$. Hence find y when $t = 2$.

14 The primitive of x^n is $\frac{x^{n+1}}{n+1}$, provided that $n \neq -1$. Why can't this rule be used when $n = -1$?

15 Find y if $y'' = 6x + 4$, given that when $x = 1$, $y' = 2$ and $y = 4$.
(Hint: Find y' and use the condition $y' = 2$ when $x = 1$ to find the constant of integration. Then find y and use the condition $y = 4$ when $x = 1$ to find the second constant).

16 A function $f(x)$ has second derivative $f''(x) = 2x - 10$. Its graph passes through the point $(3, -34)$, and at this point the tangent has a gradient of 20.

a Show that $f'(x) = x^2 - 10x + 41$.

b Hence find $f(x)$, and show that its graph cuts the y-axis at $(0, -121)$.

17 If $y'' = 8 - 6x$, show that $y = 4x^2 - x^3 + Cx + D$, for some constants C and D. Hence find the equation of the curve given that it passes through the points $(1, 6)$ and $(-1, 8)$.

ENRICHMENT

18 The gradient function of a curve is given by $f'(x) = -\dfrac{1}{x^2}$. Find the equation of the curve, given that $f(1) = f(-1) = 2$. Sketch a graph of the function.

19 **a** Prove that for a polynomial of degree n, the $(n+1)$th and higher derivatives vanish, but the nth does not.

b Prove that if the $(n+1)$th derivative of a polynomial vanishes but the nth does not, then the polynomial has degree n.

PAST-PAPER HOMEWORK · timed practice

The school trial papers used for this pack (2020–2024) have no question that can be done with this lesson's methods alone: the integration questions in these papers need the definite integral or later work. For timed practice, do Questions 11, 12, 15, 16 and 17 above in 30 minutes.

LESSON 5A

Areas and the definite integral

This lesson extends the study of areas to regions bounded by curves. Cutting a curved region into thin strips, adding up the strips, and taking a limit leads to a new symbol, the definite integral $\int_a^b f(x)\,dx$, which is defined to be an area. When the graph is made of straight lines or parts of circles, the integral is found with the area formulae. For other curves, upper and lower rectangles trap the integral, and a limiting process gives its exact value.

CONCEPT 1 · a new symbol, the definite integral

THE TWO PRINCIPLES OF AREA

All area formulae and calculations of area are based on two principles:

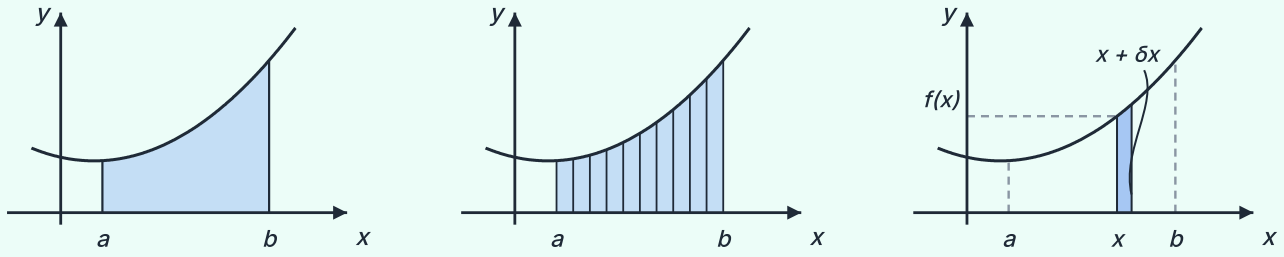
1. Area of a rectangle = length × breadth.
2. When a region is dissected, the area is unchanged.

A region bounded by straight lines, such as a triangle or a trapezium, can be cut up and rearranged into rectangles with a few well-chosen cuts. Dissecting a curved region into rectangles, however, requires an infinite number of rectangles, and so must be a limiting process, just as differentiation is.

A NEW SYMBOL: THE DEFINITE INTEGRAL

Some new notation is needed to reflect this process of infinite dissection as it applies to functions and their graphs.

The diagram on the left below shows the region contained between a given curve $y = f(x)$ and the x-axis, from $x = a$ to $x = b$, where $a \leq b$. The curve must be continuous and, for the moment, never go below the x-axis.



In the middle diagram, the region has been dissected into a number of thin strips. Each strip is approximately a rectangle, but only roughly so, because the upper boundary is curved. The area of the region is the sum of the areas of all the strips.

The third diagram shows just one of the strips, above the value x on the x-axis. Its height at the left-hand end is $f(x)$, and provided the strip is very thin, the height is still about $f(x)$ at the right-hand end. Let the width of the strip be δx, where δx is, as usual in calculus, thought of as being very small. Then, roughly,

$$\begin{aligned}\textbf{area of strip} &\doteq \textbf{height} \times \textbf{width} \\ &\doteq f(x)\,\delta x.\end{aligned}$$

Adding up all the strips, using sigma notation for the sum,

$$\begin{aligned}\textbf{area of shaded region} &\doteq \sum_{x=a}^{b}(\textbf{area of each strip}) \\ &\doteq \sum_{x=a}^{b} f(x)\,\delta x.\end{aligned}$$

Now imagine that there are infinitely many of these strips, each infinitesimally thin, so that the inaccuracy disappears. This involves taking the limit, and we might expect to see something like this

$$\textbf{area of shaded region} = \lim_{\delta x \to 0} \sum_{x=a}^{b} f(x)\,\delta x,$$

but instead, we use the brilliant and flexible notation introduced by Leibniz.

The width δx is replaced by the symbol dx, which suggests an infinitesimal width, and an old form $\int$ of the letter S is used to suggest an infinite sum under a smooth curve. The result is the strange-looking symbol $\int_a^b f(x)\,dx$. We now *define* this symbol to be the shaded area,

$$\int_a^b f(x)\,dx = \textbf{area of shaded region.}$$

This new object $\int_a^b f(x)\,dx$ is called a *definite integral*. The rest of the chapter is concerned with evaluating definite integrals and applying them.

1 THE DEFINITE INTEGRAL

Let $f(x)$ be a function that is continuous in a closed interval $[a, b]$, where $a \leq b$. For the moment, suppose that $f(x)$ is never negative in the interval.

- The *definite integral* $\int_a^b f(x)\,dx$ is defined to be the area of the region between the curve and the x-axis, from $x = a$ to $x = b$.
- The function $f(x)$ is called the *integrand*, and the values $x = a$ and $x = b$ are called the *lower* and *upper limits* (or *bounds*) of the integral.

WHY THE NAME 'INTEGRATION'

The name 'integration' suggests putting many parts together to make a whole. The notation arises from building up the region from an infinitely large number of infinitesimally thin strips. Integration is 'making a whole' from these thin slices.

READING THE SYMBOL

Read $\int_1^4 x^3\,dx$ as 'the integral from 1 to 4 of $x^3\,dx$'. The lower limit is written at the bottom of the integral sign and the upper limit at the top. The dx at the end says that x is the variable along the horizontal axis, and it closes the integral. Before using Box 1, check that the integrand is never negative between the limits.

TUTOR DEMO **Integrand and limits**

a Write the area of the region between the curve $y = x^3$ and the x-axis, from $x = 1$ to $x = 4$, as a definite integral. Name the integrand and the lower and upper limits.

b Describe the region whose area is $\int_0^3 (9 - x^2)\,dx$. Check that the integrand is never negative in the interval.

c Explain why Box 1 cannot yet be used to give a meaning to $\int_{-2}^1 (x + 1)\,dx$.

YOUR TURN **Same skill**

a Write the area of the region between the curve $y = \sqrt{x}$ and the x-axis, from $x = 4$ to $x = 9$, as a definite integral. Name the integrand and the lower and upper limits.

b Describe the region whose area is $\int_{-2}^{2} (4 - x^2)\,dx$. Check that the integrand is never negative in the interval.

c Explain why Box 1 cannot yet be used to give a meaning to $\int_{0}^{3} (x - 1)\,dx$.

CONCEPT 2 · evaluating definite integrals using area formulae

WHEN THE GRAPH IS MADE OF LINES OR CIRCLES

When the function is linear or circular, the definite integral can be calculated from the graph using well-known area formulae, although a quicker method will be developed later for linear functions.

Here are the relevant area formulae:

2 AREA FORMULAE FOR TRIANGLE, TRAPEZIUM AND CIRCLE

Triangle:	$\text{Area} = \frac{1}{2}bh$	$= \frac{1}{2} \times \text{base} \times \text{height}$
Trapezium:	$\text{Area} = \frac{1}{2}(a + b)h$	$= \text{average of parallel sides} \times \text{width}$
Circle:	$\text{Area} = \pi r^2$	$= \pi \times \text{square of the radius}$

NOTES ON THE FORMULAE

For a trapezium, h is the perpendicular distance between the parallel sides. Depending on the orientation, the word ‘height’ or ‘width’ may be more appropriate. Similarly, any side of a triangle may be taken as its ‘base’.

SEMICIRCLES AND QUADRANTS

A semicircle is half a circle, with area $\frac{1}{2}\pi r^2$, and a quadrant is a quarter of a circle, with area $\frac{1}{4}\pi r^2$. The graph of $y = \sqrt{r^2 - x^2}$ is the upper semicircle with centre the origin and radius r, because squaring gives $y^2 = r^2 - x^2$, that is, $x^2 + y^2 = r^2$, with $y \geq 0$ (see the Revision page).

METHOD

1. Sketch the graph of $y = f(x)$ and shade the region from $x = a$ to $x = b$.
2. Name the shape: a rectangle, triangle, trapezium, semicircle or quadrant, or several of these together.
3. Read the lengths from the graph. Widths are differences of x-values, such as $b - a$. Heights are values of the function, such as $f(a)$ and $f(b)$.
4. Apply the area formula, and add the pieces if there are several.

TUTOR DEMO **Lines, absolute value and a semicircle**

Evaluate using a graph and area formulae:

a $\int_2^5 (x-2)\,dx$ b $\int_3^5 (x-2)\,dx$ c $\int_{-4}^2 |x|\,dx$ d $\int_{-6}^6 \sqrt{36-x^2}\,dx$

YOUR TURN **Same method**

Evaluate using a graph and area formulae:

a $\int_2^6 (x-2)\,dx$ b $\int_4^6 (x-2)\,dx$ c $\int_{-2}^3 |x|\,dx$ d $\int_{-2}^2 \sqrt{4-x^2}\,dx$

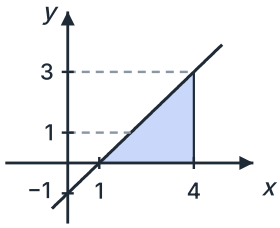
WORKED EXAMPLE 1 **From the text: keep for homework**

Evaluate using a graph and area formulae:

a $\int_1^4 (x-1)\,dx$ b $\int_2^4 (x-1)\,dx$ c $\int_{-2}^2 |x|\,dx$ d $\int_{-5}^5 \sqrt{25-x^2}\,dx$

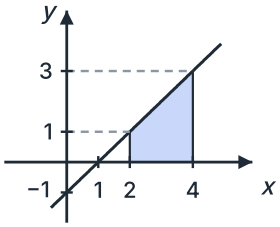
SOLUTION

a The area represented by the integral is the shaded triangle, with base $4-1=3$ and height 3.



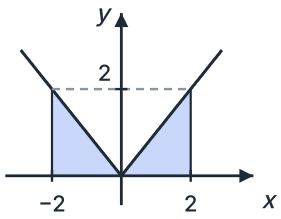
$$\begin{aligned}\text{Hence } \int_1^4 (x-1)\,dx &= \tfrac{1}{2} \times \text{base} \times \text{height} \\ &= \tfrac{1}{2} \times 3 \times 3 \\ &= 4\tfrac{1}{2}.\end{aligned}$$

b The area represented by the integral is the shaded trapezium, with width $4-2=2$ and parallel sides of length 1 and 3.



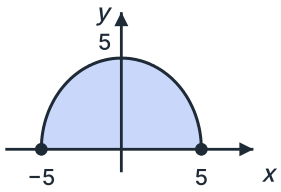
$$\begin{aligned}\text{Hence } \int_2^4 (x-1)\,dx &= \text{average of parallel sides} \times \text{width} \\ &= \frac{1+3}{2} \times 2 \\ &= 4.\end{aligned}$$

c Each shaded triangle has base 2 and height 2.



$$\begin{aligned}\text{Hence } \int_{-2}^{2} |x|\,dx &= 2 \times \left(\tfrac{1}{2} \times \text{base} \times \text{height}\right) \\ &= 2 \times \left(\tfrac{1}{2} \times 2 \times 2\right) \\ &= 4.\end{aligned}$$

d The shaded region is a semi-circle with radius 5.



$$\begin{aligned}\text{Hence } \int_{-5}^{5} \sqrt{25 - x^2}\,dx &= \frac{1}{2} \times \pi r^2 \\ &= \frac{1}{2} \times 5^2 \times \pi \\ &= \frac{25\pi}{2}.\end{aligned}$$

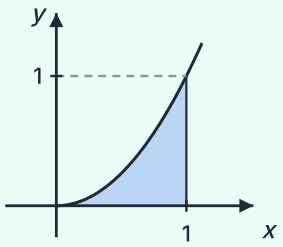
STEPS THE BOOK SKIPS: WORKED EXAMPLE 1

- **a** The height is the value of $x - 1$ at $x = 4$, which is $4 - 1 = 3$.
- **b** The parallel sides 1 and 3 are the values of $x - 1$ at $x = 2$ and at $x = 4$. Then $\dfrac{1 + 3}{2} \times 2 = 2 \times 2 = 4$.
- **c** The value of $|x|$ at $x = -2$ and at $x = 2$ is 2, so each height is 2. Then $2 \times \left(\tfrac{1}{2} \times 2 \times 2\right) = 2 \times 2 = 4$.
- **d** Squaring $y = \sqrt{25 - x^2}$ gives $x^2 + y^2 = 25$ with $y \geq 0$, the upper half of the circle with centre the origin and radius 5.

CONCEPT 3 · using upper and lower rectangles to trap an integral

UPPER AND LOWER RECTANGLES

First-principles integration calculations are more elaborate than those in first-principles differentiation. But they are used when proving the fundamental theorem of calculus, which will soon make calculations of integrals straightforward. One should therefore carry out a very few such calculations in order to understand what is happening. The technique, in geometric forms, was already highly developed by the Greeks.

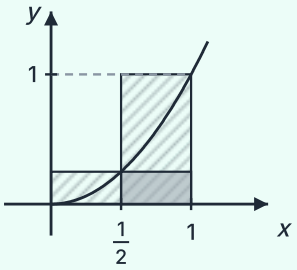


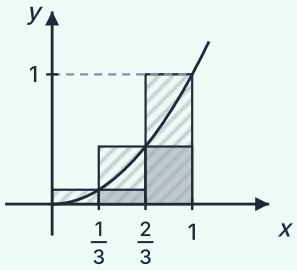
We shall find the integral $A = \int_0^1 x^2\,dx$ sketched on the right.

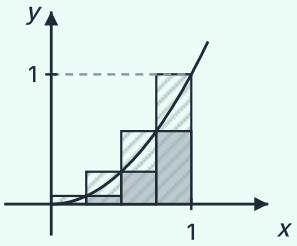
The first step in the process is to trap the integral using *upper rectangles* (or *outer rectangles*) that the shaded region lies inside, and *lower rectangles* (or *inner rectangles*) that lie inside the shaded region. To begin the process, notice that the shaded area is completely contained within the unit square, so

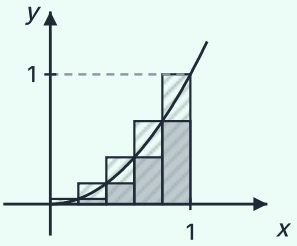
$$0 < A < 1.$$

In the four pictures below, the region has been sliced successively into two, three, four and five strips, then upper and lower rectangles have been constructed so that the region is trapped, or sandwiched, between the upper and lower rectangles. Calculating the areas of these upper and lower rectangles provides tighter and tighter bounds on the area A.









READING THE PICTURES

In the pictures, the lower rectangles are grey and the upper rectangles are hatched. Each upper rectangle covers the lower rectangle below it.

FINDING THE HEIGHTS

Because $y = x^2$ is increasing for $x \geq 0$, the lowest point of the curve over each strip is at its left-hand end, and the highest point is at its right-hand end. So each **lower** rectangle has the height of the curve at the **left** end of its strip, and each **upper** rectangle has the height at the **right** end. The first lower rectangle has height $0^2 = 0$, so it has zero area. Every rectangle has the same width, $\frac{1}{n}$ for n strips, so that width can be taken out as a common factor.

TUTOR DEMO **Trapping $\int_0^1 x^3\,dx$**

Let $A = \displaystyle\int_0^1 x^3\,dx$.

a Sketch the region and explain why $0 < A < 1$.

b Divide the interval $0 \leq x \leq 1$ into 2 strips of equal width. Find the sums of the lower rectangles and of the upper rectangles, and hence write down bounds for A.

c Repeat part **b** with 4 strips of equal width.

YOUR TURN **Same method**

Let $A = \displaystyle\int_0^2 x^2\,dx$.

a Sketch the region and explain why $0 < A < 8$.

b Divide the interval $0 \leq x \leq 2$ into 2 strips of equal width. Find the sums of the lower rectangles and of the upper rectangles, and hence write down bounds for A.

c Repeat part **b** with 4 strips of equal width.

THE BOUNDS FROM THE FOUR PICTURES

In the first picture,

$$\frac{1}{2}\times\left(\frac{1}{2}\right)^2 < A < \frac{1}{2}\times\left(\frac{1}{2}\right)^2 + \frac{1}{2}\times 1^2$$
$$\frac{1}{8} < A < \frac{5}{8}.$$

In the second picture,

$$\frac{1}{3}\times\left(\frac{1}{3}\right)^2 + \frac{1}{3}\times\left(\frac{2}{3}\right)^2 < A < \frac{1}{3}\times\left(\frac{1}{3}\right)^2 + \frac{1}{3}\times\left(\frac{2}{3}\right)^2 + \frac{1}{3}\times 1^2$$
$$\frac{5}{27} < A < \frac{14}{27}.$$

In the third picture,

$$\frac{1}{4}\left(\left(\frac{1}{4}\right)^2 + \left(\frac{2}{4}\right)^2 + \left(\frac{3}{4}\right)^2\right) < A$$
$$< \frac{1}{4}\left(\left(\frac{1}{4}\right)^2 + \left(\frac{2}{4}\right)^2 + \left(\frac{3}{4}\right)^2 + \left(\frac{4}{4}\right)^2\right)$$
$$\frac{14}{64} < A < \frac{30}{64}.$$

In the fourth picture,

$$\frac{1}{5}\left(\left(\frac{1}{5}\right)^2 + \left(\frac{2}{5}\right)^2 + \left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2\right) < A$$
$$< \frac{1}{5}\left(\left(\frac{1}{5}\right)^2 + \left(\frac{2}{5}\right)^2 + \left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 + \left(\frac{5}{5}\right)^2\right)$$
$$\frac{30}{125} < A < \frac{55}{125}.$$

STEPS THE BOOK SKIPS: THE FOUR PICTURES

In the first picture, there are two strips of width $\frac{1}{2}$. The lower rectangles have heights $0^2 = 0$ and $\left(\frac{1}{2}\right)^2$, and the upper rectangles have heights $\left(\frac{1}{2}\right)^2$ and 1^2.

$$\frac{1}{2}\times\left(\frac{1}{2}\right)^2 < A < \frac{1}{2}\times\left(\frac{1}{2}\right)^2 + \frac{1}{2}\times 1^2$$
$$\frac{1}{2}\times\frac{1}{4} < A < \frac{1}{2}\times\frac{1}{4} + \frac{1}{2}\times 1$$
$$\frac{1}{8} < A < \frac{1}{8} + \frac{4}{8}$$
$$\frac{1}{8} < A < \frac{5}{8}.$$

In the second picture, there are three strips of width $\frac{1}{3}$.

$$\frac{1}{3}\times\left(\frac{1}{3}\right)^2 + \frac{1}{3}\times\left(\frac{2}{3}\right)^2 < A < \frac{1}{3}\times\left(\frac{1}{3}\right)^2 + \frac{1}{3}\times\left(\frac{2}{3}\right)^2 + \frac{1}{3}\times 1^2$$
$$\frac{1}{3}\times\frac{1}{9} + \frac{1}{3}\times\frac{4}{9} < A < \frac{1}{3}\times\frac{1}{9} + \frac{1}{3}\times\frac{4}{9} + \frac{1}{3}\times 1$$
$$\frac{1}{27} + \frac{4}{27} < A < \frac{1}{27} + \frac{4}{27} + \frac{9}{27}$$
$$\frac{5}{27} < A < \frac{14}{27}.$$

In the third picture, there are four strips of width $\frac{1}{4}$. Taking out the common factor $\frac{1}{4}$, the lower and upper sums are

$$\tfrac{1}{4}\left(\left(\tfrac{1}{4}\right)^2+\left(\tfrac{2}{4}\right)^2+\left(\tfrac{3}{4}\right)^2\right)=\tfrac{1}{4}\times\tfrac{1+4+9}{16}=\tfrac{14}{64},$$

$$\tfrac{1}{4}\left(\left(\tfrac{1}{4}\right)^2+\left(\tfrac{2}{4}\right)^2+\left(\tfrac{3}{4}\right)^2+\left(\tfrac{4}{4}\right)^2\right)=\tfrac{1}{4}\times\tfrac{1+4+9+16}{16}=\tfrac{30}{64},$$

so $\tfrac{14}{64}<A<\tfrac{30}{64}$.

In the fourth picture, there are five strips of width $\tfrac{1}{5}$. The lower and upper sums are

$$\tfrac{1}{5}\left(\left(\tfrac{1}{5}\right)^2+\left(\tfrac{2}{5}\right)^2+\left(\tfrac{3}{5}\right)^2+\left(\tfrac{4}{5}\right)^2\right)=\tfrac{1}{5}\times\tfrac{1+4+9+16}{25}=\tfrac{30}{125},$$

$$\tfrac{1}{5}\left(\left(\tfrac{1}{5}\right)^2+\left(\tfrac{2}{5}\right)^2+\left(\tfrac{3}{5}\right)^2+\left(\tfrac{4}{5}\right)^2+\left(\tfrac{5}{5}\right)^2\right)=\tfrac{1}{5}\times\tfrac{1+4+9+16+25}{25}=\tfrac{55}{125},$$

so $\tfrac{30}{125}<A<\tfrac{55}{125}$.

As decimals, the four sandwiches are $0.125<A<0.625$, $0.185\ldots<A<0.518\ldots$, $0.21875<A<0.46875$ and $0.24<A<0.44$: the bounds are getting tighter.

CONCEPT 4 · the limiting process

TURNING THE SANDWICH INTO A LIMIT

The bounds on the area are getting tighter, but the exact value of the area can only be obtained if this sandwiching process can be turned into a limiting process. The calculations below will need the formula for the sum of the first n squares, proven in Question 2f of Exercise 2A by induction:

$$1^2+2^2+\cdots+n^2=\tfrac{1}{6}n(n+1)(2n+1).$$

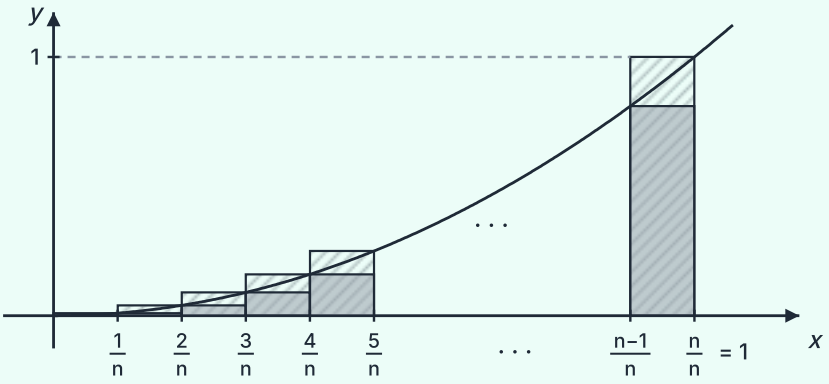
THE FOUR STEPS

- **A** Divide the interval into n subintervals of equal width. Then (sum of lower rectangles) $\le A \le$ (sum of upper rectangles), however many strips there are.
- **B** Write the sum of the upper rectangles in terms of n, using the formula, and find its limit as $n\to\infty$.
- **C** Do the same for the sum of the lower rectangles. The formula is used with $n-1$ in place of n.
- **D** Both sums have the same limit, and A is trapped between them, so A equals that limit.

THE ALGEBRA AND THE LIMITS

- Write n^3 in the denominator as $n \times n \times n$, and split the fraction into factors that each have a limit: for example $\dfrac{n+1}{n} = 1 + \dfrac{1}{n}$.
- As $n \to \infty$, $\dfrac{1}{n} \to 0$ and $\dfrac{1}{2n} \to 0$, so $1 + \dfrac{1}{n} \to 1$ and $1 - \dfrac{1}{2n} \to 1$.
- Substituting $n-1$ for n in $\frac{1}{6}n(n+1)(2n+1)$ gives $\frac{1}{6}(n-1)\big((n-1)+1\big)\big(2(n-1)+1\big) = \frac{1}{6}(n-1)n(2n-1)$.

THE LIMITING PROCESS: $\displaystyle\int_0^1 x^2\,dx = \tfrac{1}{3}$



A Divide the interval $0 \le x \le 1$ into n subintervals, each of width $\dfrac{1}{n}$.
On each subinterval form the upper rectangle and the lower rectangle.
Then the required region is entirely contained within the upper rectangles, and, in turn, the lower rectangles are entirely contained within the required region. Thus however many strips the region has been dissected into,

$$\textbf{(sum of lower rectangles)} \le A \le \textbf{(sum of upper rectangles)}.$$

B The heights of the successive upper rectangles are $\dfrac{1^2}{n^2}, \dfrac{2^2}{n^2}, \ldots, \dfrac{n^2}{n^2}$, and so, using the formula quoted above,

$$\begin{aligned}
\textbf{sum of upper rectangles} &= \frac{1}{n}\left(\frac{1^2}{n^2} + \frac{2^2}{n^2} + \cdots + \frac{n^2}{n^2}\right) \\
&= \frac{1}{n^3}(1^2 + 2^2 + \cdots + n^2) \\
&= \frac{1}{n^3} \times \frac{n(n+1)(2n+1)}{6}, \\
&\qquad \textbf{(using the formula above)} \\
&= \frac{1}{3} \times \frac{n}{n} \times \frac{n+1}{n} \times \frac{2n+1}{2n} \\
&= \frac{1}{3} \times \left(1 + \frac{1}{n}\right) \times \left(1 + \frac{1}{2n}\right),
\end{aligned}$$

hence the sum of the upper rectangles has limit $\frac{1}{3}$ as $n \to \infty$.

C The heights of the successive lower rectangles are $0,\ \dfrac{1^2}{n^2},\ \dfrac{2^2}{n^2},\ \ldots,\ \dfrac{(n-1)^2}{n^2}$, so substituting $n-1$ for n into the quoted formula,

$$\begin{aligned}\textbf{sum of lower rectangles} &= \frac{1}{n}\left(0 + \frac{1^2}{n^2} + \frac{2^2}{n^2} + \cdots + \frac{(n-1)^2}{n^2}\right)\\ &= \frac{1}{n^3}\left(1^2 + 2^2 + \cdots + (n-1)^2\right)\\ &= \frac{1}{n^3} \times \frac{(n-1)n(2n-1)}{6}\\ &= \frac{1}{3} \times \frac{n}{n} \times \frac{n-1}{n} \times \frac{2n-1}{2n}\\ &= \frac{1}{3} \times \left(1 - \frac{1}{n}\right) \times \left(1 - \frac{1}{2n}\right),\end{aligned}$$

hence the sum of the lower rectangles also has limit $\frac{1}{3}$ as $n \to \infty$.

D Finally, (sum of lower rectangles) $\leq A \leq$ (sum of upper rectangles), and both these sums have the same limit $\frac{1}{3}$, so it follows that $A = \frac{1}{3}$.

The conclusion of all this heavy machinery of limits is simply that $\displaystyle\int_0^1 x^2\,dx = \tfrac{1}{3}$. You will be relieved to know that we will soon have much quicker methods.

STEPS THE BOOK SKIPS: THE LIMITING PROCESS

- In **B**, every rectangle has width $\dfrac{1}{n}$ and every height has denominator n^2:

$$\begin{aligned}\frac{1}{n}\left(\frac{1^2}{n^2} + \frac{2^2}{n^2} + \cdots + \frac{n^2}{n^2}\right) &= \frac{1}{n} \times \frac{1}{n^2} \times \left(1^2 + 2^2 + \cdots + n^2\right)\\ &= \frac{1}{n^3}(1^2 + 2^2 + \cdots + n^2).\end{aligned}$$

- To split the fraction, write $6n^3 = 3 \times n \times n \times 2n$:

$$\begin{aligned}\frac{1}{n^3} \times \frac{n(n+1)(2n+1)}{6} &= \frac{n(n+1)(2n+1)}{3 \times n \times n \times 2n}\\ &= \frac{1}{3} \times \frac{n}{n} \times \frac{n+1}{n} \times \frac{2n+1}{2n}.\end{aligned}$$

- The limit: $\dfrac{1}{n} \to 0$ and $\dfrac{1}{2n} \to 0$ as $n \to \infty$, so the sum of the upper rectangles has limit $\frac{1}{3} \times 1 \times 1 = \frac{1}{3}$.

TUTOR DEMO **Proving $\int_0^2 x^2\,dx = \frac{8}{3}$**

Let $A = \displaystyle\int_0^2 x^2\,dx$. Divide the interval $0 \le x \le 2$ into n subintervals, each of width $\dfrac{2}{n}$. Use the formula $1^2 + 2^2 + \cdots + n^2 = \frac{1}{6}n(n+1)(2n+1)$ to find the sums of the upper and lower rectangles in terms of n. Take the limit as $n \to \infty$, and hence prove that $A = \frac{8}{3}$.

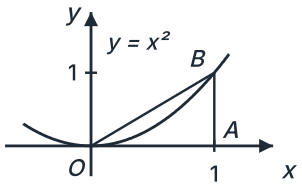
YOUR TURN **Same method**

Let $A = \displaystyle\int_0^1 x\,dx$. Divide the interval $0 \le x \le 1$ into n subintervals, each of width $\dfrac{1}{n}$. Use the formula $1 + 2 + \cdots + n = \frac{1}{2}n(n+1)$ to find the sums of the upper and lower rectangles in terms of n. Take the limit as $n \to \infty$, and hence prove that $A = \frac{1}{2}$. Check your answer with the area of a triangle.

TEXTBOOK HOMEWORK · Exercise 5A

FOUNDATION

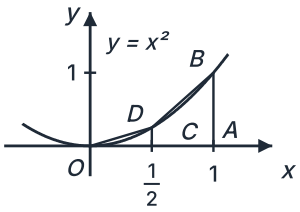
1



a Find the area of $\triangle OAB$ in the diagram above.

b Hence explain why $\int_0^1 x^2\,dx < \frac{1}{2}$.

2

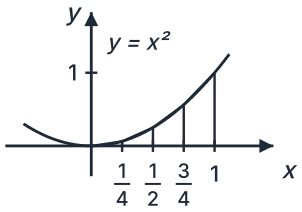


a Find the area of $\triangle OCD$ in the diagram above.

b Find the area of the trapezium $CABD$.

c Hence explain why $\int_0^1 x^2\,dx < \frac{3}{8}$.

3

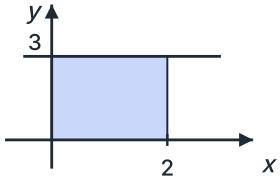


a Use the diagram above to show that $\int_0^1 x^2\,dx < \frac{11}{32}$.

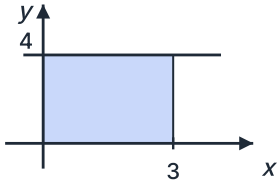
b Explain why $\frac{11}{32}$ is a better approximation to $\int_0^1 x^2\,dx$ than $\frac{3}{8}$ is.

4 Use area formulae to calculate these definite integrals (sketches are given below).

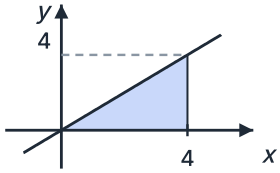
a $\int_0^2 3\,dx$



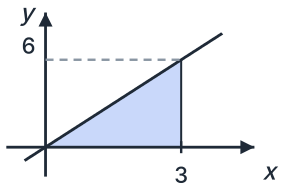
b $\int_0^3 4\,dx$



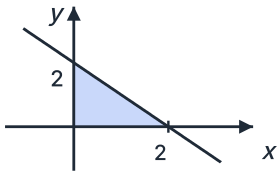
c $\int_0^4 x\,dx$



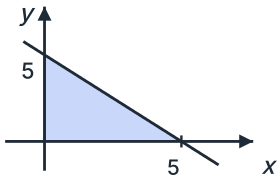
d $\int_0^3 2x\,dx$



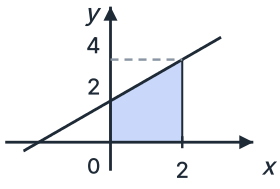
e $\int_0^2 (2-x)\,dx$



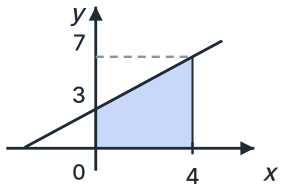
f $\int_0^5 (5-x)\,dx$



g $\int_0^2 (x+2)\,dx$

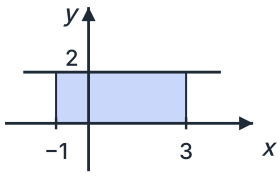


h $\int_0^4 (x+3)\,dx$

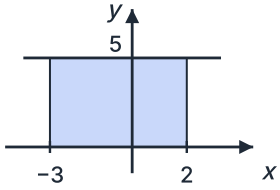


5 Use area formulae to calculate the sketched definite integrals.

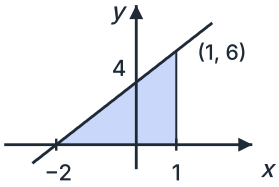
a $\int_{-1}^{3} 2\,dx$



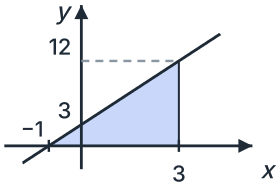
b $\int_{-3}^{2} 5\,dx$



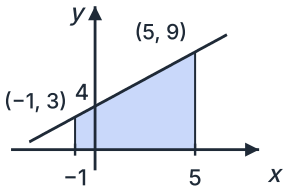
c $\int_{-2}^{1} (2x+4)\,dx$



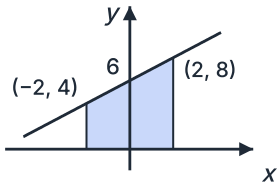
d $\int_{-1}^{3} (3x+3)\,dx$



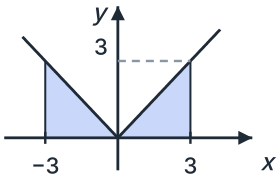
e $\int_{-1}^{5} (x+4)\,dx$



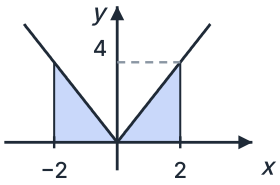
f $\int_{-2}^{2} (x+6)\,dx$



g $\int_{-3}^{3} |x|\,dx$



h $\int_{-2}^{2} |2x|\,dx$



6 Sketch a graph of each definite integral, then use an area formula to calculate it.

a $\int_{0}^{3} 5\,dx$

b $\int_{-3}^{0} 5\,dx$

c $\int_{-1}^{4} 5\,dx$

d $\int_{-2}^{6} 5\,dx$

e $\int_{-5}^{0} (x+5)\,dx$

f $\int_{0}^{2} (x+5)\,dx$

g $\int_{2}^{4} (x+5)\,dx$

h $\int_{-1}^{3} (x+5)\,dx$

i $\int_{4}^{8} (x-4)\,dx$

j $\int_{4}^{10} (x-4)\,dx$

k $\int_{5}^{7} (x-4)\,dx$

l $\int_{6}^{10} (x-4)\,dx$

m $\int_{-2}^{2} |x|\,dx$

n $\int_{-4}^{4} |x|\,dx$

o $\int_{0}^{5} |x-5|\,dx$

p $\int_{5}^{10} |x-5|\,dx$

7 Sketch a graph of each definite integral, then use an area formula to calculate it.

a $\int_{-4}^{4} \sqrt{16 - x^2}\, dx$

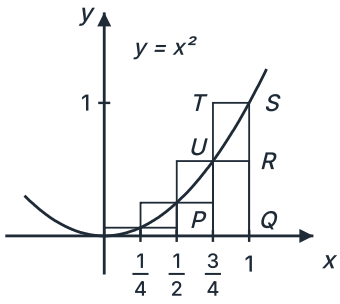
b $\int_{-5}^{0} \sqrt{25 - x^2}\, dx$

DEVELOPMENT

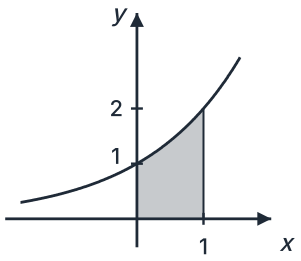
8 a In the diagram to the right, add the areas of the lower rectangles. (For example, $PQRU$ is a lower rectangle.)

b Add the areas of the upper rectangles. (For example, $PQST$ is an upper rectangle.)

c Hence explain why $\frac{7}{32} < \int_0^1 x^2\, dx < \frac{15}{32}$.



9 The area of the region in the diagram to the right is given by $\int_0^1 2^x \, dx$.



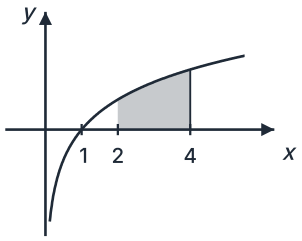
a Use one lower and one upper rectangle to show that $1 < \int_0^1 2^x \, dx < 2$.

b Use 2 lower and 2 upper rectangles of equal width to show that (with decimals rounded to one place) $1.2 < \int_0^1 2^x \, dx < 1.7$.

c Use 4 lower and 4 upper rectangles of equal width to show that $1.3 < \int_0^1 2^x \, dx < 1.6$.

d What trend can be identified in the parts above?

10 The area of the region in the diagram to the right is given by $\int_2^4 \ln x \, dx$.



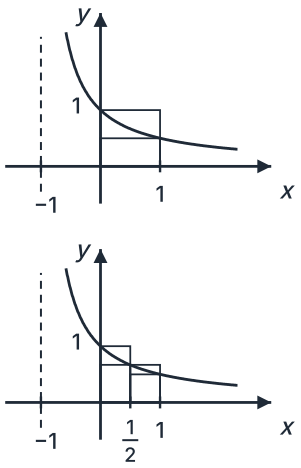
a Use 2 lower and 2 upper rectangles to show that (with decimals rounded to 2 places) $1.79 < \int_2^4 \ln x \, dx < 2.48$.

b Use 4 lower and 4 upper rectangles of equal width to show that $1.98 < \int_2^4 \ln x \, dx < 2.33$.

c Use 8 lower and 8 upper rectangles of equal width to show that $2.07 < \int_2^4 \ln x \, dx < 2.24$.

d What trend can be identified in the parts above?

11 Let $A = \int_0^1 \frac{1}{x+1}\,dx$.



a Use the areas of the lower and upper rectangles in the top diagram to show that $\frac{1}{2} < A < 1$.

b Use the areas of the 2 lower and 2 upper rectangles in the bottom diagram to show that $\frac{7}{12} < A < \frac{5}{6}$. (That is, $0.58 < A < 0.83$, correct to 2 decimal places.)

c Use 3 lower and 3 upper rectangles of equal width to show that $\frac{37}{60} < A < \frac{47}{60}$. (That is, $0.62 < A < 0.78$, correct to 2 decimal places.)

d Finally, use 4 lower and 4 upper rectangles of equal width to show that $\frac{533}{840} < A < \frac{319}{420}$. (That is, correct to 2 decimal places, $0.63 < A < 0.76$.)

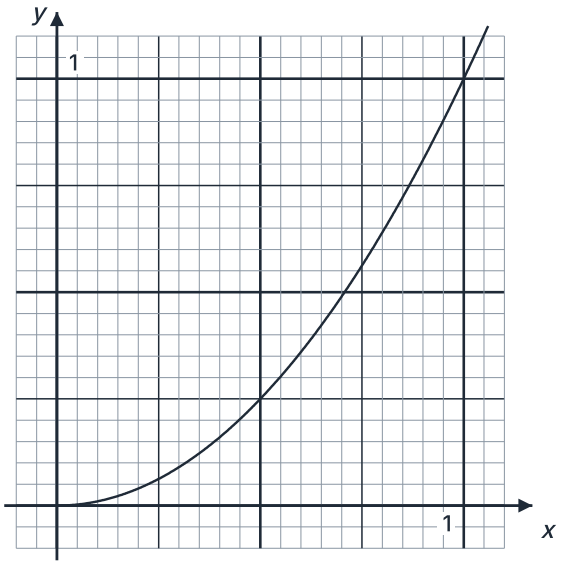
e As the number of rectangles increases, is the interval within which A lies getting bigger or smaller?

f The exact value of A is $\ln 2 = 0.693\,147\ldots$. Do the lower and upper limits of the intervals in parts **a** to **d** seem to be approaching the exact value?

12 [Technology] Some of the previous questions involve summing the areas of lower and upper rectangles to approximate a definite integral. Many software programs can do this automatically, using any prescribed number of rectangles. Steadily increasing the number of rectangles will show the sums of the lower and upper rectangles converging to the exact area, which can be checked either using area formulae or using the exact value of the definite integral as calculated later in the course.
Investigate some of the definite integrals from Questions 1–3 and 8–13 in this way.

13 The diagram to the right shows the graph of $y = x^2$ from $x = 0$ to $x = 1$, drawn on graph paper.

The scale is 20 little divisions to 1 unit. This means that 400 little squares make up 1 square unit.



a Count how many little squares there are under the graph from $x = 0$ to $x = 1$ (keeping reasonable track of fragments of squares), then divide by 400 to approximate $\int_0^1 x^2\,dx$.

Write your answer correct to 2 decimal places.

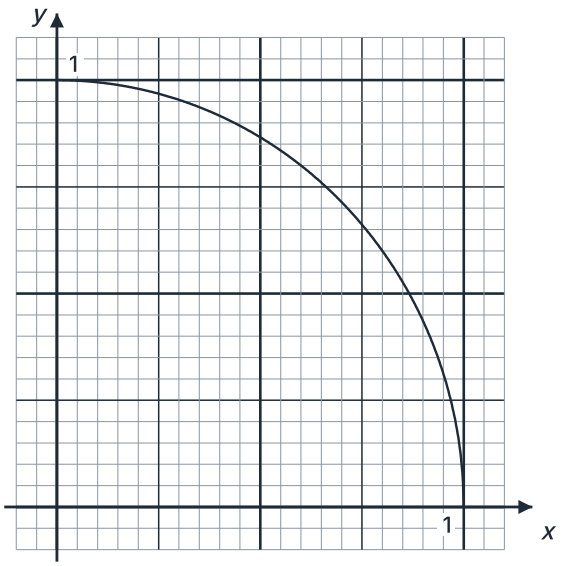
b By counting the appropriate squares, approximate:

(i) $\int_0^{\frac{1}{2}} x^2\,dx$ (ii) $\int_{\frac{1}{2}}^1 x^2\,dx$

Confirm that the sum of the answers to parts (i) and (ii) is the answer to part **a**.

14 The diagram to the right shows the quadrant $y = \sqrt{1 - x^2}$, from $x = 0$ to $x = 1$.

As before, the scale is 20 little divisions to 1 unit.



a Count how many little squares there are under the graph from $x = 0$ to $x = 1$.

b Divide by 400 to approximate $\int_0^1 \sqrt{1 - x^2}\,dx$.

Write your answer correct to 2 decimal places.

c Hence, using the fact that a quadrant has area $\frac{1}{4}\pi r^2$, find an approximation for π. Give your answer correct to 2 decimal places.

ENRICHMENT

15 Using exactly the same setting out as in the example in the notes above this exercise, prove from first principles that $\int_0^1 x^3\,dx = \frac{1}{4}$. Use upper and lower rectangles, and take the limit as the number of rectangular strips approaches infinity.

Note: This calculation will need the formula $1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{1}{4}n^2(n+1)^2$ for the sum of the first n cubes, which was proven by mathematical induction in worked Example 1 of Section 2A.

16 Prove the following two definite integrals from first principles. Use upper and lower rectangles, and take the limit as the number of rectangular strips approaches infinity.

a $\displaystyle\int_0^a x^2\,dx = \frac{a^3}{3}$

b $\displaystyle\int_0^a x^3\,dx = \frac{a^4}{4}$

17 Draw a large sketch of $y = x^2$ for $0 \leq x \leq 1$, and let U be the point $(1, 0)$. For some positive integer n, let $P_0\,(= O),\ P_1,\ P_2,\ \ldots,\ P_n$ be the points on the curve with x-coordinates $x = 0,\ x = \dfrac{1}{n},\ x = \dfrac{2}{n},\ \ldots,\ x = \dfrac{n}{n} = 1$. Join the chords $P_0P_1,\ P_1P_2,\ \ldots,\ P_{n-1}P_n$, and join P_nU.

a Use the area formula for a trapezium to find the area of the polygon $P_0P_1P_2 \cdots P_nU$.

b Explain geometrically why this area is always greater than $\int_0^1 x^2\,dx$.

c Show that its limit as $n \to \infty$ is $\frac{1}{3}$. This is half of the proof that $\int_0^1 x^2\,dx = \frac{1}{3}$.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 5 minutes **1 mark** Calculator allowed · show working · hints are below each question if you get stuck

This is the only school-paper item that needs just this lesson's ideas. For more timed practice, do Questions 5(d), 6(h), 6(o), 7(a), 8 and 11(c) above in 25 minutes.

PP 1 **James Ruse Agricultural High School, 2023, Year 12 Trial HSC Examination, Mathematics Extension 1, Q4** **1 mark**

A function is defined by $f(x) = \begin{cases} 2, & x < 3 \\ x - 1, & x \geq 3 \end{cases}$. The value of $\int_1^5 f(x)\,dx$ is

A. 2
B. 6
C. 8
D. 10

Hint: what does the graph of $y = f(x)$ look like between $x = 1$ and $x = 5$?

LESSON 5B

The fundamental theorem of calculus

There is a remarkably simple formula for evaluating definite integrals, based on taking the primitive of the function. The formula is called the *fundamental theorem of calculus*, because the whole of calculus depends on it. This lesson states the theorem and uses it: write down a primitive, substitute the upper and lower limits, and subtract. Expanding brackets, splitting fractions and negative indices prepare the integrand first. The challenging proof is delayed until Lesson 5D, when the usefulness of the theorem will have been established.

CONCEPT 1 · the fundamental theorem of calculus

PRIMITIVES

Let us first review from Lesson 4J what primitives are, and the first step in finding them.

3 PRIMITIVES

- A function $F(x)$ is called a *primitive* or *anti-derivative* of a function $f(x)$ if its derivative is $f(x)$:

  $$F(x) \text{ is a primitive of } f(x) \text{ if } F'(x) = f(x).$$

- To find the general primitive of a power x^n, where $n \neq -1$:

  $$\text{If } \frac{dy}{dx} = x^n, \text{ then } y = \frac{x^{n+1}}{n+1} + C, \text{ for some constant } C.$$

'Increase the index by 1 and divide by the new index.'

STATEMENT OF THE FUNDAMENTAL THEOREM

The fundamental theorem says that a definite integral can be evaluated by writing down any primitive $F(x)$ of $f(x)$, then substituting the upper and lower limits into it and subtracting.

4 THE FUNDAMENTAL THEOREM OF CALCULUS

Let $f(x)$ be a function that is continuous in a closed interval $[a, b]$. Then

$$\int_a^b f(x)\,dx = F(b) - F(a), \quad \text{where } F(x) \text{ is any primitive of } f(x).$$

AREAS AND TANGENTS ARE INVERSE PROCESSES

This result is extraordinary because it says that taking areas and taking tangents are inverse processes, which is not obvious.

WHY 'INVERSE PROCESSES'

The definite integral was defined in Lesson 5A as an area. Box 4 says that its value comes from a primitive, that is, from reversing differentiation, and differentiation is about the gradients of tangents.

USING THE FUNDAMENTAL THEOREM TO EVALUATE AN INTEGRAL

The conventional way to set out these calculations is to enclose the primitive in square brackets, writing the lower and upper limits as subscript and superscript respectively:

$$\int_a^b f(x)\,dx = \Big[F(x)\Big]_a^b = F(b) - F(a).$$

WHY ANY PRIMITIVE WILL DO

Every primitive of $f(x)$ has the form $F(x) + C$. With the constant included,

$$\Big[F(x) + C\Big]_a^b = \big(F(b) + C\big) - \big(F(a) + C\big) = F(b) - F(a),$$

because the two constants cancel. So the constant is left out when evaluating a definite integral.

SETTING OUT

1. Write the primitive in square brackets, with the limits at the right-hand bracket.
2. Substitute the upper limit, then substitute the lower limit, and subtract: (value at b) − (value at a).
3. Simplify the arithmetic one line at a time.

TUTOR DEMO **Evaluate, then check with areas**

Evaluate these definite integrals. Then draw diagrams to show the regions that they represent, and check the answers using area formulae.

a $\displaystyle\int_0^3 2x\,dx$

b $\displaystyle\int_2^5 (2x - 3)\,dx$

YOUR TURN **Same method**

Evaluate these definite integrals. Then draw diagrams to show the regions that they represent, and check the answers using area formulae.

a $\int_0^1 4x\,dx$

b $\int_3^5 (2x-5)\,dx$

WORKED EXAMPLE 1 **From the text: keep for homework**

Evaluate these definite integrals. Then draw diagrams to show the regions that they represent, and check the answers using area formulae.

a $\int_0^2 2x\,dx$ b $\int_2^4 (2x-3)\,dx$

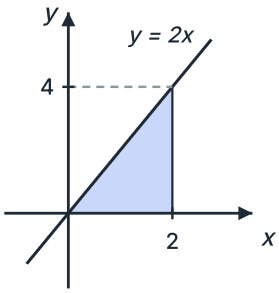
SOLUTION

a

$$\begin{aligned}\int_0^2 2x\,dx &= \Big[x^2\Big]_0^2 \qquad (x^2 \text{ is a primitive of } 2x)\\ &= 2^2 - 0^2 \qquad (\text{substitute 2, then substitute 0 and subtract})\\ &= 4\end{aligned}$$

Using areas,

$$\begin{aligned}\text{area of triangle} &= \tfrac{1}{2} \times 2 \times 4\\ &= 4.\end{aligned}$$

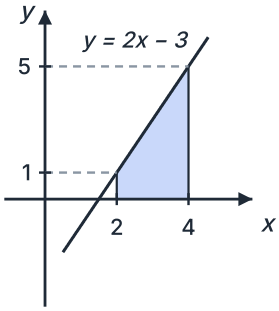


b

$$\begin{aligned}\int_2^4 (2x-3)\,dx &= \Big[x^2 - 3x\Big]_2^4 \qquad (\text{take the primitive of each term})\\ &= (16-12) - (4-6) \qquad (\text{substitute 4, then substitute 2})\\ &= 4 - (-2)\\ &= 6\end{aligned}$$

Using areas,

$$\begin{aligned}\text{area of trapezium} &= \frac{1+5}{2} \times 2\\ &= 6.\end{aligned}$$



STEPS THE BOOK SKIPS: WORKED EXAMPLE 1

- **a** x^2 is a primitive of $2x$, because $\frac{d}{dx}x^2 = 2x$. Then $2^2 - 0^2 = 4 - 0 = 4$.
- Using areas: the region is a triangle with base $2 - 0 = 2$ and height $2 \times 2 = 4$ (the value of $2x$ at $x = 2$).

- **b** Take the primitive of each term: x^2 for $2x$, and $-3x$ for -3. Substitute 4, then 2: $(4^2 - 3 \times 4) - (2^2 - 3 \times 2) = (16 - 12) - (4 - 6) = 6$.
- Using areas: the region is a trapezium with width $4 - 2 = 2$ and parallel sides $2 \times 2 - 3 = 1$ and $2 \times 4 - 3 = 5$. Then $\dfrac{1 + 5}{2} \times 2 = 3 \times 2 = 6$.

CONCEPT 2 · primitives with several terms

BRACKETS ARE NEEDED

Note: Whenever the primitive has two or more terms, brackets are needed when substituting the upper and lower limits of integration.

WHY THE BRACKETS MATTER

The minus sign between the brackets subtracts the *whole* value at the lower limit. For example, $(16 - 12) - (4 - 6) = 4 - (-2) = 6$, whereas leaving the brackets out would give $16 - 12 - 4 - 6 = -6$, which is wrong.

NEVER OMIT A SUBSTITUTION OF 0

When the lower limit is 0, still write the bracket $(0 - 0)$ for it. Many primitives give 0 at $x = 0$, but not all of them do. For example, using Lesson 4J, a primitive of $(x + 1)^2$ is $\frac{1}{3}(x + 1)^3$, and

$$\left[\tfrac{1}{3}(x + 1)^3\right]_0^1 = \tfrac{1}{3} \times 2^3 - \tfrac{1}{3} \times 1^3 = \tfrac{8}{3} - \tfrac{1}{3} = \tfrac{7}{3},$$

where the value at 0 is $\frac{1}{3}$, not 0.

POWERS OF NEGATIVE NUMBERS AND OF FRACTIONS

Put each negative number in brackets before taking a power: $(-2)^2 = 4$, $(-2)^3 = -8$, $(-1)^4 = 1$. For fractions, take the power of the numerator and of the denominator: $\left(\frac{2}{3}\right)^3 = \frac{8}{27}$.

TUTOR DEMO **Several terms, a zero and a negative limit**

Evaluate these definite integrals.

a $\int_0^1 7x^2\,dx$

b $\int_0^3 (16 - x^2)\,dx$

c $\int_1^2 (3x^2 - 2x + 4)\,dx$

d $\int_{-2}^1 (3x^2 + 1)\,dx$

YOUR TURN **Same method**

Evaluate these definite integrals.

a $\int_0^1 4x^2\,dx$

b $\int_0^2 (9 - x^2)\,dx$

c $\int_1^3 (3x^2 + 2x - 1)\,dx$

d $\int_{-1}^2 (3x^2 + 2)\,dx$

WORKED EXAMPLE 2 **From the text: keep for homework**

Evaluate these definite integrals.

a $\int_0^1 5x^2\,dx$ b $\int_0^4 (25-x^2)\,dx$

SOLUTION

a

$$\begin{aligned}\int_0^1 5x^2\,dx &= \left[\tfrac{5}{3}x^3\right]_0^1 \qquad \text{(increase the index from 2 to 3, then divide by 3)}\\ &= \tfrac{5}{3}-0 \qquad \text{(substitute 1, then substitute 0 and subtract)}\\ &= \tfrac{5}{3}\end{aligned}$$

b

$$\begin{aligned}\int_0^4 (25-x^2)\,dx &= \left[25x-\tfrac{1}{3}x^3\right]_0^4\\ &= \left(100-\tfrac{1}{3}\times 64\right)-(0-0)\\ &\qquad \text{(never omit a substitution of 0)}\\ &= 78\tfrac{2}{3}\end{aligned}$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 2

- **a** Substitute 1, then 0: $\frac{5}{3}\times 1^3-\frac{5}{3}\times 0^3=\frac{5}{3}-0=\frac{5}{3}$.
- **b** The primitive of 25 is $25x$, and the primitive of x^2 is $\frac{1}{3}x^3$. Substitute 4, then 0: $\left(25\times 4-\frac{1}{3}\times 4^3\right)-\left(25\times 0-\frac{1}{3}\times 0^3\right)=\left(100-\frac{1}{3}\times 64\right)-(0-0)$.
- Then $100-\frac{64}{3}=\frac{300}{3}-\frac{64}{3}=\frac{236}{3}=78\frac{2}{3}$.

CONCEPT 3 · expanding brackets in the integrand

EXPAND FIRST

As with differentiation, it is often necessary to expand the brackets in the integrand before finding a primitive. With integration, there is no ‘product rule’ that could avoid the expansion.

DO NOT INTEGRATE EACH FACTOR SEPARATELY

The primitive of a product is *not* the product of the primitives. For example, $x(x+1)$ is not integrated by multiplying $\frac{1}{2}x^2$ by $\frac{1}{2}x^2+x$. Expand to x^2+x first, then take the primitive of each term.

TUTOR DEMO **Expand, then evaluate**

Expand the brackets in each integral, then evaluate it.

a $\int_1^3 x(x+2)\,dx$

b $\int_0^3 (x-5)(x-6)\,dx$

YOUR TURN **Same method**

Expand the brackets in each integral, then evaluate it.

a $\int_1^2 x(x+3)\,dx$

b $\int_0^2 (x-3)(x-4)\,dx$

WORKED EXAMPLE 3 **From the text: keep for homework**

Expand the brackets in each integral, then evaluate it.

a $\int_1^6 x(x+1)\,dx$

b $\int_0^3 (x-4)(x-6)\,dx$

SOLUTION

a

$$\begin{aligned}
&\int_1^6 x(x+1)\,dx \\
&= \int_1^6 (x^2+x)\,dx \\
&= \left[\frac{x^3}{3}+\frac{x^2}{2}\right]_1^6 \\
&= (72+18)-\left(\frac{1}{3}+\frac{1}{2}\right) \\
&= 90-\tfrac{5}{6} \qquad \text{(care with the fractions)} \\
&= 89\tfrac{1}{6}
\end{aligned}$$

b

$$\begin{aligned}
&\int_0^3 (x-4)(x-6)\,dx \\
&= \int_0^3 (x^2-10x+24)\,dx \\
&= \left[\frac{x^3}{3}-5x^2+24x\right]_0^3 \\
&= (9-45+72)-(0-0+0) \\
&= 36
\end{aligned}$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 3

- **a** Expand: $x(x+1)=x^2+x$. Substitute 6: $\frac{216}{3}+\frac{36}{2}=72+18$. Then $\frac{1}{3}+\frac{1}{2}=\frac{2}{6}+\frac{3}{6}=\frac{5}{6}$, so the value is $\left(\frac{216}{3}+\frac{36}{2}\right)-\left(\frac{1}{3}+\frac{1}{2}\right)=(72+18)-\left(\frac{2}{6}+\frac{3}{6}\right)=90-\frac{5}{6}=89\frac{1}{6}$.
- **b** Expand: $(x-4)(x-6)=x^2-6x-4x+24=x^2-10x+24$. Substitute 3: $\frac{27}{3}-5\times 9+24\times 3=9-45+72$.

CONCEPT 4 · writing the integrand as separate fractions

SPLIT THE FRACTION FIRST

If the integrand is a fraction with two or more terms in the numerator, it should normally be written as separate fractions, as with differentiation. With integration, there is no 'quotient rule' that could avoid this.

DIVIDING POWERS

Divide each term of the numerator by the denominator, subtracting the indices: $\frac{x^4}{x^2} = x^{4-2} = x^2$, $\frac{x^3}{x^3} = x^0 = 1$, $\frac{x^4}{x^3} = x$. Do not divide only the first term. For example,

$$\frac{3x^4 - 2x^2}{x^2} = \frac{3x^4}{x^2} - \frac{2x^2}{x^2} = 3x^2 - 2.$$

TUTOR DEMO **Split, then evaluate**

Write each integrand as two separate fractions, then evaluate the integral.

a $\int_1^3 \frac{4x^4 - 3x^2}{x^2}\,dx$

b $\int_{-3}^{-1} \frac{x^3 - 3x^4}{x^3}\,dx$

YOUR TURN **Same method**

Write each integrand as two separate fractions, then evaluate the integral.

a $\displaystyle\int_1^2 \frac{6x^4+2x^2}{x^2}\,dx$

b $\displaystyle\int_{-2}^{-1} \frac{2x^3-x^4}{x^3}\,dx$

WORKED EXAMPLE 4 **From the text: keep for homework**

Write each integrand as two separate fractions, then evaluate the integral.

a $\displaystyle\int_1^2 \frac{3x^4-2x^2}{x^2}\,dx$

b $\displaystyle\int_{-3}^{-2} \frac{x^3-2x^4}{x^3}\,dx$

SOLUTION

a

$$\begin{aligned}\int_1^2 \frac{3x^4-2x^2}{x^2}\,dx &= \int_1^2 (3x^2-2)\,dx \\ &= \left[x^3-2x\right]_1^2 \\ &= (8-4)-(1-2) \\ &= 4-(-1) \\ &= 5\end{aligned}$$

b

$$\begin{aligned}\int_{-3}^{-2} \frac{x^3-2x^4}{x^3}\,dx &= \int_{-3}^{-2} (1-2x)\,dx \\ &= \left[x-x^2\right]_{-3}^{-2} \\ &= (-2-4)-(-3-9) \\ &= -6-(-12) \\ &= 6\end{aligned}$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 4

- **a** Divide each term by x^2: $\dfrac{3x^4}{x^2}=3x^2$ and $\dfrac{2x^2}{x^2}=2$.
- **b** Divide each term by x^3: $\dfrac{x^3}{x^3}=1$ and $\dfrac{2x^4}{x^3}=2x$. Substitute -2, then -3: $\left(-2-(-2)^2\right)-\left(-3-(-3)^2\right)=(-2-4)-(-3-9)=6$.

CONCEPT 5 · negative indices, and asymptotes

NEGATIVE INDICES

The fundamental theorem also works when the indices are negative. Care is needed when converting between negative powers of x and fractions.

CONVERTING NEGATIVE POWERS

$$\frac{1}{x^4} = x^{-4}, \qquad \frac{x^{-3}}{-3} = -\frac{1}{3x^3} \quad \text{(the 3 stays with } x^3 \text{ in the denominator).}$$

THE INDEX -1 CANNOT BE HANDLED BY THIS RULE

Note: The negative index -1 cannot be handled by this rule, because it would generate division by 0, which is nonsense:

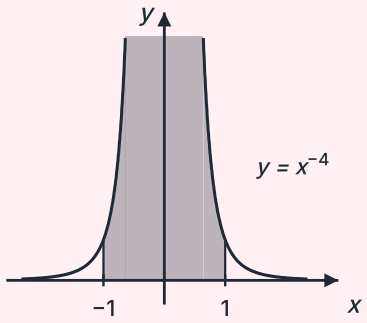
$$\int_1^2 \frac{1}{x}\,dx = \int_1^2 x^{-1}\,dx = \left[\frac{x^0}{0}\right]_1^2 = \text{nonsense.}$$

Chapter 6 on exponential and logarithmic functions will handle this integral.

WARNING: DO NOT INTEGRATE ACROSS AN ASYMPTOTE

The following calculation seems just as valid as part **b** of Worked example 5 below:

$$\begin{aligned}\int_{-1}^1 x^{-4}\,dx &= \left[\frac{x^{-3}}{-3}\right]_{-1}^1 \\ &= -\frac{1}{3} - \frac{1}{3} \\ &= -\frac{2}{3}.\end{aligned}$$



But in fact the calculation is nonsense. The function has an asymptote at $x = 0$, so it is not even defined there. (And the function $y = x^{-4}$ is always positive, so how could it give a negative answer for an integral?) You cannot integrate across an asymptote, and you always need to be on the lookout for such meaningless integrals.

STEPS THE BOOK SKIPS: THE WARNING

Substituting 1, then -1, into $\frac{x^{-3}}{-3} = -\frac{1}{3x^3}$ gives

$$-\frac{1}{3\times 1^3} - \left(-\frac{1}{3\times(-1)^3}\right) = -\frac{1}{3} - \frac{1}{3} = -\frac{2}{3}.$$

CHECK FOR AN ASYMPTOTE FIRST

Box 4 needs $f(x)$ to be continuous in the *whole* closed interval $[a, b]$. Before using it, check that the denominator is never zero anywhere from a to b.

TUTOR DEMO **Negative indices**

Use negative indices to evaluate these definite integrals.

a $\int_1^4 x^{-2}\,dx$

b $\int_1^2 \frac{1}{x^3}\,dx$

c Explain why $\int_{-2}^{1} \frac{1}{x^2}\,dx$ is meaningless.

YOUR TURN **Same method**

Use negative indices to evaluate these definite integrals.

a $\int_1^3 x^{-2}\,dx$

b $\int_1^2 \frac{1}{x^5}\,dx$

c Explain why $\int_{-1}^{2} x^{-3}\,dx$ is meaningless.

WORKED EXAMPLE 5 **From the text: keep for homework**

Use negative indices to evaluate these definite integrals.

a $\int_1^5 x^{-2}\,dx$

b $\int_1^2 \frac{1}{x^4}\,dx$

SOLUTION

a Increase the index from -2 to -1, and divide by -1.

$$\begin{aligned}\int_1^5 x^{-2}\,dx &= \left[\frac{x^{-1}}{-1}\right]_1^5 \\ &= \left[-\frac{1}{x}\right]_1^5 \\ &= -\frac{1}{5} - (-1) \\ &= -\frac{1}{5} + 1 \\ &= \frac{4}{5}\end{aligned}$$

b Increase the index from -4 to -3, and divide by -3.

$$\begin{aligned}\int_1^2 \frac{1}{x^4}\,dx &= \int_1^2 x^{-4}\,dx \\ &= \left[\frac{x^{-3}}{-3}\right]_1^2 \\ &= \left[-\frac{1}{3x^3}\right]_1^2 \\ &= -\frac{1}{24} - \left(-\frac{1}{3}\right) \\ &= \frac{7}{24}\end{aligned}$$

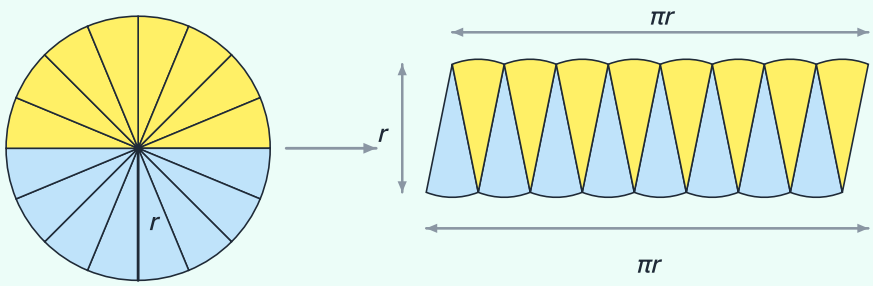
STEPS THE BOOK SKIPS: WORKED EXAMPLE 5

- **a** Substitute 5, then 1, into $-\frac{1}{x}$: $-\frac{1}{5} - \left(-\frac{1}{1}\right) = -\frac{1}{5} - (-1) = \frac{4}{5}$.
- **b** Substitute 2, then 1, into $-\frac{1}{3x^3}$:
  $-\frac{1}{3\times 8} - \left(-\frac{1}{3\times 1}\right) = -\frac{1}{24} - \left(-\frac{1}{3}\right) = -\frac{1}{24} + \frac{8}{24} = \frac{7}{24}$.

CONCEPT 6 · the area of a circle

AN OLD LIMITING ARGUMENT

In earlier years, the formula $A = \pi r^2$ for the area of a circle was developed. Because the boundary is a curve, some limiting process had to be used in its proof. For comparison with the notation for the definite integral explained at the start of this section, here is the most common version of that argument. It is a little rough in its logic, but very quick. It involves dissecting the circle into infinitesimally thin sectors and then rearranging them into a rectangle.



The height of the rectangle in the diagram is r. Because the circumference $2\pi r$ is divided equally between the top and bottom sides, the length of the rectangle is πr. Hence the rectangle has area πr^2, which is therefore the area of the circle.

STEPS THE BOOK SKIPS: THE AREA OF THE RECTANGLE

The height r is the radius. The area of the rectangle is

$$\text{length} \times \text{height} = \pi r \times r = \pi r^2.$$

WHERE THE ROUGH LOGIC IS

With finitely many sectors, the top and bottom of the shape are made of small arcs, not straight lines, and the sides slope a little. As the sectors become thinner, the shape becomes closer and closer to a rectangle. This is a limiting process, just like the strips of Lesson 5A.

TUTOR DEMO **Rearranging the sectors**

A circle has radius 4. It is cut into a large number of equal thin sectors, which are rearranged into a shape close to a rectangle, as in the diagram above.

a Find the circumference of the circle, and hence the length of the rectangle.

b Write down the height of the rectangle, and hence find its area. Check your answer with $A = \pi r^2$.

YOUR TURN **Same argument**

A circle has radius 7. It is cut into a large number of equal thin sectors, which are rearranged into a shape close to a rectangle, as in the diagram above.

a Find the circumference of the circle, and hence the length of the rectangle.

b Write down the height of the rectangle, and hence find its area. Check your answer with $A = \pi r^2$.

TEXTBOOK HOMEWORK · Exercise 5B

FOUNDATION

Technology: Many programs allow definite integrals to be calculated automatically. This allows not just quick checking of the answers, but experimentation with further definite integrals. It would be helpful to generate screen sketches of the graphs and the regions associated with the integrals.

1 Evaluate these definite integrals using the fundamental theorem.

a $\int_0^1 2x\,dx$ b $\int_1^4 2x\,dx$ c $\int_1^3 4x\,dx$

d $\int_2^5 8x\,dx$ e $\int_2^3 3x^2\,dx$ f $\int_0^3 5x^4\,dx$

g $\int_1^2 10x^4\,dx$ h $\int_0^1 12x^5\,dx$ i $\int_0^1 11x^{10}\,dx$

2 a Evaluate these definite integrals using the fundamental theorem.

i $\int_0^1 4\,dx$ ii $\int_2^7 5\,dx$ iii $\int_4^5 dx$

b Check your answers by sketching the graph of the region involved.

3 Evaluate these definite integrals using the fundamental theorem.

a $\int_3^6 (2x+1)\,dx$

b $\int_2^4 (2x-3)\,dx$

c $\int_0^3 (4x+5)\,dx$

d $\int_2^3 (3x^2-1)\,dx$

e $\int_1^4 (6x^2+2)\,dx$

f $\int_0^1 (3x^2+2x)\,dx$

g $\int_1^2 (4x^3+3x^2+1)\,dx$

h $\int_0^2 (2x+3x^2+8x^3)\,dx$

i $\int_3^5 (3x^2-6x+5)\,dx$

4 Evaluate these definite integrals using the fundamental theorem. You will need to take care when finding powers of negative numbers.

a $\int_{-1}^{0} (1-2x)\,dx$ b $\int_{-1}^{0} (2x+3)\,dx$ c $\int_{-2}^{1} 3x^2\,dx$

d $\int_{-1}^{2} (4x^3+5)\,dx$ e $\int_{-2}^{2} (5x^4+6x^2)\,dx$ f $\int_{-2}^{-1} (4x^3+12x^2-3)\,dx$

5 Evaluate these definite integrals using the fundamental theorem. You will need to take care when adding and subtracting fractions.

a $\int_{1}^{4} (x+2)\,dx$ b $\int_{0}^{2} (x^2+x)\,dx$ c $\int_{0}^{3} (x^3+x^2)\,dx$

d $\int_{-1}^{1} (x^3-x+1)\,dx$ e $\int_{-2}^{3} (2x^2-3x+1)\,dx$ f $\int_{-4}^{-2} (16-x^3-x)\,dx$

DEVELOPMENT

6 By expanding the brackets first, evaluate these definite integrals.

a $\int_2^3 x(2+3x)\,dx$ b $\int_0^2 (x+1)(3x+1)\,dx$ c $\int_{-1}^1 x^2(5x^2+1)\,dx$

d $\int_{-1}^2 (x-3)^2\,dx$ e $\int_{-1}^0 x(x-1)(x+1)\,dx$ f $\int_{-1}^0 (1-x^2)^2\,dx$

7 Write each integrand as separate fractions, then evaluate the integral.

a $\int_1^3 \dfrac{3x^3+4x^2}{x}\,dx$ b $\int_1^2 \dfrac{4x^4-x}{x}\,dx$ c $\int_2^3 \dfrac{5x^2+9x^4}{x^2}\,dx$

d $\int_1^2 \dfrac{x^3+4x^2}{x}\,dx$ e $\int_1^3 \dfrac{x^3-x^2+x}{x}\,dx$ f $\int_{-2}^{-1} \dfrac{x^3-2x^5}{x^2}\,dx$

8 Evaluate these definite integrals using the fundamental theorem. You will need to take care when finding powers of fractions.

a $\int_0^{\frac{1}{2}} x^2 \, dx$ b $\int_0^{\frac{2}{3}} (2x + 3x^2) \, dx$ c $\int_{\frac{1}{2}}^{\frac{3}{4}} (6 - 4x) \, dx$

9 a Evaluate these definite integrals.

i $\int_5^{10} x^{-2} \, dx$ ii $\int_2^3 2x^{-3} \, dx$ iii $\int_{\frac{1}{2}}^1 4x^{-5} \, dx$

b By writing them with negative indices, evaluate these definite integrals.

i $\int_1^2 \frac{1}{x^2} \, dx$ ii $\int_1^4 \frac{1}{x^3} \, dx$ iii $\int_{\frac{1}{2}}^1 \frac{3}{x^4} \, dx$

10 a i Show that $\int_2^k 3\,dx = 3k - 6$.

ii Hence find the value of k, given that $\int_2^k 3\,dx = 18$.

b i Show that $\int_0^k x\,dx = \frac{1}{2}k^2$.

ii Hence find the positive value of k, given that $\int_0^k x\,dx = 18$.

11 Find the value of k if $k > 0$ and:

a $\int_k^3 2\,dx = 4$

b $\int_k^8 3\,dx = 12$

c $\int_2^3 (k-3)\,dx = 5$

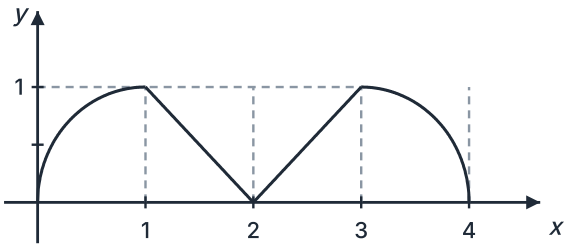
d $\int_3^k (x-3)\,dx = 0$

e $\int_1^k (x+1)\,dx = 6$

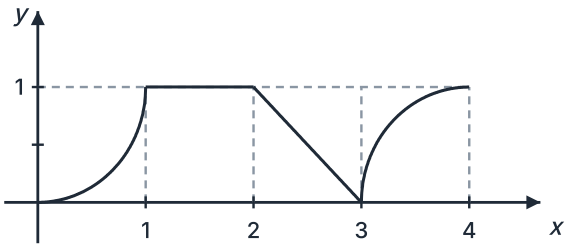
f $\int_1^k (k+3x)\,dx = \dfrac{13}{2}$

12 Use area formulae to find $\int_0^4 f(x)\,dx$ in each sketch of $f(x)$.

a



b



13 Evaluate each definite integral.

a $\int_1^2 \dfrac{1+x^2}{x^2}\,dx$

b $\int_{-2}^{-1} \dfrac{1+2x}{x^3}\,dx$

c $\int_{-3}^{-1} \dfrac{1-x^3-4x^5}{2x^2}\,dx$

14 Evaluate these definite integrals.

a $\displaystyle\int_1^3 \left(x+\frac{1}{x}\right)^2 dx$ b $\displaystyle\int_1^2 \left(x^2+\frac{1}{x^2}\right)^2 dx$ c $\displaystyle\int_{-2}^{-1} \left(\frac{1}{x^2}+\frac{1}{x}\right)^2 dx$

15 a Explain why the function $y=\dfrac{1}{x^2}$ is never negative.

b Sketch the integrand and explain why the argument below is invalid.

$$\int_{-1}^{1} \frac{dx}{x^2} = \left[-\frac{1}{x}\right]_{-1}^{1} = -1-1 = -2.$$

c Without evaluating any integrals, say which of these integrals are meaningless.

i $\displaystyle\int_0^2 \frac{1}{(3-x)^2}\,dx$ ii $\displaystyle\int_2^4 \frac{1}{(3-x)^2}\,dx$ iii $\displaystyle\int_4^6 \frac{1}{(3-x)^2}\,dx$

16 a Use the differential form of the fundamental theorem $\dfrac{d}{dx}\displaystyle\int_a^x f(t)\,dt = f(x)$ to find:

i $\dfrac{d}{dx}\displaystyle\int_1^x t^2\,dt$

ii $\dfrac{d}{dx}\displaystyle\int_2^x (t^3+3t)\,dt$

iii $\dfrac{d}{dx}\displaystyle\int_a^x \dfrac{1}{t}\,dt$

iv $\dfrac{d}{dx}\displaystyle\int_a^x (t^3-3)^4\,dt$

b Confirm your answers to parts (i) and (ii) above by evaluating the definite integral and then differentiating with respect to x.

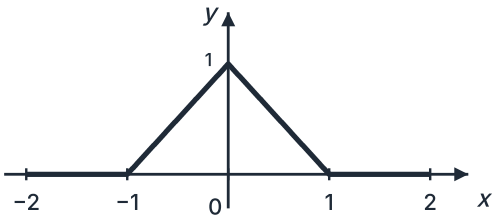
PAST-PAPER HOMEWORK · mini exam

Suggested time: 5 minutes **1 mark** Calculator allowed · show working · hints are below each question if you get stuck

This is the only school-paper item that needs just this lesson's ideas. For more timed practice, do Questions 4(f), 6(f), 7(f), 11(f), 13(c) and 14(b) above in 25 minutes.

PP 1 **North Sydney Boys High School, 2022, HSC Course Assessment Task 3-Trial Examination, Mathematics Extension 1, Q10** 1 mark

The graph of the function $y = f(x)$ is sketched below. The value of $\int_{-1}^{1} f(x^2 - 1)\,dx$ equals



A. 2 B. $\frac{1}{3}$ C. $\sqrt{2}$ D. $\frac{2}{3}$

Hint: what is being fed into f?

LESSON 5C

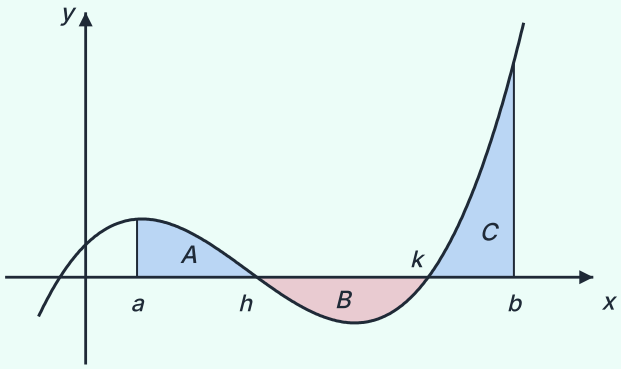
The definite integral and its properties

This lesson first extends the definite integral to functions with negative values: areas below the x-axis count as negative, so a definite integral is a sum of signed areas. Then some simple properties of the definite integral are established using arguments about the dissection of regions: odd and even functions, dissecting the interval, intervals of zero width, running an integral backwards, sums and multiples of functions, and inequalities. Each property is checked with the fundamental theorem from Lesson 5B.

CONCEPT 1 · integrating functions with negative values

AREAS BELOW THE x-AXIS COUNT AS NEGATIVE

When a function has negative values, its graph is below the x-axis, so the 'heights' of the little rectangles in the dissection are negative numbers. This means that any areas below the x-axis should contribute negative values to the value of the final integral.



For example, in the diagram above, where $a < h < k < b$, the region B is below the x-axis because the function $f(x)$ is negative in $[h, k]$. So the heights of all the infinitesimal strips making up B are negative, and B therefore contributes a negative number to the definite integral,

$$\int_a^b f(x)\,dx = +\text{area } A - \text{area } B + \text{area } C.$$

Thus we attach the sign $+$ or $-$ to each area, depending on whether the curve, and therefore the region, is above or below the x-axis. For this reason, the three terms in the sum above are often referred to as *signed areas under the curve*, because a sign has been attached to the area of each region.

5 THE DEFINITE INTEGRAL AS THE SUM OF SIGNED AREAS

Let $f(x)$ be a function continuous in the closed interval $[a, b]$, where $a \leq b$, and suppose that we are taking the definite integral over $[a, b]$.

- For regions where the curve is above the x-axis, we attach $+$ to the area.
  For regions where the curve is below the x-axis, we attach $-$ to the area.
  These areas, with signs attached, are called *signed areas under the curve*.
- The *definite integral* $\int_a^b f(x)\,dx$ is the sum of these signed areas under the curve in the interval $[a, b]$.

The whole definite integral $\int_a^b f(x)\,dx$ is often also referred to as *the signed area under the curve*.

READING THE SIGN OF AN ANSWER

- The fundamental theorem (Lesson 5B) attaches the signs automatically. You do not need to split the interval to evaluate the integral.
- A negative answer means that there is more area below the x-axis than above it. A zero answer means that the areas above and below are equal, so they cancel out.
- To find a signed area from a sketch, find the area of each region with an area formula, then attach $+$ above the axis and $-$ below it.

TUTOR DEMO **Signed areas of a line**

Evaluate these definite integrals.

a $\int_0^3 (x-3)\,dx$ b $\int_3^5 (x-3)\,dx$ c $\int_0^5 (x-3)\,dx$

Sketch the graph of $y = x - 3$ and shade the regions associated with these integrals. Then explain how each result is related to the shaded regions.

YOUR TURN **Same method**

Evaluate these definite integrals.

a $\int_0^2 (x-2)\,dx$ b $\int_2^5 (x-2)\,dx$ c $\int_0^5 (x-2)\,dx$

Sketch the graph of $y = x - 2$ and shade the regions associated with these integrals. Then explain how each result is related to the shaded regions.

WORKED EXAMPLE 1 **From the text: keep for homework**

Evaluate these definite integrals.

a $\int_0^4 (x-4)\,dx$ b $\int_4^6 (x-4)\,dx$ c $\int_0^6 (x-4)\,dx$

Sketch the graph of $y = x - 4$ and shade the regions associated with these integrals. Then explain how each result is related to the shaded regions.

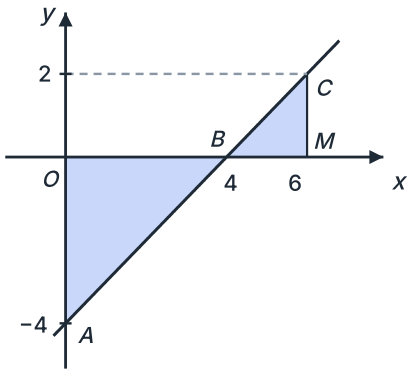
SOLUTION

a
$$\begin{aligned}\int_0^4 (x-4)\,dx &= \left[\tfrac{1}{2}x^2 - 4x\right]_0^4 \\ &= (8-16) - (0-0) \\ &= -8\end{aligned}$$

b
$$\begin{aligned}\int_4^6 (x-4)\,dx &= \left[\tfrac{1}{2}x^2 - 4x\right]_4^6 \\ &= (18-24) - (8-16) \\ &= -6 - (-8) \\ &= 2\end{aligned}$$

c
$$\begin{aligned}\int_0^6 (x-4)\,dx &= \left[\tfrac{1}{2}x^2 - 4x\right]_0^6 \\ &= (18-24) - (0-0) \\ &= -6\end{aligned}$$

Triangle OAB is below the x-axis and has area 8, and triangle BMC is above the x-axis, and has area 2, so the answers -8 in part **a** and $+2$ in part **b** are expected. This integral in part **c** represents the area of $\triangle BMC$ minus the area of $\triangle OAB$, so the value of the integral is $2 - 8 = -6$ is expected.



STEPS THE BOOK SKIPS: WORKED EXAMPLE 1

- a $\left(\frac{1}{2} \times 4^2 - 4 \times 4\right) - \left(\frac{1}{2} \times 0^2 - 4 \times 0\right) = (8-16) - (0-0) = -8$
- b $\left(\frac{1}{2} \times 6^2 - 4 \times 6\right) - \left(\frac{1}{2} \times 4^2 - 4 \times 4\right) = (18-24) - (8-16) = 2$
- c $\left(\frac{1}{2} \times 6^2 - 4 \times 6\right) - \left(\frac{1}{2} \times 0^2 - 4 \times 0\right) = (18-24) - (0-0) = -6$
- The areas: triangle OAB has area $\frac{1}{2} \times 4 \times 4 = 8$, and triangle BMC has area $\frac{1}{2} \times 2 \times 2 = 2$.

CONCEPT 2 · odd and even functions

RECALL: ODD AND EVEN FUNCTIONS (YEAR 11)

- A function $f(x)$ is *odd* if $f(-x) = -f(x)$ for all x in its domain. Its graph has *point symmetry in the origin*: a rotation of $180°$ about O leaves it unchanged.
- A function $f(x)$ is *even* if $f(-x) = f(x)$ for all x in its domain. Its graph has *line symmetry in the y-axis*: a reflection in the y-axis leaves it unchanged.

INTEGRATING ODD AND EVEN FUNCTIONS

In Worked example 2a below, the function $y = x^3 - 4x$ is an odd function, with point symmetry in the origin. Thus the area of each shaded hump is the same. Hence the whole integral from $x = -2$ to $x = 2$ is zero, because the equal humps above and below the x-axis cancel out.

In the second diagram, in Worked example 2b, the function $y = x^2 + 1$ is even, with line symmetry in the y-axis. Thus the areas to the left and right of the y-axis are equal, so there is a doubling instead of a cancelling.

6 INTEGRATING ODD AND EVEN FUNCTIONS

- If $f(x)$ is odd, then $\int_{-a}^{a} f(x)\,dx = 0$.
- If $f(x)$ is even, then $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$.

TESTING FOR ODD AND EVEN

- Substitute $-x$ for x and simplify. For example, $(-x)^3 - 4(-x) = -x^3 + 4x = -(x^3 - 4x)$, so $x^3 - 4x$ is odd. Also $(-x)^2 + 1 = x^2 + 1$, so $x^2 + 1$ is even.
- A polynomial with only odd powers of x is odd. A polynomial with only even powers of x is even (a constant term counts as an even power, x^0). A polynomial with both odd and even powers, such as $x^2 + x$, is neither odd nor even.
- Box 6 needs the limits to be opposites, $-a$ and a. It says nothing about $\int_{-1}^{3} x^3\,dx$, for example.

TUTOR DEMO **Using symmetry**

Sketch these integrals, then evaluate them using symmetry.

a $\int_{-3}^{3}(x^3-9x)\,dx$ b $\int_{-3}^{3}(x^2+1)\,dx$

In part **a**, also do the calculation without using symmetry.

YOUR TURN **Same method**

Sketch these integrals, then evaluate them using symmetry.

a $\int_{-1}^{1}(x^3-x)\,dx$ b $\int_{-1}^{1}(3x^2+1)\,dx$

In part **a**, also do the calculation without using symmetry.

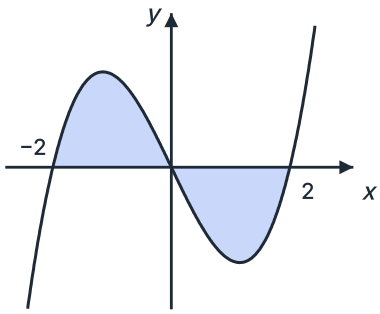
WORKED EXAMPLE 2 **From the text: keep for homework**

Sketch these integrals, then evaluate them using symmetry.

a $\int_{-2}^{2}(x^3-4x)\,dx$ b $\int_{-2}^{2}(x^2+1)\,dx$

SOLUTION

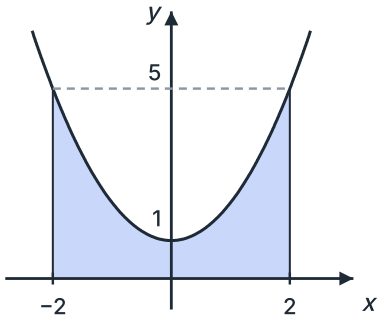
a $\int_{-2}^{2}(x^3-4x)\,dx=0$, because the integrand is odd.



Without this simplification, the calculation is:

$$\begin{aligned}\int_{-2}^{2}(x^3-4x)\,dx &= \left[\tfrac{1}{4}x^4-2x^2\right]_{-2}^{2}\\ &=(4-8)-(4-8)\\ &=0,\ \text{as before.}\end{aligned}$$

b Because the integrand is even,



$$\begin{aligned}\int_{-2}^{2}(x^2+1)\,dx &= 2\int_{0}^{2}(x^2+1)\,dx\\ &=2\left[\tfrac{1}{3}x^3+x\right]_{0}^{2}\\ &=2\left(\left(2\tfrac{2}{3}+2\right)-(0+0)\right)\\ &=9\tfrac{1}{3}.\end{aligned}$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 2

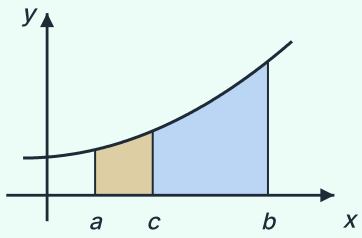
- **a** Check that the integrand is odd: $(-x)^3-4(-x)=-x^3+4x=-(x^3-4x)$.
  Without symmetry, substitute 2, then -2:
  $\left(\tfrac{1}{4}\times 2^4-2\times 2^2\right)-\left(\tfrac{1}{4}\times(-2)^4-2\times(-2)^2\right)=(4-8)-(4-8)=0.$
- **b** Check that the integrand is even: $(-x)^2+1=x^2+1$. Then
  $2\left(\left(\tfrac{1}{3}\times 2^3+2\right)-\left(\tfrac{1}{3}\times 0^3+0\right)\right)=2\left(\left(2\tfrac{2}{3}+2\right)-(0+0)\right)$, and
  $2\left(\left(2\tfrac{2}{3}+2\right)-(0+0)\right)=2\times 4\tfrac{2}{3}=9\tfrac{1}{3}.$

CONCEPT 3 · dissecting the interval, zero width, running backwards

DISSECTION OF THE INTERVAL

When a region is dissected, its area remains the same. In particular, we can always dissect the region by dissecting the interval $a \leq x \leq b$ over which we are integrating.

Thus if $f(x)$ is being integrated over the closed interval $[a, b]$ and the number c lies in this interval, then:

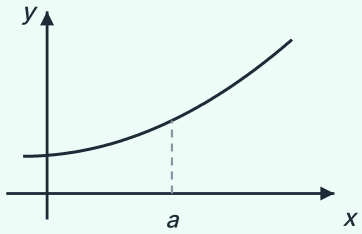


7 DISSECTION OF THE INTERVAL

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

INTERVALS OF ZERO WIDTH

Suppose that a function is integrated over an interval $a \leq x \leq a$ of width zero, and that the function is defined at $x = a$. In this situation, the region also has width zero, so the integral is zero.



8 INTERVALS OF ZERO WIDTH

$$\int_a^a f(x)\,dx = 0$$

RUNNING AN INTEGRAL BACKWARDS FROM RIGHT TO LEFT

A further small qualification must be made to the definition of the definite integral. Suppose that the limits of the integral are reversed, so that the integral 'runs backwards' from right to left over the interval. Then its value reverses in sign:

9 REVERSING THE INTERVAL

Let $f(x)$ be continuous in the closed interval $[a, b]$, where $a \leq b$. Then we define

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx.$$

This agrees perfectly with calculations using the fundamental theorem, because $F(a) - F(b) = -\big(F(b) - F(a)\big)$.

THE FUNDAMENTAL THEOREM AGREES WITH BOXES 7, 8 AND 9

Let $F(x)$ be a primitive of $f(x)$, so that $\int_a^b f(x)\,dx = F(b) - F(a)$.

- **Box 7:** $\big(F(c) - F(a)\big) + \big(F(b) - F(c)\big) = F(b) - F(a)$, because the two $F(c)$ terms cancel.
- **Box 8:** $\int_a^a f(x)\,dx = F(a) - F(a) = 0$.
- **Box 9:** $\int_b^a f(x)\,dx = F(a) - F(b) = -\big(F(b) - F(a)\big) = -\int_a^b f(x)\,dx$.
- Box 7 also holds when the limits are not in increasing order. For example, going from a to b and then back to a gives $\int_a^b f(x)\,dx + \int_b^a f(x)\,dx = \int_a^a f(x)\,dx = 0$, which is Box 9 again.

TUTOR DEMO **Dissecting, zero width and reversing**

a By evaluating both sides, show that $\int_0^3 x^2\,dx = \int_0^1 x^2\,dx + \int_1^3 x^2\,dx$.

b Write down the value of $\int_2^2 (x^3+5)\,dx$, giving a reason.

c Evaluate and compare $\int_1^3 (x+1)\,dx$ and $\int_3^1 (x+1)\,dx$ using the fundamental theorem.

YOUR TURN **Same method**

a By evaluating both sides, show that $\int_0^4 x\,dx = \int_0^1 x\,dx + \int_1^4 x\,dx$.

b Write down the value of $\int_{-3}^{-3} (x^2 - 4x)\,dx$, giving a reason.

c Evaluate and compare $\int_2^5 (x-2)\,dx$ and $\int_5^2 (x-2)\,dx$ using the fundamental theorem.

WORKED EXAMPLE 3 **From the text: keep for homework**

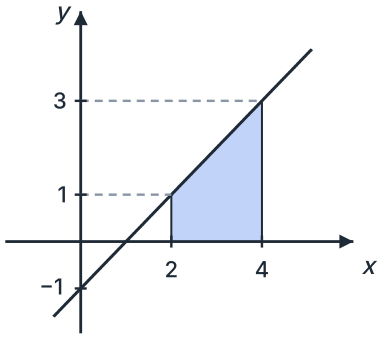
Evaluate and compare these two definite integrals using the fundamental theorem.

a $\int_2^4 (x-1)\,dx$ b $\int_4^2 (x-1)\,dx$

SOLUTION

a
$$\begin{aligned}\int_2^4 (x-1)\,dx &= \left[\frac{x^2}{2} - x\right]_2^4 \\ &= (8-4) - (2-2) \\ &= 4,\end{aligned}$$

which is positive, because the region is above the x-axis.



b
$$\begin{aligned}\int_4^2 (x-1)\,dx &= \left[\frac{x^2}{2} - x\right]_4^2 \\ &= (2-2) - (8-4) \\ &= -4,\end{aligned}$$

which is the opposite of part **a**, because the integral runs backwards from right to left, from $x = 4$ to $x = 2$.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 3

- **a** Substitute 4, then 2: $\left(\frac{4^2}{2} - 4\right) - \left(\frac{2^2}{2} - 2\right) = (8-4) - (2-2) = 4.$
- **b** Substitute 2, then 4: $\left(\frac{2^2}{2} - 2\right) - \left(\frac{4^2}{2} - 4\right) = (2-2) - (8-4) = -4.$

CONCEPT 4 · sums and multiples of functions

SUMS OF FUNCTIONS

When two functions are added, the two regions are piled on top of each other, so that:

10 INTEGRAL OF A SUM

$$\int_a^b \big(f(x) + g(x)\big)\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$

MULTIPLES OF FUNCTIONS

Similarly, when a function is multiplied by a constant, the region is stretched vertically by that constant, so that:

11 INTEGRAL OF A MULTIPLE

$$\int_a^b kf(x)\,dx = k\int_a^b f(x)\,dx$$

THE FUNDAMENTAL THEOREM AGREES WITH BOXES 10 AND 11

- If $F'(x) = f(x)$ and $G'(x) = g(x)$, then $F(x) + G(x)$ is a primitive of $f(x) + g(x)$, so
$$\int_a^b \big(f(x) + g(x)\big)\,dx = \big(F(b) + G(b)\big) - \big(F(a) + G(a)\big)$$
$$= \big(F(b) - F(a)\big) + \big(G(b) - G(a)\big).$$
- Similarly, $kF(x)$ is a primitive of $kf(x)$, so
$$\int_a^b kf(x)\,dx = kF(b) - kF(a) = k\big(F(b) - F(a)\big).$$
- Use Box 10 to split an integrand into simpler pieces, and Box 11 to take a constant factor outside the integral.

TUTOR DEMO **Sums and multiples**

Evaluate each pair of expressions and show that they are equal.

a $\displaystyle\int_0^2 (x^2 + x + 1)\,dx$

b $\displaystyle\int_0^2 x^2\,dx + \int_0^2 x\,dx + \int_0^2 1\,dx$

c $\displaystyle\int_1^2 6x^3\,dx$

d $\displaystyle 6\int_1^2 x^3\,dx$

YOUR TURN **Same method**

Evaluate each pair of expressions and show that they are equal.

a $\int_0^3 (x^2+1)\,dx$

b $\int_0^3 x^2\,dx + \int_0^3 1\,dx$

c $\int_1^2 4x^3\,dx$

d $4\int_1^2 x^3\,dx$

WORKED EXAMPLE 4 **From the text: keep for homework**

Evaluate these two expressions and show that they are equal.

a $\int_0^1 (x^2+x+1)\,dx$

b $\int_0^1 x^2\,dx + \int_0^1 x\,dx + \int_0^1 1\,dx$

SOLUTION

a $$\int_0^1 (x^2+x+1)\,dx = \left[\frac{x^3}{3}+\frac{x^2}{2}+x\right]_0^1$$
$$= \left(\tfrac{1}{3}+\tfrac{1}{2}+1\right)-(0+0+0)$$
$$= 1\tfrac{5}{6}.$$

b $$\int_0^1 x^2\,dx + \int_0^1 x\,dx + \int_0^1 1\,dx = \left[\frac{x^3}{3}\right]_0^1 + \left[\frac{x^2}{2}\right]_0^1 + \left[x\right]_0^1$$
$$= \left(\tfrac{1}{3}-0\right)+\left(\tfrac{1}{2}-0\right)+(1-0)$$
$$= 1\tfrac{5}{6}, \text{ the same as in part a.}$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 4

Use the common denominator 6: $\frac{1}{3}+\frac{1}{2}+1=\frac{2}{6}+\frac{3}{6}+\frac{6}{6}=1\frac{5}{6}$.

WORKED EXAMPLE 5 **From the text: keep for homework**

Evaluate these two expressions and show that they are equal.

a $\int_1^3 10x^3\,dx$ b $10\int_1^3 x^3\,dx$

SOLUTION

a
$$\begin{aligned}\int_1^3 10x^3\,dx &= \left[\frac{10x^4}{4}\right]_1^3 \\ &= \frac{810}{4} - \frac{10}{4} \\ &= \frac{800}{4} \\ &= 200\end{aligned}$$

b
$$\begin{aligned}10\int_1^3 x^3\,dx &= 10 \times \left[\frac{x^4}{4}\right]_1^3 \\ &= 10 \times \left(\frac{81}{4} - \frac{1}{4}\right) \\ &= 10 \times \frac{80}{4} \\ &= 200\end{aligned}$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 5

- **a** Substitute 3, then 1: $\frac{10 \times 3^4}{4} - \frac{10 \times 1^4}{4} = \frac{810}{4} - \frac{10}{4} = 200.$
- **b** $10 \times \frac{80}{4} = 10 \times 20 = 200.$

CONCEPT 5 · inequalities with definite integrals

A LOWER CURVE HAS A SMALLER INTEGRAL

Suppose that a curve $y = f(x)$ is always underneath another curve $y = g(x)$ in an interval $a \le x \le b$. Then the area under the curve $y = f(x)$ from $x = a$ to $x = b$ is less than the area under the curve $y = g(x)$.

In the language of definite integrals:

12 INEQUALITY

If $f(x) \le g(x)$ in the closed interval $[a, b]$, then

$$\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx.$$

BOUNDS FROM A RECTANGLE

Use Box 12 with constant functions. If $m \le f(x) \le M$ for $a \le x \le b$, then

$$\int_a^b m\,dx \le \int_a^b f(x)\,dx \le \int_a^b M\,dx,$$

that is,

$$m(b-a) \le \int_a^b f(x)\,dx \le M(b-a),$$

because $\int_a^b M\,dx$ is the area of a rectangle with width $b-a$ and height M. In a sketch, the region lies inside that rectangle.

TUTOR DEMO **Trapping an integral in a rectangle**

a Sketch the graph of $f(x) = 9 - x^2$, for $-3 \le x \le 3$.

b Explain why $0 \le \int_{-3}^{3} (9 - x^2)\,dx \le 54$.

c Evaluate the integral, and check that its value lies between these bounds.

YOUR TURN **Same method**

a Sketch the graph of $f(x) = 1 - x^2$, for $-1 \leq x \leq 1$.

b Explain why $0 \leq \int_{-1}^{1} (1 - x^2)\,dx \leq 2$.

c Evaluate the integral, and check that its value lies between these bounds.

WORKED EXAMPLE 6 **From the text: keep for homework**

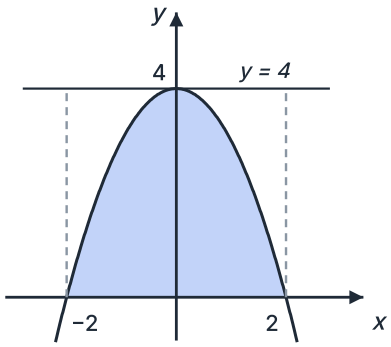
a Sketch the graph of $f(x) = 4 - x^2$, for $-2 \leq x \leq 2$.

b Explain why $0 \leq \int_{-2}^{2} (4 - x^2)\,dx \leq 16$.

SOLUTION

a The parabola and line are sketched opposite.

b Clearly $0 \leq 4 - x^2 \leq 4$ over the interval $-2 \leq x \leq 2$. Hence the region associated with the integral is inside the square of side length 4 in the diagram opposite.



THE STEPS BEHIND WORKED EXAMPLE 6

- The parabola $y = 4 - x^2$ has y-intercept 4 and x-intercepts -2 and 2, and the line is $y = 4$.
- Why $0 \leq 4 - x^2 \leq 4$: since $x^2 \geq 0$, $4 - x^2 \leq 4$. In the interval $-2 \leq x \leq 2$, $x^2 \leq 4$, so $4 - x^2 \geq 0$.
- The square has area $4 \times 4 = 16$. In the language of Box 12,

  $$\int_{-2}^{2} 0\,dx \leq \int_{-2}^{2} (4 - x^2)\,dx \leq \int_{-2}^{2} 4\,dx,$$

  that is,

  $$0 \leq \int_{-2}^{2} (4 - x^2)\,dx \leq 16.$$

TEXTBOOK HOMEWORK · Exercise 5C

FOUNDATION

Technology: All the properties of the definite integral discussed in this section have been justified visually from sketches of the graphs. Computer sketches of the graphs in this exercise would be helpful in reinforcing these explanations. The simplification of integrals of even and odd functions is particularly important and is easily demonstrated visually by graphing programs.

1 Evaluate $\int_4^5 (2x-3)\,dx$ and $\int_5^4 (2x-3)\,dx$. What do you notice?

2 Show, by evaluating the definite integrals, that:

a $\int_0^1 6x^2\,dx = 6\int_0^1 x^2\,dx$

b $\int_{-1}^2 (x^3+x^2)\,dx = \int_{-1}^2 x^3\,dx + \int_{-1}^2 x^2\,dx$

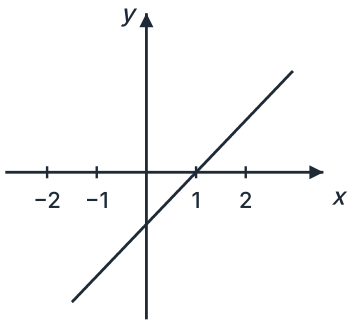
c $\int_0^3 (x^2-4x+3)\,dx = \int_0^2 (x^2-4x+3)\,dx + \int_2^3 (x^2-4x+3)\,dx$

3 Without evaluating the definite integrals, explain why:

a $\int_2^2 (x^2 - 3x)\,dx = 0$

b $\int_{-2}^2 x\,dx = 0$

4 The diagram to the right shows the line $y = x - 1$.
Without evaluating the definite integrals, explain why:



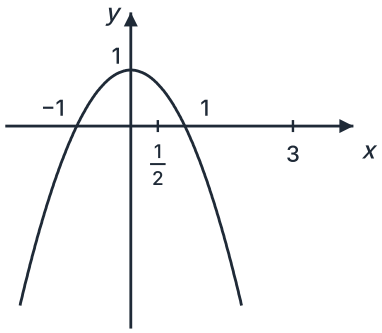
a $\int_0^1 (x - 1)\,dx$ is negative,

b $\int_1^2 (x - 1)\,dx$ is positive,

c $\int_0^2 (x - 1)\,dx$ is zero,

d $\int_{-2}^2 (x - 1)\,dx$ is negative.

5 The diagram to the right shows the parabola $y = 1 - x^2$.
Without evaluating the definite integrals, explain why:



a $\int_{-1}^1 (1 - x^2)\,dx$ is positive,

b $\int_1^3 (1 - x^2)\,dx$ is negative,

c $\int_{-1}^0 (1 - x^2)\,dx = \int_0^1 (1 - x^2)\,dx$,

d $\int_0^{\frac{1}{2}} (1 - x^2)\,dx > \int_{\frac{1}{2}}^1 (1 - x^2)\,dx$.

6

a If $\int_1^3 f(x)\,dx = 7$, then what is the value of $\int_3^1 f(x)\,dx$?

b If $\int_2^1 g(x)\,dx = -5$, then what is the value of $\int_1^2 g(x)\,dx$?

7 Use a diagram to explain why $\int_0^1 2x\,dx > \int_0^1 x\,dx$.

8 Write $\int_{-2}^{0} x^3\,dx + \int_0^1 x^3\,dx$ as a single integral, and then use a diagram to explain why this definite integral is negative.

DEVELOPMENT

9 In each part, evaluate the definite integrals. Then use the properties of the definite integral to explain the relationships amongst the integrals within that part.

a i $\displaystyle\int_0^2 (3x^2 - 1)\,dx$ ii $\displaystyle\int_2^0 (3x^2 - 1)\,dx$

b i $\displaystyle\int_0^1 20x^3\,dx$ ii $\displaystyle 20\int_0^1 x^3\,dx$

c i $\displaystyle\int_1^4 (4x + 5)\,dx$ ii $\displaystyle\int_1^4 4x\,dx$ iii $\displaystyle\int_1^4 5\,dx$

d i $\displaystyle\int_0^2 12x^3\,dx$ ii $\displaystyle\int_0^1 12x^3\,dx$ iii $\displaystyle\int_1^2 12x^3\,dx$

e i $\displaystyle\int_3^3 (4 - 3x^2)\,dx$ ii $\displaystyle\int_{-2}^{-2} (4 - 3x^2)\,dx$

10 Use the properties of the definite integral to evaluate each integral without using a primitive function. Give reasons.

a $\displaystyle\int_3^3 \sqrt{9-x^2}\,dx$ b $\displaystyle\int_4^4 (x^3-3x^2+5x-7)\,dx$ c $\displaystyle\int_{-1}^1 x^3\,dx$

d $\displaystyle\int_{-5}^5 (x^3-25x)\,dx$ e $\displaystyle\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x\,dx$ f $\displaystyle\int_{-2}^2 \frac{x}{1+x^2}\,dx$

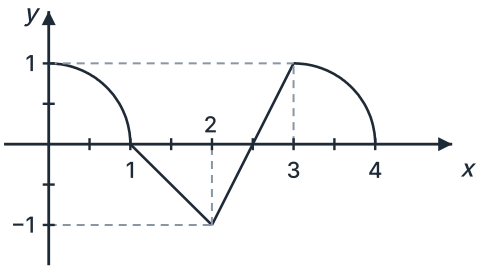
11 a On one set of axes sketch $y=x^2$ and $y=x^3$, clearly showing the points of intersection.

b Hence explain why $0 < \displaystyle\int_0^1 x^3\,dx < \int_0^1 x^2\,dx < 1$.

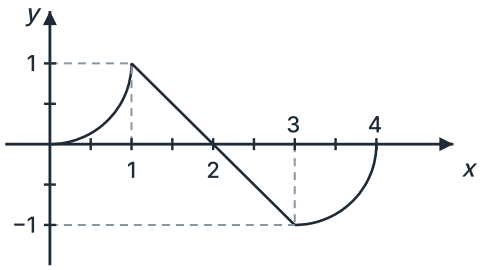
c Check the inequality in part **b** by evaluating each integral.

12 Use area formulae to find $\int_0^4 f(x)\,dx$, given the following sketches of $f(x)$.

a



b



13 **a** Calculate using a graph and an area formula.

i $\int_0^5 1\,dx$ ii $\int_0^5 x\,dx$

b Using these results, and the properties of integrals of sums and multiples, evaluate:

i $\int_0^5 2x\,dx$ ii $\int_0^5 (x+1)\,dx$ iii $\int_0^5 (3x-2)\,dx$

14 a Calculate these definite integrals using graphs and area formulae.

i $\int_{-\frac{1}{4}}^{\frac{1}{4}} (1-4x)\,dx$ ii $\int_{-5}^{1} |x+5|\,dx$ iii $\int_{0}^{2} (|x|+3)\,dx$

b Hence write down the value of:

i $\int_{\frac{1}{4}}^{-\frac{1}{4}} (1-4x)\,dx$ ii $\int_{1}^{-5} |x+5|\,dx$ iii $\int_{2}^{0} (|x|+3)\,dx$

15 Using the properties of the definite integral, explain why:

b $\int_{-k}^{k} (bx^4+dx^2+f)\,dx = 2\int_{0}^{k} (bx^4+dx^2+f)\,dx$

16 Sketch a graph of each integral and hence determine whether each statement is true or false.

a $\int_{-1}^{1} 2^x\,dx = 0$ c $\int_{-2}^{-1} \frac{1}{x}\,dx > 0$ d $\int_{2}^{1} \frac{1}{x}\,dx > 0$

17

a i Show that $\int_3^4 dx = \int_2^3 dx = \int_1^2 dx = 1.$

ii Show that $\frac{2}{7}\int_3^4 x\,dx = \frac{2}{5}\int_2^3 x\,dx = \frac{2}{3}\int_1^2 x\,dx = 1.$

iii Show that $\frac{3}{37}\int_3^4 x^2\,dx = \frac{3}{19}\int_2^3 x^2\,dx = \frac{3}{7}\int_1^2 x^2\,dx = 1.$

b Use the results above and the theorems on the definite integral to calculate:

i $\int_1^4 dx$

ii $\int_1^3 x\,dx$

iii $\int_2^1 x^2\,dx$

iv $\int_1^2 (x^2+1)\,dx$

v $\int_1^3 7x^2\,dx$

vi $\int_1^4 (3x^2-6x+5)\,dx$

ENRICHMENT

18 State with reasons whether each statement is true or false.

a $\displaystyle\int_{-90}^{90} \sin^3 x^\circ \, dx = 0$

b $\displaystyle\int_{-30}^{30} \sin 4x^\circ \cos 2x^\circ \, dx = 0$

c $\displaystyle\int_{-1}^{1} 2^{-x^2} \, dx = 0$

d $\displaystyle\int_{0}^{1} 2^x \, dx < \int_{0}^{1} 3^x \, dx$

e $\displaystyle\int_{-1}^{0} 2^x \, dx < \int_{-1}^{0} 3^x \, dx$

f $\displaystyle\int_{0}^{1} \frac{dt}{1+t^n} \leq \int_{0}^{1} \frac{dt}{1+t^{n+1}}$, where $n = 1, 2, 3, \ldots$.

19 Evaluate each definite integral, then establish the result that follows. Provide a sketch of each situation.

a $\displaystyle\int_1^N \frac{1}{x^2}\,dx$ converges to 1 as $N \to \infty$.

b $\displaystyle\int_\varepsilon^1 \frac{1}{x^2}\,dx$ diverges to ∞ as $\varepsilon \to 0^+$.

c $\displaystyle\int_1^N \frac{1}{\sqrt{x}}\,dx$ diverges to ∞ as $N \to \infty$.

d $\displaystyle\int_\varepsilon^1 \frac{1}{\sqrt{x}}\,dx$ converges to 2 as $\varepsilon \to 0^+$.

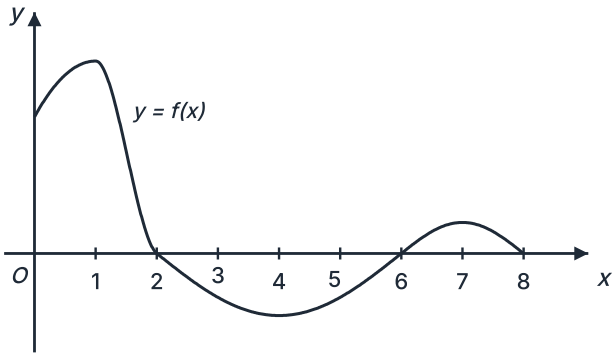
PAST-PAPER HOMEWORK · mini exam

Suggested time: 5 minutes **3 marks** Calculator allowed · show working · hints are below each question if you get stuck

These are the only school-paper items that need just this lesson's ideas. For more timed practice, do Questions 9(d), 10(f), 12(b), 14, 17(b) and 18(b) above in 25 minutes.

PP 1 **Sydney Technical High School, 2021, Extension 1 Mathematics, Trial HSC, Q6** 1 mark

The graph of $y = f(x)$ has been drawn to scale for $0 \leq x \leq 8$.



Which of the following integrals has the greatest value?

(A) $\int_0^1 f(x)\,dx$ (B) $\int_0^2 f(x)\,dx$

(C) $\int_0^7 f(x)\,dx$ (D) $\int_0^8 f(x)\,dx$

Hint: what does the graph of $y = f(x)$ look like across $[0, 8]$?

PP 2 **Abbotsleigh, 2023, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q10** 1 mark

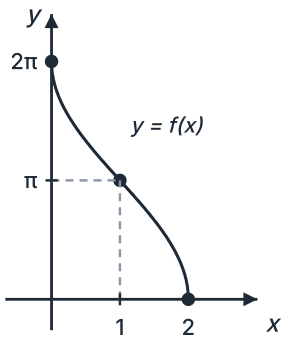
Which of the following is equal to $\int_{-1}^{1} \cos^{-1} x\,dx$?

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{2}$ (C) π (D) 2π

Hint: what does the graph of $y = \cos^{-1} x$ look like between $x = -1$ and $x = 1$?

PP 3 **Epping Boys High School, 2024, Trial Higher School Certificate Examination, Mathematics Extension 1, Q12(c)(ii)** **1 mark**

Part (i), not included here, asks for a sketch of the graph of $f(x) = 2\cos^{-1}(x-1)$. The graph is shown to the right.



(ii) Hence, explain why $\int_0^2 \left[2\cos^{-1}(x-1) - \pi\right] dx = 0$ 1

Hint: what does the graph of $y = 2\cos^{-1}(x-1) - \pi$ look like?

LESSON 5D

Proving the fundamental theorem

This lesson proves the fundamental theorem of calculus, which was stated and used in Lessons 5B and 5C. The key idea is to let the upper limit of a definite integral vary. This gives a new function, the *signed area function* $A(x) = \int_a^x f(t)\,dt$. The differential form of the theorem says that $A'(x) = f(x)$. It is proven by trapping a thin strip between a lower and an upper rectangle, as in Lesson 5A. The integral form $\int_a^b f(x)\,dx = F(b) - F(a)$ then follows, because any two primitives differ by a constant. The lesson ends with a note on continuous functions.

CONCEPT 1 · the signed area function

WHERE THIS SECTION FITS

This section develops a proof of the fundamental theorem of calculus, as stated and used in Sections 5B and 5C. This would be the time to review the account of first-principles integration in Section 5A, because the approach used there of trapping an integral between upper and lower rectangles is used again in this proof.

IN THIS PACK

Sections 5A, 5B and 5C of the textbook are Lessons 5A, 5B and 5C of this pack.

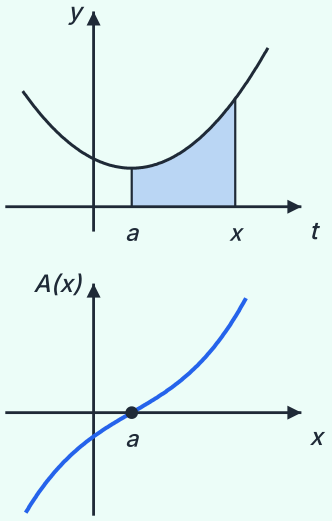
THE DEFINITE INTEGRAL AS A FUNCTION OF ITS UPPER LIMIT

The value of a definite integral $\int_a^b f(x)\,dx$ changes when the value of b changes. This means that it is a function of its upper limit b. When we want to emphasise the functional relationship with the upper limit, we usually replace the letter b by the letter x, which is conventionally the variable of a function.

In turn, the original letter x needs to be replaced by some other letter, usually t. Then the definite integral is clearly represented as a function of its upper limit x, as in the first diagram above. This function is called the *signed area function* for $f(x)$ starting at $x = a$, and is defined by

$$A(x) = \int_a^x f(t)\,dt.$$

The function $A(x)$ is always zero at $x = a$. For the function sketched above, $A(x)$ increases at an increasing rate (see the second sketch). $A(x)$ is also defined to the left of a, where it is negative because the integrals run backwards.



READING THE TWO SKETCHES LINE BY LINE

- At $x = a$: $A(a) = \int_a^a f(t)\,dt = 0$, because the region from a to a has zero width.
- To the right of a: when x moves a little to the right, a thin new strip of height about $f(x)$ is added to the shaded region. Here $f(x)$ is positive, so $A(x)$ increases. Also $f(x)$ gets larger as x moves right, so each new strip is taller than the one before, and $A(x)$ increases at an increasing rate.
- To the left of a: when $x < a$, the integral runs backwards, from a down to x. Reversing the limits changes the sign (Lesson 5C), so

  $$A(x) = \int_a^x f(t)\,dt = -\int_x^a f(t)\,dt.$$

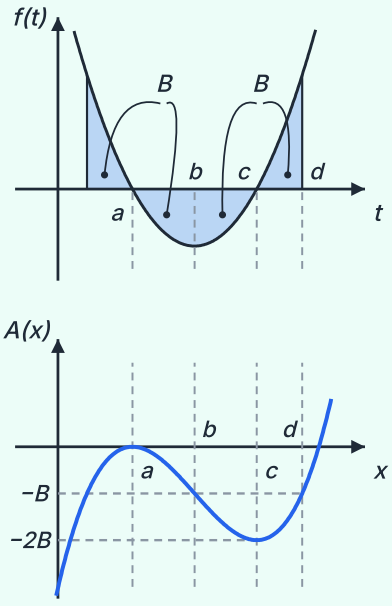
  The integral $\int_x^a f(t)\,dt$ is a positive area, because the curve is above the t-axis. Hence $A(x)$ is negative.

THE SIGNED AREA FUNCTION

The function in the sketch above was never negative. But the definite integral is the *signed area* under the curve, meaning that a negative sign is attached to areas of regions below the x-axis (provided that the integral is not running backwards).

The upper graph of $f(t)$ to the right is a parabola with axis of symmetry $x = b$. The four areas marked B are all equal. Here are some properties of the signed area function

$$A(x) = \int_a^x f(t)\,dt.$$



- $A(a) = 0$, as always.
- In the interval $[a, b]$, $f(t)$ is negative and decreasing, so $A(x)$ decreases at an increasing rate.
- In the interval $[b, c]$, $f(t)$ is negative and increasing, so $A(x)$ decreases at a decreasing rate.
- In the interval $[c, \infty)$, $f(t)$ is positive and increasing, so $A(x)$ increases at an increasing rate.
- $A(b) = -B$ and $A(c) = -2B$ and $A(d) = -B$.
- $A(x)$ is also defined in the interval $(-\infty, a]$ to the left of a, where it is negative because the integrals run backwards and the curve is above the x-axis.

The lower diagram above is a sketch of the signed area function $A(x)$ of $f(x)$.

WHERE THE VALUES AND THE RATES COME FROM

The region from a to b is below the axis and has area B, so its signed area is $-B$. Add the regions one at a time:

$$A(b) = \int_a^b f(t)\,dt = -B,$$

$$A(c) = \int_a^b f(t)\,dt + \int_b^c f(t)\,dt = -B + (-B) = -2B,$$

$$A(d) = \int_a^c f(t)\,dt + \int_c^d f(t)\,dt = -2B + B = -B.$$

Think of the thin strips again. In $[a, b]$ each new strip has a negative signed area, and the strips get deeper because $f(t)$ is decreasing, so $A(x)$ goes down faster and faster. In $[b, c]$ the strips are still below the axis, but they get shallower, so $A(x)$ goes down more and more slowly. At c the curve crosses the axis, so $A(x)$ stops decreasing and starts increasing: it has a local minimum $A(c) = -2B$.

The fourth area B, to the left of a, is the mirror image in the axis of symmetry of the region from c to d. It is above the axis, and the integral from a back to its left-hand end runs backwards, so at that end $A(x) = -B$ again.

13 THE SIGNED AREA FUNCTION

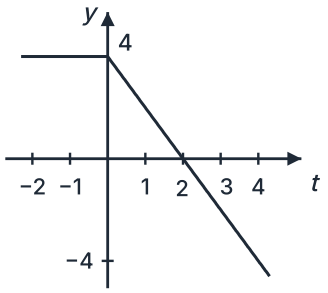
Suppose that $f(x)$ is a function defined in some interval I containing a. The *signed area function* for $f(x)$ starting at a is the function defined by the definite integral

$$A(x) = \int_a^x f(t)\,dt, \qquad \textbf{for all } x \textbf{ in the interval } I.$$

In Chapter 16 on continuous probability distributions, the cumulative distribution function of a continuous distribution will be defined in this way.

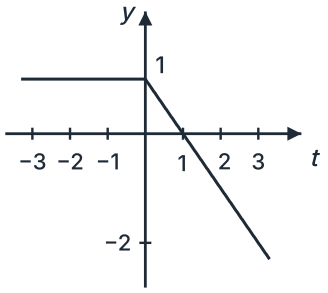
TUTOR DEMO A table of values of $A(x)$

Let $A(x) = \int_0^x f(t)\,dt$ be the signed area function starting at $t = 0$ for the graph sketched to the right. Use area formulae to draw up a table of values for $y = A(x)$ in the interval $[-2, 4]$, then sketch $y = A(x)$.



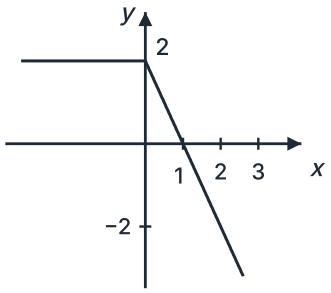
YOUR TURN Same method

Let $A(x) = \int_0^x f(t)\,dt$ be the signed area function starting at $t = 0$ for the graph sketched to the right. Use area formulae to draw up a table of values for $y = A(x)$ in the interval $[-3, 3]$, then sketch $y = A(x)$.



WORKED EXAMPLE 13 **From the text: keep for homework**

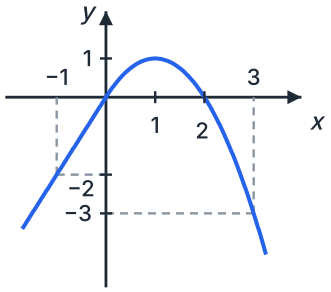
Let $A(x) = \int_0^x f(t)\,dt$ be the signed area function starting at $t = 0$ for the graph sketched to the right. Use area formulae to draw up a table of values for $y = A(x)$ in the interval $[-3, 3]$, then sketch $y = A(x)$.



SOLUTION

Use triangles for $x > 0$ and rectangles for $x < 0$.

x	-3	-2	-1	0	1	2	3
$A(x)$	-6	-4	-2	0	1	0	-3



For $x = -2$ and $x = -1$, $A(x)$ is negative because the integrals run backwards and the curve is above the x-axis. The area function $A(x)$ is increasing for $t < 1$ because $y > 0$, and is decreasing for $x > 1$ because $y < 0$.

WHERE THE TABLE ENTRIES COME FROM

The graph is the horizontal line $y = 2$ for $t < 0$, and the line through $(0, 2)$ and $(1, 0)$ for $t \geq 0$, which is at height -2 when $t = 2$ and at height -4 when $t = 3$.

- $x = -3,\ -2,\ -1$: each is a rectangle of height 2 above the axis, and the integral runs backwards, so $A(-3) = -(3 \times 2) = -6$, $A(-2) = -(2 \times 2) = -4$ and $A(-1) = -(1 \times 2) = -2$.
- $x = 0$: $A(0) = 0$, as always.
- $x = 1$: a triangle above the axis, with base 1 and height 2, so $A(1) = \frac{1}{2} \times 1 \times 2 = 1$.
- $x = 2$: from 1 to 2 there is a triangle below the axis, with base 1 and height 2, so $A(2) = 1 - \frac{1}{2} \times 1 \times 2 = 1 - 1 = 0$.
- $x = 3$: from 1 to 3 there is a triangle below the axis, with base 2 and height 4, so $A(3) = 1 - \frac{1}{2} \times 2 \times 4 = 1 - 4 = -3$.

To sketch $y = A(x)$, plot the seven points and join them smoothly. For $x \leq 0$ the points lie on a straight line, because each step of 1 to the left takes away the same area 2. The highest point is at $x = 1$, where $A(x)$ changes from increasing to decreasing.

CONCEPT 2 · the fundamental theorem, differential form

STATING THE DIFFERENTIAL FORM

We can now state and prove the *differential form* of the fundamental theorem of calculus, from which we will derive the *integral form* used already in Sections 5B and 5C.

THE FUNDAMENTAL THEOREM: DIFFERENTIAL FORM

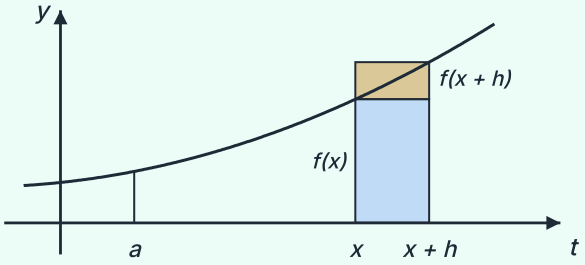
Theorem: If $f(x)$ is continuous, then the signed area function for $f(x)$ is a primitive of $f(x)$. That is,

$$A'(x) = \frac{d}{dx}\int_a^x f(t)\,dt = f(x).$$

PROOF

Proof: Because the theorem is so fundamental, its proof must begin with the definition of the derivative as a limit,

$$A'(x) = \lim_{h\to 0}\frac{A(x+h)-A(x)}{h}.$$



Subtracting areas in the diagram above,

$$A(x+h)-A(x) = \int_x^{x+h} f(t)\,dt,$$

$$\text{so}\qquad A'(x) = \lim_{h\to 0}\frac{1}{h}\int_x^{x+h} f(t)\,dt.$$

This limit is handled by means of a clever sandwiching technique.
Suppose that $f(t)$ is increasing in the interval $[x, x+h]$, as in the diagram above.
Then the lower rectangle on the interval $[x, x+h]$ has height $f(x)$,
and the upper rectangle on the interval $[x, x+h]$ has height $f(x+h)$,

$$\textbf{so using areas,}\qquad h\times f(x) \le \int_x^{x+h} f(t)\,dt \le h\times f(x+h)$$

$$\div h\qquad f(x) \le \frac{1}{h}\int_x^{x+h} f(t)\,dt \le f(x+h). \qquad (1)$$

Thus the middle expression is 'sandwiched' between $f(x)$ and $f(x+h)$.
Because $f(x)$ is continuous, $f(x+h)\to f(x)$ as $h\to 0$,

$$\textbf{so by (1),}\qquad \lim_{h\to 0}\frac{1}{h}\int_x^{x+h} f(t)\,dt = f(x),$$

meaning that $A'(x) = f(x)$, as required.

If $f(x)$ is decreasing in the interval $[x, x+h]$, the same argument applies, but with the inequalities reversed.

Note: This theorem shows that the signed area function $A(x) = \int_a^x f(t)\,dt$ is a primitive of $f(x)$. It is therefore often written as $F(x)$ rather than $A(x)$.

THE STEPS OF THE PROOF, FILLED IN

- **Subtracting areas.** The region from a to $x+h$ is the region from a to x together with the thin strip from x to $x+h$:

$$\begin{aligned} A(x+h) &= \int_a^{x+h} f(t)\,dt \\ &= \int_a^x f(t)\,dt + \int_x^{x+h} f(t)\,dt \\ &= A(x) + \int_x^{x+h} f(t)\,dt. \end{aligned}$$

  Subtracting $A(x)$ from both sides gives $A(x+h) - A(x) = \int_x^{x+h} f(t)\,dt$.

- **The rectangles.** Both rectangles have width h. The blue lower rectangle lies inside the strip under the curve from x to $x+h$, and the strip lies inside the upper rectangle (blue and amber together). Area of a rectangle = width × height, which gives $h \times f(x)$ and $h \times f(x+h)$.
- **Dividing by h.** In the diagram $h > 0$, so dividing by h keeps the inequality signs the same way round. (When $h < 0$, the strip runs from $x+h$ to x, and the same sandwich applied to that strip gives the same limit.)
- **The sandwich.** As $h \to 0$, the left-hand side $f(x)$ stays fixed and the right-hand side $f(x+h)$ approaches $f(x)$. The middle expression is trapped between two quantities that both approach $f(x)$, so it approaches $f(x)$ too. This is the same reasoning as step **D** in Lesson 5A.
- **Decreasing f.** Then the lower rectangle has height $f(x+h)$ and the upper one has height $f(x)$, so $f(x+h) \le \frac{1}{h}\int_x^{x+h} f(t)\,dt \le f(x)$, with the same limit $f(x)$.
- **The lower limit a does not matter.** Starting at another number a_1 instead of a gives $\int_{a_1}^x f(t)\,dt = \int_{a_1}^a f(t)\,dt + \int_a^x f(t)\,dt$. This differs from $A(x)$ by a constant, and the derivative of a constant is zero.

TUTOR DEMO **Using the differential form**

Use the differential form of the fundamental theorem to simplify these expressions. Do not try to evaluate the integral and then differentiate it.

a $\dfrac{d}{dx}\displaystyle\int_0^x (t^3+2)^4\,dt$ b $\dfrac{d}{dx}\displaystyle\int_1^x \sqrt{1+t^3}\,dt$ c $\dfrac{d}{dx}\displaystyle\int_{-2}^x e^{-3t^2}\,dt$

YOUR TURN **Same method**

Use the differential form of the fundamental theorem to simplify these expressions. Do not try to evaluate the integral and then differentiate it.

a $\dfrac{d}{dx}\displaystyle\int_2^x (t^4-1)^5\,dt$ b $\dfrac{d}{dx}\displaystyle\int_3^x \log_e(t^2+1)\,dt$ c $\dfrac{d}{dx}\displaystyle\int_{-1}^x e^{t^3}\,dt$

WORKED EXAMPLE 14 **From the text: keep for homework**

Use the differential form of the fundamental theorem to simplify these expressions. Do not try to evaluate the integral and then differentiate it.

a $\dfrac{d}{dx}\displaystyle\int_0^x (t^2+1)^3\,dt$ b $\dfrac{d}{dx}\displaystyle\int_4^x (\log_e t)\,dt$ c $\dfrac{d}{dx}\displaystyle\int_{-3}^x e^{-t^2}\,dt$

SOLUTION

The differential form says that $\dfrac{d}{dx}\displaystyle\int_a^x f(t)\,dt = f(x)$. Hence:

a $\dfrac{d}{dx}\displaystyle\int_0^x (t^2+1)^3\,dt = (x^2+1)^3$

b $\dfrac{d}{dx}\displaystyle\int_4^x (\log_e t)\,dt = \log_e x$

c $\dfrac{d}{dx}\displaystyle\int_{-3}^x e^{-t^2}\,dt = e^{-x^2}$

READING EXAMPLE 14

In each part, read off the integrand $f(t)$ and replace t by x: in **a** $f(t) = (t^2+1)^3$, in **b** $f(t) = \log_e t$, and in **c** $f(t) = e^{-t^2}$. The lower limits 0, 4 and -3 do not appear in the answers.

CONCEPT 3 · the fundamental theorem, integral form

THE INTEGRAL FORM

The integral form of the fundamental theorem is the familiar form that we have been using in Sections 5B and 5C.

THE FUNDAMENTAL THEOREM: INTEGRAL FORM

Theorem: Suppose that $f(x)$ is continuous in the closed interval $[a, b]$, and that $F(x)$ is a primitive of $f(x)$. Then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

PROOF

Proof: We now know that $F(x)$ and $\int_a^x f(t)\,dt$ are both primitives of $f(x)$.

Because any two primitives differ only by a constant,

$$\int_a^x f(t)\,dt = F(x) + C, \qquad \text{for some constant } C.$$

$$\text{Substituting } x = a, \qquad \int_a^a f(t)\,dt = F(a) + C,$$

but $\int_a^a f(t)\,dt = 0$, because the area in this definite integral has zero width,

$$\text{so} \qquad 0 = F(a) + C$$
$$C = -F(a).$$

$$\text{Thus} \qquad \int_a^x f(t)\,dt = F(x) - F(a),$$

and changing letters from x to b and from t to x gives

$$\int_a^b f(x)\,dx = F(b) - F(a), \qquad \text{as required.}$$

THE REASONS IN THE PROOF

- $\int_a^x f(t)\,dt$ is a primitive of $f(x)$ by the differential form, proven above.
- 'Any two primitives differ only by a constant' is Box 14B in Lesson 4J.
- The line 'Thus ...' comes from putting $C = -F(a)$ into the first line, $\int_a^x f(t)\,dt = F(x) + C$.

THE CHANGE OF LETTERS

Putting $x = b$ into $\int_a^x f(t)\,dt = F(x) - F(a)$ gives $\int_a^b f(t)\,dt = F(b) - F(a)$. The letter t inside the integral only names the horizontal axis: $\int_a^b f(t)\,dt$ and $\int_a^b f(x)\,dx$ are the same area, so t may be changed back to x. In the examples below, the primitive is written in square brackets, $\big[F(t)\big]_a^x = F(x) - F(a)$, exactly as in Lesson 5B.

TUTOR DEMO **The two forms agree**

Use the integral form of the fundamental theorem to evaluate each integral. Then differentiate your result, thus confirming the consistency of the discussion above.

a $\dfrac{d}{dx}\int_2^x 12t^3\,dt$ b $\dfrac{d}{dx}\int_{-1}^x (t^3 - 6t^2 + 4)\,dt$ c $\dfrac{d}{dx}\int_2^x \dfrac{1}{t^3}\,dt$

YOUR TURN **Same method**

Use the integral form of the fundamental theorem to evaluate each integral. Then differentiate your result, thus confirming the consistency of the discussion above.

a $\dfrac{d}{dx}\displaystyle\int_1^x 3t^2\,dt$

b $\dfrac{d}{dx}\displaystyle\int_{-1}^x (t^3+3t^2-2)\,dt$

c $\dfrac{d}{dx}\displaystyle\int_1^x \dfrac{2}{t^2}\,dt$

WORKED EXAMPLE 15 **From the text: keep for homework**

Use the integral form of the fundamental theorem to evaluate each integral. Then differentiate your result, thus confirming the consistency of the discussion above.

a $\dfrac{d}{dx}\displaystyle\int_1^x 6t^2\,dt$ b $\dfrac{d}{dx}\displaystyle\int_{-2}^x (t^3-9t^2+5)\,dt$ c $\dfrac{d}{dx}\displaystyle\int_4^x \dfrac{1}{t^2}\,dt$

SOLUTION

a

$$\begin{aligned}\int_1^x 6t^2\,dt &= \Big[2t^3\Big]_1^x \\ &= 2x^3-2,\end{aligned}$$

so $$\begin{aligned}\frac{d}{dx}\int_1^x 6t^2\,dt &= \frac{d}{dx}(2x^3-2) \\ &= 6x^2,\end{aligned}$$

consistent with the differential form.

b

$$\begin{aligned}\int_{-2}^x (t^3-9t^2+5)\,dt &= \Big[\tfrac{1}{4}t^4-3t^3+5t\Big]_{-2}^x \\ &= \left(\tfrac{1}{4}x^4-3x^3+5x\right)-(4+24-10) \\ &= \tfrac{1}{4}x^4-3x^3+5x-18,\end{aligned}$$

so $$\begin{aligned}\frac{d}{dx}\int_{-2}^x (t^3-9t^2+5)\,dt &= \frac{d}{dx}\left(\tfrac{1}{4}x^4-3x^3+5x-18\right) \\ &= x^3-9x^2+5,\end{aligned}$$

consistent with the differential form.

c

$$\begin{aligned}\int_4^x \frac{1}{t^2}\,dt &= \int_4^x t^{-2}\,dt \\ &= \Big[-t^{-1}\Big]_4^x \\ &= -x^{-1}+\tfrac{1}{4},\end{aligned}$$

so $$\begin{aligned}\frac{d}{dx}\int_4^x \frac{1}{t^2}\,dt &= \frac{d}{dx}\left(-x^{-1}+\tfrac{1}{4}\right) \\ &= x^{-2},\end{aligned}$$

consistent with the differential form.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 15

- **a** The primitive of $6t^2$ is $\dfrac{6t^3}{3}=2t^3$. Substitute x, then 1: $2x^3-2\times 1^3=2x^3-2$.
- **b** The primitive of t^3-9t^2+5 is $\dfrac{t^4}{4}-\dfrac{9t^3}{3}+5t=\tfrac{1}{4}t^4-3t^3+5t$. Substituting -2 gives $\tfrac{1}{4}(-2)^4-3(-2)^3+5(-2)=\tfrac{1}{4}\times 16-3\times(-8)-10=4+24-10$.

- **c** Write $\frac{1}{t^2} = t^{-2}$. Its primitive is $\frac{t^{-1}}{-1} = -t^{-1}$. Substitute x, then 4: $-x^{-1} - \left(-4^{-1}\right) = -x^{-1} + \frac{1}{4}$. Then $\frac{d}{dx}\left(-x^{-1} + \frac{1}{4}\right) = -(-1)x^{-2} = x^{-2}$.

READING EXAMPLE 15

In each part the final answer is the integrand with t replaced by x (in part **c**, $x^{-2} = \frac{1}{x^2}$), just as the differential form says. The constant $-F(a)$ disappears when we differentiate.

CONCEPT 4 · super challenge, continuous functions

CONTINUOUS AT A POINT, IN AN INTERVAL, AND AS A FUNCTION

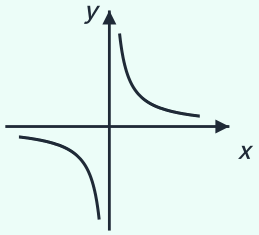
In Section 9K of the Year 11 volume, we defined continuity at a point: you can draw the curve through the point without lifting the pencil off the paper.

Then throughout this chapter, we have been using the phrase, 'continuous in the closed interval $[a, b]$'. This idea is also straightforward, and informally means that you can place the pencil on the left-hand endpoint $\big(a, f(a)\big)$ and draw the curve to the right-hand endpoint $\big(b, f(b)\big)$ without lifting the pencil off the paper.

There is, however, a global notion of a *continuous function*:

A continuous function is a function continuous at every value in its domain.

This may look obvious, like so many definitions in mathematics, but it is not. It means, for example, that the reciprocal function $y = \frac{1}{x}$ is a continuous function. This is because $x = 0$ is not in its domain, so the function is continuous at every value of x in its domain.



Thus $y = \frac{1}{x}$ is a 'continuous function with a discontinuity'. (Or perhaps the word 'discontinuity' is redefined.) We recommend avoiding the concept completely unless some question specifically requires it.

HOW TO CLASSIFY A FUNCTION

1. Find the domain.
2. Look for any value **in the domain** where the pencil must be lifted to draw the graph through that point.
3. If there is none, it is a continuous function. A value that is not in the domain, such as $x = 0$ for $y = \frac{1}{x}$, does not count.

TUTOR DEMO **Continuous or not?**

Classify these functions as continuous or not continuous according to the definition above.

a $y = x^2 + 1$

b $y = \dfrac{1}{x+3}$

c $y = \begin{cases} \dfrac{1}{x+3}, & \text{for } x \neq -3, \\ 0, & \text{for } x = -3. \end{cases}$

d $y = \sqrt{x+3}$

YOUR TURN **Same method**

Classify these functions as continuous or not continuous according to the definition above.

a $y = 3 - x$

b $y = \dfrac{1}{x^2}$

c $y = \begin{cases} \dfrac{1}{x^2}, & \text{for } x \neq 0, \\ 0, & \text{for } x = 0. \end{cases}$

d $y = \dfrac{1}{\sqrt{x+1}}$

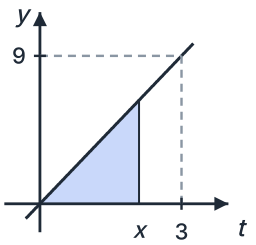
PRACTICE FOR THIS PART

The textbook has no worked example for this part. Exercise 5D Question 10 practises it.

TEXTBOOK HOMEWORK · Exercise 5D

FOUNDATION

1 The graph to the right shows $y = 3t$, for $0 \leq t \leq 3$.



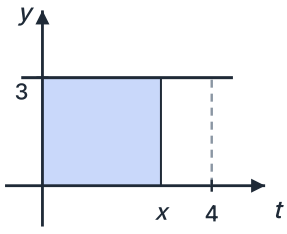
a Use the triangle area formula to find the signed area function

$$A(x) = \int_0^x 3t\,dt, \text{ for } 0 \leq x \leq 3.$$

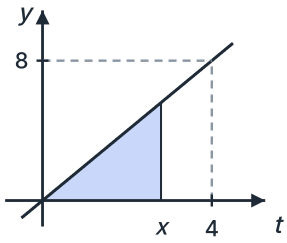
b Differentiate $A(x)$ to show that $A'(x)$ is the original function, apart from a change of letter.

2 Write down the equation of each function, then use area formulae, not integration, to calculate the signed area function $A(x) = \int_0^x f(t)\,dt$, for $0 \leq t \leq 4$. Then differentiate $A(x)$ to confirm that $A'(x)$ is the original function, apart from a change of letter.

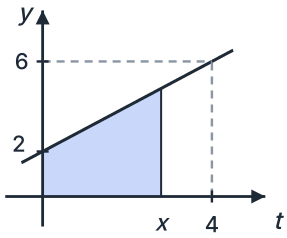
a



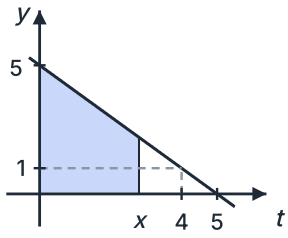
b



c



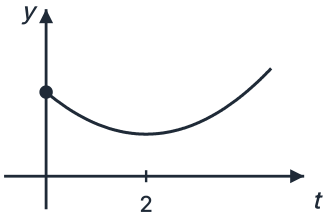
d



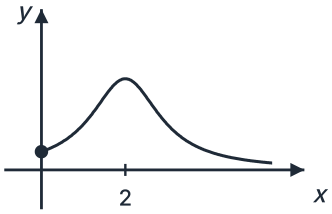
DEVELOPMENT

3 For each function sketched below, describe the behaviour of the signed area function $A(x) = \int_0^x f(t)\,dt$, for all $x \geq 0$, in the interval $[0, 2]$ and in the interval $[2, \infty)$. Then draw a freehand sketch of $y = A(x)$.

a



b



4 The differential form $\frac{d}{dx}\int_a^x f(t)\,dt$ of the fundamental theorem tells us that the derivative of the integral is the original function, with a change of letter. Use this to simplify these expressions. Do not attempt to find primitives.

a $\frac{d}{dx}\int_1^x \frac{1}{t}\,dt$ b $\frac{d}{dx}\int_0^x \frac{1}{1+t^3}\,dt$ c $\frac{d}{dx}\int_0^x e^{-\frac{1}{2}t^2}\,dt$

5 Use the differential form of the fundamental theorem to simplify these expressions. Then confirm the consistency of the discussion in this section by performing the integration and then differentiating.

a $\dfrac{d}{dx}\displaystyle\int_1^x (3t^2 - 12)\,dt$ b $\dfrac{d}{dx}\displaystyle\int_2^x (t^3 + 4t)\,dt$ c $\dfrac{d}{dx}\displaystyle\int_2^x \dfrac{1}{t^2}\,dt$

6

a Use the differential form of the fundamental theorem $\dfrac{d}{dx}\displaystyle\int_a^x f(t)\,dt = f(x)$ to find:

i $\dfrac{d}{dx}\displaystyle\int_1^x t^2\,dt$

ii $\dfrac{d}{dx}\displaystyle\int_2^x (t^3+3t)\,dt$

iii $\dfrac{d}{dx}\displaystyle\int_a^x \dfrac{1}{t}\,dt$

iv $\dfrac{d}{dx}\displaystyle\int_a^x (t^3-3)^4\,dt$

b Confirm your answers to parts (i) and (ii) above by evaluating the definite integral and then differentiating with respect to x.

7

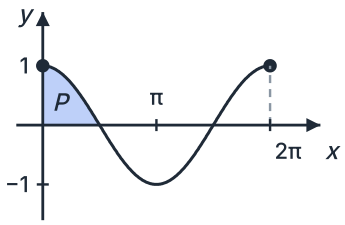
a Sketch $y = e^t$, then sketch the signed area function $A(x) = \int_0^x e^t \, dt$. How would you describe the behaviour of $y = A(x)$?

b Sketch $y = \log_e t$, then sketch the signed area function $A(x) = \int_1^x \log_e t \, dt$. How would you describe the behaviour of $y = A(x)$?

c Sketch $y = \dfrac{1}{t}$, then sketch the signed area function $A(x) = \int_1^x \dfrac{1}{t} \, dt$. How would you describe the behaviour of $y = A(x)$?

ENRICHMENT

8 **a** Sketched to the right is $y = \cos x$, for $0 \leq x \leq 2\pi$. Copy and complete the table of values for the signed area function $A(x) = \int_0^x \cos t \, dt$, for $0 \leq x \leq 2\pi$, given that the region marked P has area exactly 1 (this is proven in Chapter 7). Then sketch $y = A(x)$.

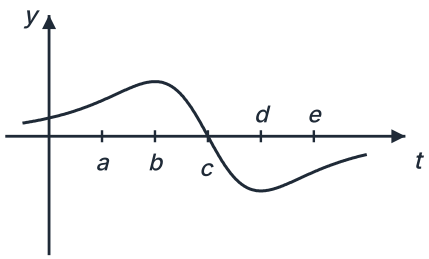


x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$A(x)$					

What is your guess for the equation of $A(x)$, and what does this suggest the derivative of $\sin x$ is?

b Sketch $y = \sin x$, for $0 \leq x \leq 2\pi$, and repeat the procedures in part **a**.

9 The function $y = f(t)$ sketched to the right has point symmetry in $(c, 0)$.



Let $A(x) = \int_a^x f(t)\,dt$.

a Where is $A(x)$ increasing, and when is it decreasing?

b Where does $A(x)$ have a maximum turning point, and where does $A(x)$ have a minimum turning point?

c Where does $A(x)$ have inflections?

d Where are the zeroes of $A(x)$?

e Where is $A(x)$ positive, and where is it negative?

f Sketch $y = A(x)$.

10 This 'super challenge' question may illuminate the definition of a continuous function:

A continuous function is a function that is continuous at every number in its domain.

Classify these functions as continuous or not continuous according to the definition above.

a $y = x - 2$

b $y = \dfrac{1}{x-2}$

c $y = \begin{cases} \dfrac{1}{x-2}, & \text{for } x \neq 2, \\ 0, & \text{for } x = 2. \end{cases}$

d $y = \sqrt{x}$

e $y = \dfrac{1}{\sqrt{x}}$

f $y = \begin{cases} \dfrac{1}{\sqrt{x}}, & \text{for } x > 0, \\ 0, & \text{for } x = 0. \end{cases}$

PAST-PAPER HOMEWORK · timed practice

None of the integration questions in the school trial papers used for this pack (2020–2024) first needs this lesson: each one can be done with the methods of an earlier lesson, or needs work from a later lesson. For timed practice, do Questions 2(d), 5(a), 5(c), 6 and 9 above in 30 minutes.

LESSON 5E

The indefinite integral

Now that primitives have been established as the key to calculating definite integrals, this lesson turns again to the task of finding primitives. A new and convenient notation, the *indefinite integral* $\int f(x)\,dx$, is introduced for the general primitive, and the constant of integration must always be included. The two standard forms for integration then deal with powers of x and powers of $ax+b$, with positive, negative and fractional indices. Brackets are expanded first where needed, using the special expansions.

CONCEPT 1 · the indefinite integral and the constant of integration

BACK TO PRIMITIVES

Now that primitives have been established as the key to calculating definite integrals, this section turns again to the task of finding primitives. First, a new and convenient notation for the primitive is introduced.

THE INDEFINITE INTEGRAL

Because of the close connection established by the fundamental theorem between primitives and definite integrals, the term *indefinite integral* is often used for the general primitive. The usual notation for the indefinite integral of a function $f(x)$ is an integral sign without any upper or lower limits. For example, *the primitive* or *the indefinite integral* of x^2+1 is

$$\int (x^2+1)\,dx = \frac{x^3}{3} + x + C, \qquad \text{for some constant } C.$$

The word 'indefinite' suggests that the integral cannot be evaluated further because no limits for the integral have yet been specified.

CHECK BY DIFFERENTIATING

An indefinite integral can always be checked by differentiating the answer. The derivative of each constant is zero, so

$$\frac{d}{dx}\left(\frac{x^3}{3} + x + C\right) = \frac{3x^2}{3} + 1 + 0 = x^2 + 1,$$

which is the function that was integrated.

THE CONSTANT OF INTEGRATION

A definite integral ends up as a pure number. An indefinite integral, on the other hand, is a function of x. The pronumeral x is carried across to the answer.

It also contains an unknown constant C (or c, as it is often written) and the indefinite integral can also be regarded as a function of C (or of c). The constant is called a 'constant of integration' and is an important part of the answer. It must always be included.

The only exception to including the constant of integration is when calculating definite integrals, because in that situation any primitive can be used.

Note: Strictly speaking, the words 'for some constant C' or 'where C is a constant' should follow the first mention of the new pronumeral C, because no pronumeral should be used without having been formally introduced. There is a limit to one's patience, however (in this book there is often no room), and usually it is quite clear that C is the constant of integration. If another pronumeral such as D is used, it would be wise to introduce it formally.

A NUMBER OR A FUNCTION

Compare the definite integral and the indefinite integral of the same function $x^2 + 1$.

$$\begin{aligned}\int_0^3 (x^2+1)\,dx &= \left[\frac{x^3}{3} + x\right]_0^3 \\ &= (9+3) - (0+0) \\ &= 12\end{aligned}$$

This is a number. The indefinite integral $\displaystyle\int (x^2+1)\,dx = \frac{x^3}{3} + x + C$ is a function of x and of C. If the constant were included in the definite integral, it would cancel: $(9+3+C) - (0+0+C) = 12$. This is why any primitive can be used for a definite integral (Lesson 5B).

STANDARD FORMS FOR INTEGRATION

The two rules for finding primitives given in the last section of Chapter 4 can now be restated in this new notation.

14 STANDARD FORMS FOR INTEGRATION

Suppose that $n \neq -1$. Then

- $\displaystyle\int x^n\,dx = \frac{x^{n+1}}{n+1} + C,$ for some constant C.
- $\displaystyle\int (ax+b)^n\,dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C,$ for some constant C.

INTEGRATION AND INTEGRAL

The word ‘integration’ is commonly used to refer both to the finding of a primitive, and to the evaluating of a definite integral. Similarly, the unqualified term ‘integral’ is used to refer both to the indefinite integral and to the definite integral.

SUMS, MULTIPLES AND CONSTANTS (FROM LESSON 4J)

The last section of Chapter 4 is Lesson 4J of this pack. Integrate a sum term by term, and keep each constant multiple in front. A constant k is kx^0, so its integral is $kx + C$. For example,

$$\begin{aligned}\int (3x^2 + 4x - 5)\,dx &= 3 \times \frac{x^3}{3} + 4 \times \frac{x^2}{2} - 5x + C \\ &= x^3 + 2x^2 - 5x + C.\end{aligned}$$

Only one constant C is needed for the whole answer.

TUTOR DEMO **The first standard form**

Use the standard form $\displaystyle\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$ to find:

a $\displaystyle\int 7\,dx$

b $\displaystyle\int 15x^4\,dx$

c $\displaystyle\int (6x^2 - 4x + 3)\,dx$

YOUR TURN **Same method**

Use the standard form $\int x^n\,dx = \dfrac{x^{n+1}}{n+1} + C$ to find:

a $\int 5\,dx$ b $\int 20x^3\,dx$ c $\int (9x^2 + 2x - 4)\,dx$

WORKED EXAMPLE 16 **From the text: keep for homework**

Use the standard form $\int x^n\,dx = \dfrac{x^{n+1}}{n+1} + C$ to find:

a $\int 9\,dx$ b $\int 12x^3\,dx$

SOLUTION

a $\int 9\,dx = 9x + C$, for some constant C.

Note: We know that $9x$ is the primitive of 9, because $\dfrac{d}{dx}(9x) = 9$. But the formula still gives the correct answer, because $9 = 9x^0$, so increasing the index to 1 and dividing by this new index 1,

$$\int 9x^0\,dx = \frac{9x^1}{1} + C, \qquad \text{for some constant } C$$
$$= 9x + C.$$

b

$$\int 12x^3\,dx = 12 \times \frac{x^4}{4} + C, \qquad \text{for some constant } C$$
$$= 3x^4 + C$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 16

b Increase the index from 3 to 4, then divide by 4. Then $12 \div 4 = 3$, so $12 \times \dfrac{x^4}{4} + C = \dfrac{12}{4}x^4 + C = 3x^4 + C$.

CONCEPT 2 · the second standard form, for powers of ax + b

READING THE SECOND STANDARD FORM

To use $\int (ax+b)^n \, dx = \dfrac{(ax+b)^{n+1}}{a(n+1)} + C$:

1. Write down n, a and b. The number a is the coefficient of x inside the bracket, with its sign.
2. Keep the bracket $ax+b$ as it is, and increase the index to $n+1$.
3. Divide by the new index $n+1$ *and* by a, then simplify the fraction in front.

CHECK BY DIFFERENTIATING

Differentiating $(ax+b)^{n+1}$ gives $(n+1)(ax+b)^n \times a$, because the derivative of $ax+b$ is a. That is why the answer divides by $a(n+1)$. For example,

$$\frac{d}{dx}\left(\frac{(3x+1)^6}{18}\right) = \frac{6(3x+1)^5 \times 3}{18} = (3x+1)^5.$$

A CONSTANT IN FRONT, AND A FRACTION FOR A

A constant multiple stays in front: $\int 3(2x-1)^{10} \, dx = 3 \times \dfrac{(2x-1)^{11}}{2 \times 11} + C$. When a is a fraction, dividing by a fraction is the same as multiplying by its reciprocal. For example, with $a = \frac{1}{3}$ and $n+1 = 5$, dividing by $\frac{1}{3} \times 5 = \frac{5}{3}$ is the same as multiplying by $\frac{3}{5}$.

TUTOR DEMO **Powers of a linear function**

Use the standard form $\int (ax+b)^n \, dx = \dfrac{(ax+b)^{n+1}}{a(n+1)} + C$ to find:

a $\int (2x+1)^4 \, dx$ b $\int (7-3x)^2 \, dx$ c $\int 4(3x-2)^5 \, dx$

YOUR TURN **Same method**

Use the standard form $\int (ax+b)^n\,dx = \dfrac{(ax+b)^{n+1}}{a(n+1)} + C$ to find:

a $\int (5x+2)^3\,dx$ b $\int (4-3x)^5\,dx$ c $\int 2(4x-1)^7\,dx$

WORKED EXAMPLE 17 **From the text: keep for homework**

Use the standard form $\int (ax+b)^n\,dx = \dfrac{(ax+b)^{n+1}}{a(n+1)} + C$ to find:

a $\int (3x+1)^5\,dx$ b $\int (5-2x)^2\,dx$

SOLUTION

a

$$\int (3x+1)^5\,dx = \frac{(3x+1)^6}{3\times 6} + C \qquad (\text{here } n=5 \text{ and } a=3 \text{ and } b=1)$$
$$= \tfrac{1}{18}(3x+1)^6 + C, \qquad \text{for some constant } C$$

b

$$\int (5-2x)^2\,dx = \frac{(5-2x)^3}{(-2)\times 3} + C \qquad (\text{here } n=2 \text{ and } a=-2 \text{ and } b=5)$$
$$= -\tfrac{1}{6}(5-2x)^3 + C, \qquad \text{for some constant } C$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 17

a The denominator is $a(n+1)$ with $n=5$, $a=3$ and $b=1$: $3\times 6 = 18$, so $\dfrac{(3x+1)^6}{3\times 6} + C = \dfrac{(3x+1)^6}{18} + C = \tfrac{1}{18}(3x+1)^6 + C$.

b Here $n=2$, $a=-2$ and $b=5$: $(-2)\times 3 = -6$, so $\dfrac{(5-2x)^3}{(-2)\times 3} + C = \dfrac{(5-2x)^3}{-6} + C = -\tfrac{1}{6}(5-2x)^3 + C$.

CONCEPT 3 · negative indices

NEGATIVE INDICES

Both standard forms apply with negative indices as well as positive indices, as in the next worked example. The exception is the index -1, where the rule is nonsense because it results in division by zero. We shall deal with the integration of x^{-1} in Chapter 6.

NEGATIVE INDICES, STEP BY STEP

1. Rewrite each fraction with a negative index. The whole bracket moves up: $\dfrac{1}{(3x+4)^2} = (3x+4)^{-2}$.
2. Increase the index by 1. For example, -3 becomes -2, not -4.
3. Divide by the new index, which is negative, so a minus sign appears.
4. Write the answer as a fraction again, with a positive index: $\dfrac{x^{-2}}{-2} = -\dfrac{1}{2x^2}$.

Chapter 6 of the textbook is Lesson 6I in Part 2 of this course.

TUTOR DEMO **Negative indices**

Use negative indices to find these indefinite integrals.

a $\displaystyle\int \frac{8}{x^5}\,dx$

b $\displaystyle\int \frac{dx}{(2x+5)^3}$

YOUR TURN **Same method**

Use negative indices to find these indefinite integrals.

a $\int \frac{6}{x^4}\,dx$

b $\int \frac{dx}{(4x-1)^2}$

WORKED EXAMPLE 18 **From the text: keep for homework**

Use negative indices to find these indefinite integrals.

a $\int \frac{12}{x^3}\,dx$

b $\int \frac{dx}{(3x+4)^2}$

SOLUTION

a

$$\int \frac{12}{x^3}\,dx = \int 12x^{-3}\,dx$$
$$= 12 \times \frac{x^{-2}}{-2} + C$$
$$= -\frac{6}{x^2} + C$$

b

$$\int \frac{dx}{(3x+4)^2} = \int (3x+4)^{-2}\,dx$$
$$= \frac{(3x+4)^{-1}}{3 \times (-1)} + C$$
$$= -\frac{1}{3(3x+4)} + C$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 18

a Increase the index from -3 to -2, and divide by -2. Then $12 \div (-2) = -6$ and $x^{-2} = \frac{1}{x^2}$, so $12 \times \frac{x^{-2}}{-2} + C = -6x^{-2} + C = -\frac{6}{x^2} + C$.

b Here $n = -2$, $a = 3$ and $b = 4$, so the new index is -1 and $3 \times (-1) = -3$. Then $\frac{(3x+4)^{-1}}{3 \times (-1)} + C = -\frac{1}{3}(3x+4)^{-1} + C = -\frac{1}{3(3x+4)} + C$.

CONCEPT 4 · special expansions

SPECIAL EXPANSIONS

In many integrals, brackets must be expanded before the indefinite integral can be found. The next worked example uses the special expansions. Part **b** also requires negative indices.

THE SPECIAL EXPANSIONS

$$(A+B)^2 = A^2 + 2AB + B^2$$
$$(A-B)^2 = A^2 - 2AB + B^2$$
$$(A-B)(A+B) = A^2 - B^2$$

A and B can be any terms, including powers of x, negative numbers and fractions such as $\frac{1}{x^2}$. Note that $\left(\frac{1}{x^2}\right)^2 = \frac{1}{x^4}$.

WHEN THE SECOND STANDARD FORM DOES NOT APPLY

The second standard form needs a linear function $ax + b$ inside the bracket. In $(x^3 - 1)^2$, the inside $x^3 - 1$ is not linear, so the brackets must be expanded first.

TUTOR DEMO **Expand, then integrate**

Find these indefinite integrals.

a $\int (x^2 + 2)^2 \, dx$

b $\int \left(2 - \frac{1}{x^2}\right)\left(2 + \frac{1}{x^2}\right) dx$

YOUR TURN **Same method**

Find these indefinite integrals.

a $\int (x^2 - 3)^2 \, dx$

b $\int \left(\frac{1}{x} - 2\right)\left(\frac{1}{x} + 2\right) dx$

WORKED EXAMPLE 19 **From the text: keep for homework**

Find these indefinite integrals.

a $\int (x^3 - 1)^2 \, dx$

b $\int \left(3 - \frac{1}{x^2}\right)\left(3 + \frac{1}{x^2}\right) dx$

SOLUTION

a

$$\int (x^3 - 1)^2 \, dx = \int (x^6 - 2x^3 + 1) \, dx \qquad (\text{using } (A+B)^2 = A^2 + 2AB + B^2)$$

$$= \frac{x^7}{7} - \frac{x^4}{2} + x + C$$

b

$$\int \left(3 - \frac{1}{x^2}\right)\left(3 + \frac{1}{x^2}\right) dx$$

$$= \int \left(9 - \frac{1}{x^4}\right) dx \qquad (\text{using } (A-B)(A+B) = A^2 - B^2)$$

$$= \int (9 - x^{-4}) \, dx$$

$$= 9x - \frac{x^{-3}}{-3} + C$$

$$= 9x + \frac{1}{3x^3} + C$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 19

a Use $(A+B)^2 = A^2 + 2AB + B^2$ with $A = x^3$ and $B = -1$:

$$(x^3 - 1)^2 = (x^3)^2 + 2 \times x^3 \times (-1) + (-1)^2 = x^6 - 2x^3 + 1.$$

Then the primitive of $2x^3$ is $2 \times \dfrac{x^4}{4} = \dfrac{x^4}{2}$, so

$$\frac{x^7}{7} - 2 \times \frac{x^4}{4} + x + C = \frac{x^7}{7} - \frac{x^4}{2} + x + C.$$

b Use $(A-B)(A+B) = A^2 - B^2$ with $A = 3$ and $B = \dfrac{1}{x^2}$:

$$\left(3 - \frac{1}{x^2}\right)\left(3 + \frac{1}{x^2}\right) = 3^2 - \left(\frac{1}{x^2}\right)^2 = 9 - \frac{1}{x^4}.$$

Dividing by -3 changes the sign, and $x^{-3} = \dfrac{1}{x^3}$, so

$$9x - \frac{x^{-3}}{-3} + C = 9x + \tfrac{1}{3}x^{-3} + C = 9x + \frac{1}{3x^3} + C.$$

CONCEPT 5 · fractional indices

FRACTIONAL INDICES

The standard forms for finding primitives of powers also apply to fractional indices. These calculations require quick conversions between fractional indices and surds. The next worked example finds each indefinite integral and then uses it to evaluate a definite integral.

SURDS AND FRACTIONAL INDICES

$$\sqrt{x} = x^{\frac{1}{2}}, \qquad \sqrt[3]{x} = x^{\frac{1}{3}}, \qquad \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}, \qquad \sqrt[3]{x^2} = x^{\frac{2}{3}}, \qquad x\sqrt{x} = x^{\frac{3}{2}}$$

To evaluate a fractional power, take the root first and then the power:

$$4^{\frac{3}{2}} = \left(\sqrt{4}\right)^3 = 2^3 = 8, \qquad 9^{\frac{3}{2}} = \left(\sqrt{9}\right)^3 = 3^3 = 27, \qquad 8^{\frac{2}{3}} = \left(\sqrt[3]{8}\right)^2 = 2^2 = 4.$$

DIVIDING BY A FRACTIONAL INDEX

Dividing by $\frac{3}{2}$ is the same as multiplying by $\frac{2}{3}$, and dividing by $\frac{1}{2}$ is the same as multiplying by 2. So $\dfrac{x^{\frac{3}{2}}}{\frac{3}{2}} = \frac{2}{3}x^{\frac{3}{2}}$ and $\dfrac{x^{\frac{1}{2}}}{\frac{1}{2}} = 2x^{\frac{1}{2}}$.

TUTOR DEMO **Fractional and negative indices**

Use fractional and negative indices to evaluate:

a $\int_1^9 \sqrt{x}\,dx$

b $\int_1^9 \frac{1}{\sqrt{x}}\,dx$

YOUR TURN **Same method**

Use fractional and negative indices to evaluate:

a $\int_0^4 \sqrt{x}\,dx$

b $\int_1^{16} \frac{1}{\sqrt{x}}\,dx$

WORKED EXAMPLE 20 **From the text: keep for homework**

Use fractional and negative indices to evaluate:

a $\int_1^4 \sqrt{x}\,dx$ b $\int_1^4 \frac{1}{\sqrt{x}}\,dx$

SOLUTION

a Increase the index to $\frac{3}{2}$, and divide by $\frac{3}{2}$.

$$\begin{aligned}\int_1^4 \sqrt{x}\,dx &= \int_1^4 x^{\frac{1}{2}}\,dx \\ &= \tfrac{2}{3}\left[x^{\frac{3}{2}}\right]_1^4 \\ &= \tfrac{2}{3} \times (8-1) \\ &= 4\tfrac{2}{3}\end{aligned}$$

b Increase the index to $\frac{1}{2}$, and divide by $\frac{1}{2}$.

$$\begin{aligned}\int_1^4 \frac{1}{\sqrt{x}}\,dx &= \int_1^4 x^{-\frac{1}{2}}\,dx \\ &= 2\left[x^{\frac{1}{2}}\right]_1^4 \\ &= 2 \times (2-1) \\ &= 2\end{aligned}$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 20

a Dividing by $\frac{3}{2}$ is multiplying by $\frac{2}{3}$. Then substitute 4 and 1, where $4^{\frac{3}{2}} = \left(\sqrt{4}\right)^3 = 8$ and every power of 1 is 1:

$$\left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_1^4 = \tfrac{2}{3}\left[x^{\frac{3}{2}}\right]_1^4 = \tfrac{2}{3} \times \left(4^{\frac{3}{2}} - 1^{\frac{3}{2}}\right) = \tfrac{2}{3} \times (8-1) = \tfrac{14}{3} = 4\tfrac{2}{3}.$$

b Dividing by $\frac{1}{2}$ is multiplying by 2, and $4^{\frac{1}{2}} = \sqrt{4} = 2$:

$$\left[\frac{x^{\frac{1}{2}}}{\frac{1}{2}}\right]_1^4 = 2\left[x^{\frac{1}{2}}\right]_1^4 = 2 \times \left(4^{\frac{1}{2}} - 1^{\frac{1}{2}}\right) = 2 \times (2-1) = 2.$$

CONCEPT 6 · fractional powers of ax + b

A SURD OF A LINEAR FUNCTION

The second standard form also applies with fractional indices. Write the surd as a power of the linear function, then use the standard form with n a fraction. For example, $\frac{1}{\sqrt{9-2x}} = (9-2x)^{-\frac{1}{2}}$, so $n = -\frac{1}{2}$, $a = -2$ and $b = 9$, and the new index is $n+1 = \frac{1}{2}$.

TUTOR DEMO **A surd of a linear function**

a Use index notation to express $\dfrac{1}{\sqrt{5-4x}}$ as a power of $5-4x$.

b Hence find the indefinite integral $\displaystyle\int \frac{dx}{\sqrt{5-4x}}$.

YOUR TURN **Same method**

a Use index notation to express $\dfrac{1}{\sqrt{2x+7}}$ as a power of $2x+7$.

b Hence find the indefinite integral $\displaystyle\int \frac{dx}{\sqrt{2x+7}}$.

WORKED EXAMPLE 21 **From the text: keep for homework**

a Use index notation to express $\dfrac{1}{\sqrt{9-2x}}$ as a power of $9-2x$.

b Hence find the indefinite integral $\displaystyle\int \frac{dx}{\sqrt{9-2x}}$.

SOLUTION

a $\dfrac{1}{\sqrt{9-2x}} = (9-2x)^{-\frac{1}{2}}$.

b Hence

$$\int \frac{1}{\sqrt{9-2x}}\,dx = \int (9-2x)^{-\frac{1}{2}}\,dx$$

$$= \frac{(9-2x)^{\frac{1}{2}}}{-2\times\frac{1}{2}} + C, \qquad \text{using } \int (ax+b)^n\,dx = \frac{(ax+b)^{n+1}}{a(n+1)},$$

$$= -\sqrt{9-2x} + C.$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 21

b Here $n=-\frac{1}{2}$, $a=-2$ and $b=9$, so the new index is $n+1=\frac{1}{2}$, and $-2\times\frac{1}{2}=-1$. Then $\dfrac{(9-2x)^{\frac{1}{2}}}{-2\times\frac{1}{2}} + C = \dfrac{(9-2x)^{\frac{1}{2}}}{-1} + C = -\sqrt{9-2x} + C$.

TEXTBOOK HOMEWORK · Exercise 5E

FOUNDATION

Technology: Many programs that can perform algebraic manipulation are also able to deal with indefinite integrals. They can be used to check the questions in this exercise and to investigate the patterns arising in such calculations.

1 Find these indefinite integrals.

a $\displaystyle\int 4\,dx$ b $\displaystyle\int 1\,dx$ c $\displaystyle\int 0\,dx$ d $\displaystyle\int (-2)\,dx$

e $\displaystyle\int x\,dx$ f $\displaystyle\int x^2\,dx$ g $\displaystyle\int x^3\,dx$ h $\displaystyle\int x^7\,dx$

2 Find the indefinite integral of each function. Use the notation of the previous question.

a $2x$ b $4x$ c $3x^2$ d $4x^3$

e $10x^9$ f $2x^3$ g $4x^5$ h $3x^8$

3 Find these indefinite integrals.

a $\int (x + x^2)\,dx$ b $\int (x^4 - x^3)\,dx$ c $\int (x^7 + x^{10})\,dx$

d $\int (2x + 5x^4)\,dx$ e $\int (9x^8 - 11)\,dx$ f $\int (7x^{13} + 3x^8)\,dx$

g $\int (4 - 3x)\,dx$ h $\int (1 - x^2 + x^4)\,dx$ i $\int (3x^2 - 8x^3 + 7x^4)\,dx$

4 Find the indefinite integral of each function. (Leave negative indices in your answers.)

a x^{-2} b x^{-3} c x^{-8}

d $3x^{-4}$ e $9x^{-10}$ f $10x^{-6}$

5 Find these indefinite integrals. (Leave fractional indices in your answers.)

a $\int x^{\frac{1}{2}}\,dx$ b $\int x^{\frac{1}{3}}\,dx$ c $\int x^{\frac{1}{4}}\,dx$

d $\int x^{\frac{2}{3}}\,dx$ e $\int x^{-\frac{1}{2}}\,dx$ f $\int 4x^{\frac{1}{2}}\,dx$

6 By first expanding the brackets, find these indefinite integrals.

a $\int x(x+2)\,dx$ b $\int x(4-x^2)\,dx$ c $\int x^2(5-3x)\,dx$

d $\int x^3(x-5)\,dx$ e $\int (x-3)^2\,dx$ f $\int (2x+1)^2\,dx$

g $\int (1-x^2)^2\,dx$ h $\int (2-3x)(2+3x)\,dx$ i $\int (x^2-3)(1-2x)\,dx$

7 Write each integrand as separate fractions, then perform the integration.

a $\int \dfrac{x^2+2x}{x}\,dx$ b $\int \dfrac{x^7+x^8}{x^6}\,dx$ c $\int \dfrac{2x^3-x^4}{4x}\,dx$

DEVELOPMENT

8 Write these functions with negative indices and hence find their indefinite integrals.

a $\dfrac{1}{x^2}$ b $\dfrac{1}{x^3}$ c $\dfrac{1}{x^5}$ d $\dfrac{1}{x^{10}}$

e $\dfrac{3}{x^4}$ f $\dfrac{5}{x^6}$ g $\dfrac{7}{x^8}$ h $\dfrac{1}{3x^2}$

i $\dfrac{1}{7x^5}$ j $-\dfrac{1}{5x^3}$ k $\dfrac{1}{x^2}-\dfrac{1}{x^5}$ l $\dfrac{1}{x^3}+\dfrac{1}{x^4}$

9 Write these functions with fractional indices and hence find their indefinite integrals.

a $\sqrt{x}$ b $\sqrt[3]{x}$ c $\dfrac{1}{\sqrt{x}}$ d $\sqrt[3]{x^2}$

10 Use the indefinite integrals of the previous question to evaluate:

a $\displaystyle\int_0^9 \sqrt{x}\,dx$ b $\displaystyle\int_0^8 \sqrt[3]{x}\,dx$ c $\displaystyle\int_{25}^{49} \frac{1}{\sqrt{x}}\,dx$ d $\displaystyle\int_0^1 \sqrt[3]{x^2}\,dx$

11 By using the rule $\int (ax+b)^n \, dx = \dfrac{(ax+b)^{n+1}}{a(n+1)} + C$, find:

a $\int (x+1)^5 \, dx$ b $\int (x+2)^3 \, dx$ c $\int (4-x)^4 \, dx$

d $\int (3-x)^2 \, dx$ e $\int (3x+1)^4 \, dx$ f $\int (4x-3)^7 \, dx$

g $\int (5-2x)^6 \, dx$ h $\int (1-5x)^7 \, dx$ i $\int (2x+9)^{11} \, dx$

j $\int 3(2x-1)^{10} \, dx$ k $\int 4(5x-4)^6 \, dx$ l $\int 7(3-2x)^3 \, dx$

12 Find:

a $\int \left(\frac{1}{3}x-7\right)^4 \, dx$ b $\int \left(\frac{1}{4}x-7\right)^6 \, dx$ c $\int \left(1-\frac{1}{5}x\right)^3 \, dx$

13 Find:

a $\int \frac{1}{(x+1)^3}\,dx$ b $\int \frac{1}{(x-5)^4}\,dx$ c $\int \frac{1}{(3x-4)^2}\,dx$

d $\int \frac{1}{(2-x)^5}\,dx$ e $\int \frac{3}{(x-7)^6}\,dx$ f $\int \frac{8}{(4x+1)^5}\,dx$

g $\int \frac{2}{(3-5x)^4}\,dx$ h $\int \frac{4}{5(1-4x)^2}\,dx$ i $\int \frac{7}{8(3x+2)^5}\,dx$

14 By expanding the brackets, find:

a $\int \sqrt{x}\left(3\sqrt{x}-x\right)dx$ b $\int \left(\sqrt{x}-2\right)\left(\sqrt{x}+2\right)dx$ c $\int \left(2\sqrt{x}-1\right)^2 dx$

15 a Evaluate these definite integrals.

i $\int_0^1 x^{\frac{1}{2}}\,dx$ ii $\int_1^4 x^{-\frac{1}{2}}\,dx$ iii $\int_0^8 x^{\frac{1}{3}}\,dx$

b By writing them with fractional indices, evaluate these definite integrals.

i $\int_0^4 \sqrt{x}\,dx$ ii $\int_1^9 x\sqrt{x}\,dx$ iii $\int_1^9 \dfrac{dx}{\sqrt{x}}$

16 Expand the brackets and hence find:

a $\int_2^4 \left(2-\sqrt{x}\right)\left(2+\sqrt{x}\right)\,dx$ b $\int_0^1 \sqrt{x}\left(\sqrt{x}-4\right)\,dx$ c $\int_4^9 \left(\sqrt{x}-1\right)^2\,dx$

17 Explain why the indefinite integral $\int \dfrac{1}{x}\,dx$ cannot be found in the usual way using the standard form $\int x^n\,dx = \dfrac{x^{n+1}}{n+1} + C$.

18 Find each indefinite integral.

a $\int \sqrt{2x-1}\,dx$

b $\int \sqrt{7-4x}\,dx$

c $\int \sqrt[3]{4x-1}\,dx$

d $\int \dfrac{1}{\sqrt{3x+5}}\,dx$

19 Evaluate these definite integrals.

a $\int_0^2 (x+1)^4\,dx$

b $\int_2^3 (2x-5)^3\,dx$

c $\int_{-2}^2 (1-x)^5\,dx$

d $\int_0^5 \left(1-\dfrac{x}{5}\right)^4\,dx$

e $\int_0^1 \sqrt{9-8x}\,dx$

f $\int_2^7 \dfrac{1}{\sqrt{x+2}}\,dx$

g $\int_{-2}^0 \sqrt[3]{x+1}\,dx$

h $\int_1^5 \sqrt{3x+1}\,dx$

i $\int_{-3}^0 \sqrt{1-5x}\,dx$

ENRICHMENT

20 a If u and v are differentiable functions of x, prove that $\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx$.

b Hence find:

i $\int x(x-1)^4\,dx$

ii $\int x\sqrt{1+x}\,dx$

PAST-PAPER HOMEWORK · timed practice

The school trial papers used for this pack (2020–2024) have no question that first needs this lesson's methods: each integration question in the catalogue for these papers can be done with the methods of an earlier lesson, or needs later work. For timed practice, do Questions 8(l), 11(l), 12(c), 13(i), 14(c), 16(c), 18(d) and 19(i) above in 30 minutes.

LESSON 5F

Finding areas by integration

An area is always positive, but a definite integral is negative when the curve is below the x-axis. This lesson turns definite integrals into areas: sketch the region first, integrate separately over the parts above and below the x-axis, use symmetry when the function is odd or even, and write the conclusion in square units. It ends by integrating with respect to y, which finds the area between a curve and the y-axis.

CONCEPT 1 · areas and definite integrals, and regions above the x-axis

FINDING AREAS BY INTEGRATION

The aim of this lesson and the next is to use definite integrals to find the areas of regions bounded by curves, lines and the coordinate axes.

Lessons 5F–5G ignore integrals that run backwards. Running an integral backwards reverses its sign, which would confuse the discussion of areas in these lessons. When finding areas, we decide what integrals to create, and we naturally avoid integrals that run backwards.

AREAS AND DEFINITE INTEGRALS

Areas and definite integrals are closely related, but they are not the same thing.

- An area is always positive, whereas a definite integral may be positive or negative, depending on whether the curve is above or below the x-axis.

Problems on areas require care when finding the required integral or combination of integrals. Some particular techniques are listed below, but the general rule is to draw a diagram first to see which pieces need to be added or subtracted.

15 FINDING AN AREA

When using integrals to find the area of a region:

1. Draw a sketch of the curves, showing relevant intercepts and intersections.
2. Create and evaluate the necessary definite integral or integrals.
3. Write a conclusion, giving the required area in square units.

REGIONS ABOVE THE x-AXIS

When a curve lies entirely above the x-axis, the relevant integral will be positive, and the area of the region will be equal to the integral, apart from needing units.

WHERE THE LIMITS COME FROM

'The region bounded by the curve and the x-axis' gives no limits. Find them from the sketch: they are the x-intercepts, found by solving $y = 0$. For $y = 4 - x^2$, solve $4 - x^2 = 0$, so $x^2 = 4$ and $x = -2$ or $x = 2$. Areas are written in square units, shortened to u^2.

TUTOR DEMO **A region above the x-axis**

Find the area of the region bounded by the curve $y = 9 - x^2$ and the x-axis.

YOUR TURN **Same method**

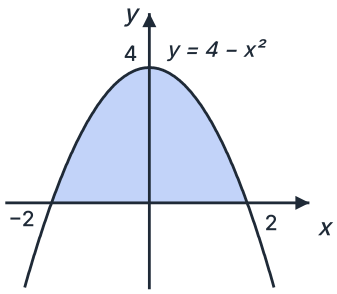
Find the area of the region bounded by the curve $y = 1 - x^2$ and the x-axis.

WORKED EXAMPLE 1 **From the text: keep for homework**

Find the area of the region bounded by the curve $y = 4 - x^2$ and the x-axis. (This was the example sketched in the introduction to this chapter.)

SOLUTION

The curve meets the x-axis at $(2, 0)$ and $(-2, 0)$.
The region lies entirely above the x-axis and the relevant integral is



$$\begin{aligned}\int_{-2}^{2}(4-x^2)\,dx &= \left[4x - \frac{x^3}{3}\right]_{-2}^{2} \\ &= \left(8 - \tfrac{8}{3}\right) - \left(-8 + \tfrac{8}{3}\right) \\ &= 5\tfrac{1}{3} - \left(-5\tfrac{1}{3}\right) \\ &= 10\tfrac{2}{3},\end{aligned}$$

which is positive because the curve lies entirely above the x-axis.
Hence the required area is $10\frac{2}{3}$ square units.

WORKED EXAMPLE 1, LINE BY LINE

Upper limit 2: $4 \times 2 - \frac{2^3}{3} = 8 - \frac{8}{3}$. Lower limit -2: $4 \times (-2) - \frac{(-2)^3}{3} = -8 - \frac{-8}{3} = -8 + \frac{8}{3}$. Then $8 - \frac{8}{3} = \frac{24}{3} - \frac{8}{3} = \frac{16}{3} = 5\frac{1}{3}$ and $-8 + \frac{8}{3} = -\frac{16}{3} = -5\frac{1}{3}$, so the integral is $5\frac{1}{3} + 5\frac{1}{3} = 10\frac{2}{3}$.

CONCEPT 2 · regions below the x-axis

REGIONS BELOW THE x-AXIS

When a curve lies entirely below the x-axis, the relevant integral will be negative, and the area will be the opposite of this.

WHY THE INTEGRAL IS NEGATIVE

Below the x-axis, every height $f(x)$ is negative, so every strip $f(x)\,\delta x$ has a negative value. This is the signed area of Lesson 5C (Box 5). The area is the integral with its sign changed: write the integral first, then give the area as a positive number.

TUTOR DEMO **A region below the x-axis**

Find the area of the region bounded by the curve $y = x^2 - 4$ and the x-axis.

YOUR TURN **Same method**

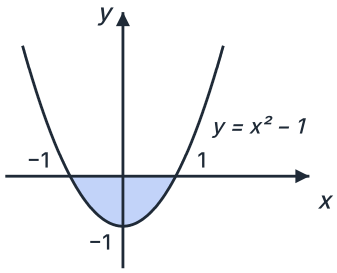
Find the area of the region bounded by the curve $y = x^2 - 9$ and the x-axis.

WORKED EXAMPLE 2 **From the text: keep for homework**

Find the area of the region bounded by the curve $y = x^2 - 1$ and the x-axis.

SOLUTION

The curve meets the x-axis at $(1, 0)$ and $(-1, 0)$.
The region lies entirely below the x-axis, and the relevant integral is



$$\begin{aligned}\int_{-1}^{1} (x^2 - 1)\,dx &= \left[\frac{x^3}{3} - x\right]_{-1}^{1} \\ &= \left(\tfrac{1}{3} - 1\right) - \left(-\tfrac{1}{3} + 1\right) \\ &= -\tfrac{2}{3} - \tfrac{2}{3} \\ &= -1\tfrac{1}{3},\end{aligned}$$

which is negative, because the curve lies entirely below the x-axis.
Hence the required area is $1\frac{1}{3}$ square units.

WORKED EXAMPLE 2, LINE BY LINE

At $x = -1$: $\frac{(-1)^3}{3} - (-1) = -\frac{1}{3} + 1$. Then $\frac{1}{3} - 1 = -\frac{2}{3}$ and $-\frac{1}{3} + 1 = \frac{2}{3}$, so the integral is $-\frac{2}{3} - \frac{2}{3} = -\frac{4}{3} = -1\frac{1}{3}$. The area is the opposite, $1\frac{1}{3}$.

CONCEPT 3 · curves that cross the x-axis

CURVES THAT CROSS THE x-AXIS

When a curve crosses the x-axis, the area of the region between the curve and the x-axis cannot usually be found by means of a single integral. This is because integrals representing regions below the x-axis have negative values.

SPLIT AT EVERY x-INTERCEPT

1. Find the x-intercepts and sketch the curve.
2. Write one integral for each piece, from one intercept to the next.
3. Evaluate each integral. A negative value means the piece is below the axis: its area is the opposite.
4. Add the areas (all positive).

A single integral over the whole interval adds the signed areas, so the parts below the axis cancel some of the parts above.

TUTOR DEMO **Above and below**

a Sketch the cubic curve $y = x(x+1)(x-3)$, showing the x-intercepts.
b Shade the region enclosed between the x-axis and the curve, and find its area.
c Find $\int_{-1}^{3} x(x+1)(x-3)\,dx$ and explain why this integral does not represent the area of the region described in part **b**.

YOUR TURN **Same method**

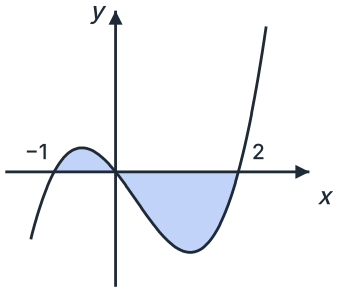
a Sketch the cubic curve $y = x(x+1)(x-4)$, showing the x-intercepts.
b Shade the region enclosed between the x-axis and the curve, and find its area.
c Find $\int_{-1}^{4} x(x+1)(x-4)\,dx$ and explain why this integral does not represent the area of the region described in part **b**.

WORKED EXAMPLE 3 **From the text: keep for homework**

a Sketch the cubic curve $y = x(x+1)(x-2)$, showing the x-intercepts.
b Shade the region enclosed between the x-axis and the curve, and find its area.
c Find $\int_{-1}^{2} x(x+1)(x-2)\,dx$ and explain why this integral does not represent the area of the region described in part **b**.

SOLUTION

a The curve has x-intercepts $x = -1$, $x = 0$ and $x = 2$, and is graphed on the right.



b Expanding the cubic,

$$\begin{aligned} y &= x(x+1)(x-2) \\ &= x(x^2 - x - 2) \\ &= x^3 - x^2 - 2x. \end{aligned}$$

For the region above the x-axis,

$$\begin{aligned}\int_{-1}^{0}(x^3-x^2-2x)\,dx &= \left[\frac{x^4}{4}-\frac{x^3}{3}-x^2\right]_{-1}^{0}\\ &= (0-0-0)-\left(\tfrac{1}{4}+\tfrac{1}{3}-1\right)\\ &= \tfrac{5}{12},\end{aligned}$$

so area above $= \frac{5}{12}$ square units.

For the region below the x-axis,

$$\begin{aligned}\int_{0}^{2}(x^3-x^2-2x)\,dx &= \left[\frac{x^4}{4}-\frac{x^3}{3}-x^2\right]_{0}^{2}\\ &= \left(4-2\tfrac{2}{3}-4\right)-(0-0-0)\\ &= -2\tfrac{2}{3},\end{aligned}$$

so area below $= 2\frac{2}{3}$ square units.

Adding these, $\textbf{total area} = \frac{5}{12}+2\frac{2}{3} = 3\frac{1}{12}$ square units.

c

$$\begin{aligned}\int_{-1}^{2}x(x+1)(x-2)\,dx &= \left[\frac{x^4}{4}-\frac{x^3}{3}-x^2\right]_{-1}^{2}\\ &= \left(4-2\tfrac{2}{3}-4\right)-\left(\tfrac{1}{4}+\tfrac{1}{3}-1\right)\\ &= -2\tfrac{2}{3}+\tfrac{5}{12}\\ &= -2\tfrac{1}{4}.\end{aligned}$$

This integral represents the difference $2\frac{2}{3}-\frac{5}{12} = 2\frac{1}{4}$ of the two areas, and is negative because the area below is larger than the area above.

WORKED EXAMPLE 3, LINE BY LINE

At $x=-1$: $\frac{(-1)^4}{4}=\frac{1}{4}$, $-\frac{(-1)^3}{3}=+\frac{1}{3}$ and $-(-1)^2=-1$. So the bracket is $\frac{3}{12}+\frac{4}{12}-\frac{12}{12}=-\frac{5}{12}$, and $0-\left(-\frac{5}{12}\right)=\frac{5}{12}$.

At $x=2$: $\frac{2^4}{4}=4$, $\frac{2^3}{3}=\frac{8}{3}=2\frac{2}{3}$ and $2^2=4$. So the bracket is $4-2\frac{2}{3}-4=-2\frac{2}{3}$.

Total: $\frac{5}{12}+\frac{32}{12}=\frac{37}{12}=3\frac{1}{12}$. Part **c**: $-\frac{32}{12}+\frac{5}{12}=-\frac{27}{12}=-\frac{9}{4}=-2\frac{1}{4}$.

CONCEPT 4 · areas associated with odd and even functions

AREAS ASSOCIATED WITH ODD AND EVEN FUNCTIONS

As always in mathematics, calculations are often much easier if symmetries can be recognised.

THE SYMMETRY, FROM LESSON 5C

An odd function, $f(-x) = -f(x)$, has point symmetry in the origin. The region on the left of the origin is the region on the right rotated by $180°$, so the two regions have equal areas, one above and one below the x-axis. An even function, $f(-x) = f(x)$, has line symmetry in the y-axis, so the regions on the two sides have equal areas on the same side of the x-axis. In both cases, find one area and double it.

TUTOR DEMO **Doubling the area**

a Show that $y = x^3 - 9x$ is an odd function.
b Using part **a**, find the area between the curve $y = x^3 - 9x$ and the x-axis.

YOUR TURN **Same method**

a Show that $y = x^3 - 4x$ is an odd function.
b Using part **a**, find the area between the curve $y = x^3 - 4x$ and the x-axis.

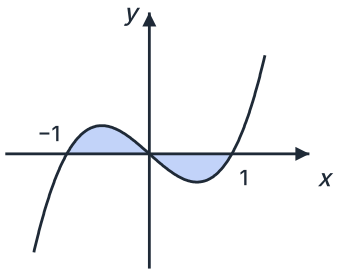
WORKED EXAMPLE 4 **From the text: keep for homework**

a Show that $y = x^3 - x$ is an odd function.
b Using part **a**, find the area between the curve $y = x^3 - x$ and the x-axis.

SOLUTION

a Let $f(x) = x^3 - x$.

$$\begin{aligned}\text{Then } f(-x) &= (-x)^3 - (-x) \\ &= -x^3 + x \\ &= -f(x), \text{ so } f(x) \text{ is odd.}\end{aligned}$$



b Factoring,

$$\begin{aligned}y &= x(x^2 - 1) \\ &= x(x-1)(x+1),\end{aligned}$$

so the x-intercepts are $x = -1$, $x = 0$ and $x = 1$.
The two shaded regions have equal areas because the function is odd.

$$\begin{aligned}\text{First, } \int_0^1 (x^3 - x)\,dx &= \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_0^1 \\ &= \left(\tfrac{1}{4} - \tfrac{1}{2}\right) - (0 - 0) \\ &= -\tfrac{1}{4},\end{aligned}$$

so area below the x-axis $= \frac{1}{4}$ square units.
Doubling, total area $= \frac{1}{2}$ square units.

WORKED EXAMPLE 4, LINE BY LINE

$\frac{1}{4} - \frac{1}{2} = \frac{1}{4} - \frac{2}{4} = -\frac{1}{4}$. The region from $x = 0$ to $x = 1$ is below the axis, so its area is $\frac{1}{4}$. The region from $x = -1$ to $x = 0$ is above the axis and has the same area, so the total is $2 \times \frac{1}{4} = \frac{1}{2}$. (The single integral $\int_{-1}^{1} (x^3 - x)\,dx$ is 0, by Lesson 5C, so it is not the area.)

CONCEPT 5 · area between a graph and the y-axis

AREA BETWEEN A GRAPH AND THE y-AXIS

Integration with respect to y rather than x can often give a result quickly, and avoid subtraction of areas.

Suppose then that we can find the inverse function, and write x is a function of y.

- A definite integral with respect to y represents the signed area of the region between the curve and the y-axis.
- This means that the definite integral is the sum of areas of regions to the right of the y-axis, minus the sum of areas of regions to the left of the y-axis.
- The limits of integration are values of y rather than values of x.

16 THE DEFINITE INTEGRAL AND INTEGRATION WITH RESPECT TO y

Let x be a continuous function of y in some closed interval $a \leq y \leq b$.

Then the definite integral $\int_a^b x\,dy$ is the sum of the areas of regions to the right of the y-axis, from $y = a$ to $y = b$, minus the sum of the areas of the regions to the left of the y-axis.

WORKING WITH THE INVERSE FUNCTION

When the function is given with y as a function of x, we first need to solve the equation for x. We are thus working with the inverse function.

THE SAME RULES WITH THE AXES SWAPPED

Everything in Concepts 1–3 still works, with y in place of x: the strips are horizontal, of length x and width δy; 'above the x-axis' becomes 'to the right of the y-axis'; and the limits are read off the y-axis. Integrate the new variable as usual:

$$\int (y - 1)\,dy = \frac{y^2}{2} - y + C.$$

TUTOR DEMO **Integrating with respect to y**

a Sketch the lines $y = 2x + 1$ and $y = 7$, and shade the region between these lines to the right of the y-axis.
b Write the equation of the line so that x is a function of y.
c Use integration with respect to y to find the area of this region.
d Confirm the result by area formulae.

YOUR TURN **Same method**

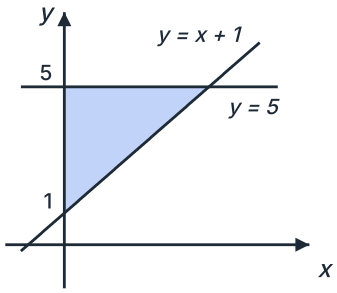
a Sketch the lines $y = x + 2$ and $y = 5$, and shade the region between these lines to the right of the y-axis.
b Write the equation of the line so that x is a function of y.
c Use integration with respect to y to find the area of this region.
d Confirm the result by area formulae.

WORKED EXAMPLE 5 **From the text: keep for homework**

a Sketch the lines $y = x + 1$ and $y = 5$, and shade the region between these lines to the right of the y-axis.
b Write the equation of the line so that x is a function of y.
c Use integration with respect to y to find the area of this region.
d Confirm the result by area formulae.

SOLUTION

a The lines are sketched on the right. They meet at $(4, 5)$.



b The given equation is $y = x + 1$.
Solving for x, $\quad x = y - 1$.

c The required integral is

$$\begin{aligned}\int_1^5 (y-1)\,dy &= \left[\frac{y^2}{2} - y\right]_1^5 \\ &= \left(\frac{25}{2} - 5\right) - \left(\frac{1}{2} - 1\right) \\ &= 7\tfrac{1}{2} - \left(-\tfrac{1}{2}\right) \\ &= 8,\end{aligned}$$

which is positive, because the region is to the right of the y-axis.
Hence the required area is $8\ \text{u}^2$.

d $\quad \text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$

$$\begin{aligned}&= \tfrac{1}{2} \times 4 \times 4 \\ &= 8\ \text{u}^2.\end{aligned}$$

WORKED EXAMPLE 5, LINE BY LINE

The lines meet where $x + 1 = 5$, so $x = 4$ and the point is $(4, 5)$. The region runs up the y-axis from $y = 1$ (where the line $y = x + 1$ crosses the y-axis) to $y = 5$, so these are the limits. In part **c**, $\frac{25}{2} - 5 = \frac{15}{2} = 7\frac{1}{2}$ and $\frac{1}{2} - 1 = -\frac{1}{2}$, so the integral is $7\frac{1}{2} + \frac{1}{2} = 8$. In part **d**, the triangle has vertices $(0, 1)$, $(0, 5)$ and $(4, 5)$: its base is $4 - 0 = 4$ along the line $y = 5$, and its height is $5 - 1 = 4$.

CONCEPT 6 · regions on both sides of the y-axis

REGIONS TO THE LEFT OF THE y-AXIS

By Box 16, an integral with respect to y over a region to the *left* of the y-axis is negative, because there x is negative. As in Concept 3, split the y-interval where the curve crosses the y-axis, evaluate each integral, and add the areas.

FRACTIONAL POWERS OF NEGATIVE NUMBERS

Take the cube root first, then the power: $(-8)^{\frac{4}{3}} = \left((-8)^{\frac{1}{3}}\right)^4 = (-2)^4 = 16$, and $(-1)^{\frac{4}{3}} = \left((-1)^{\frac{1}{3}}\right)^4 = (-1)^4 = 1$. A primitive of $y^{\frac{1}{3}}$ is $\dfrac{y^{\frac{4}{3}}}{\frac{4}{3}} = \frac{3}{4}y^{\frac{4}{3}}$ (increase the index by 1 and divide by the new index, Lesson 5E).

TUTOR DEMO **Right and left of the y-axis**

Consider the cubic $y = x^3$. One region lies between the curve and the y-axis from $y = 0$ to $y = 27$, to the right of the y-axis. A second region lies between the curve and the y-axis from $y = -8$ to $y = 0$, to the left of the y-axis.

a Write the equation of the cubic so that x is a function of y.

b Use integration with respect to y to find the areas of the regions to the right and left of the y-axis.

c Find the total area of the two regions.

YOUR TURN **Same method**

Consider the curve $y = x^5$. One region lies between the curve and the y-axis from $y = 0$ to $y = 1$, to the right of the y-axis. A second region lies between the curve and the y-axis from $y = -1$ to $y = 0$, to the left of the y-axis.

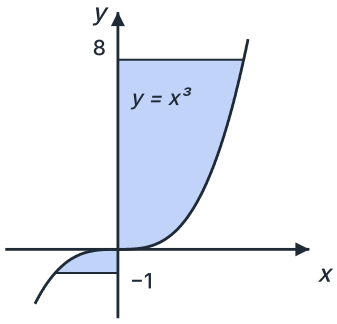
a Write the equation of the curve so that x is a function of y.

b Use integration with respect to y to find the areas of the regions to the right and left of the y-axis.

c Find the total area of the two regions.

WORKED EXAMPLE 6 **From the text: keep for homework**

The curve in the diagram is the cubic $y = x^3$.
a Write the equation of the cubic so that x is a function of y.
b Use integration with respect to y to find the areas of the shaded regions to the right and left of the y-axis.
c Find the total area of the two shaded regions.



SOLUTION

a The given equation is $y = x^3$.

$$\text{Solving for } x, \quad x^3 = y$$
$$x = y^{\frac{1}{3}}.$$

b For the region to the right of the y-axis,

$$\int_0^8 y^{\frac{1}{3}}\,dy = \tfrac{3}{4}\left[y^{\frac{4}{3}}\right]_0^8$$
$$= \tfrac{3}{4} \times (16 - 0) \qquad \left(\text{because } 8^{\frac{4}{3}} = 2^4 = 16\right)$$
$$= 12,$$

so area $= 12$ square units.

For the region to the left of the y-axis,

$$\int_{-1}^0 y^{\frac{1}{3}}\,dy = \tfrac{3}{4}\left[y^{\frac{4}{3}}\right]_{-1}^0$$
$$= \tfrac{3}{4} \times (0 - 1) \qquad \left(\text{because } (-1)^{\frac{4}{3}} = (-1)^4 = 1\right)$$
$$= -\tfrac{3}{4},$$

so area $= \frac{3}{4}$ square units.

c Adding these, total area $= 12\frac{3}{4}$ square units.

WORKED EXAMPLE 6, LINE BY LINE

The right-hand region runs from $y = 0$ to $y = 8$ (the point $(2, 8)$ on the curve), and the left-hand region from $y = -1$ (the point $(-1, -1)$) to $y = 0$. The $\frac{3}{4}$ is taken out of the bracket: $\left[\frac{3}{4}y^{\frac{4}{3}}\right]_0^8 = \frac{3}{4}\left[y^{\frac{4}{3}}\right]_0^8$. Then $\frac{3}{4} \times 16 = 12$, and $12 + \frac{3}{4} = 12\frac{3}{4}$.

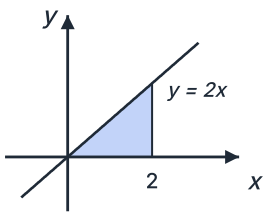
TEXTBOOK HOMEWORK · Exercise 5F

FOUNDATION

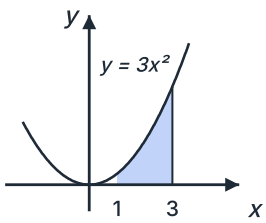
Technology: Graphing software would help in identifying the definite integrals that need to be evaluated to find the area of a given region.

1 Find the area of each shaded region below by evaluating the appropriate integral.

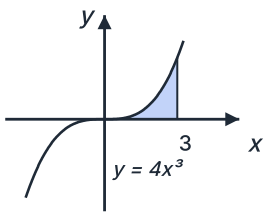
a



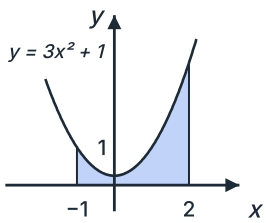
b



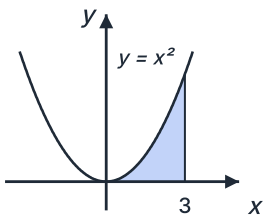
c



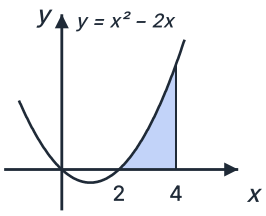
d



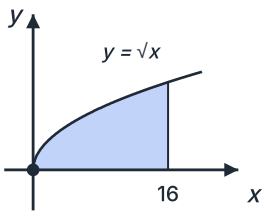
e



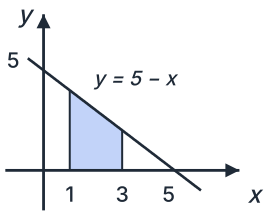
f



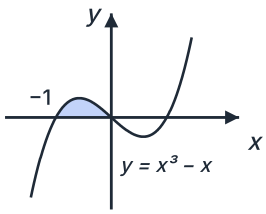
g



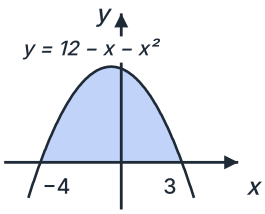
h



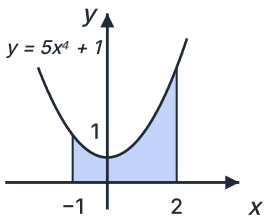
i



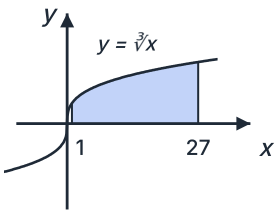
j



k

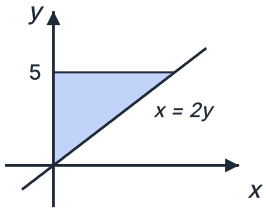


l

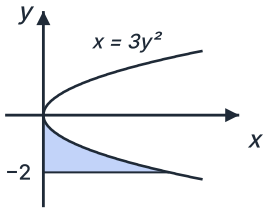


2 Find the area of each shaded region below by evaluating an integral of the form $\int_{y_1}^{y_2} g(y)\,dy$.

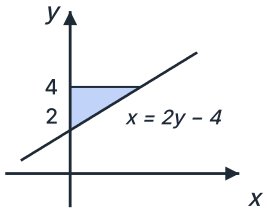
a



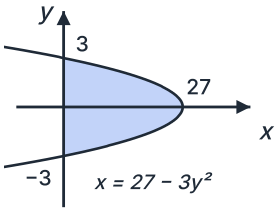
b



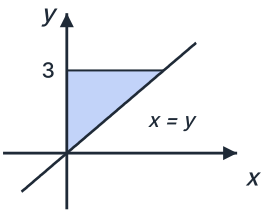
c



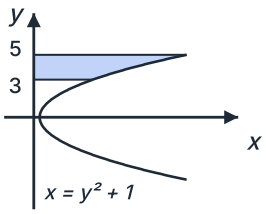
d



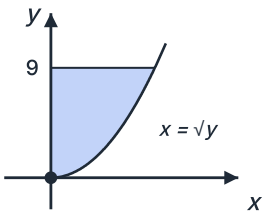
e



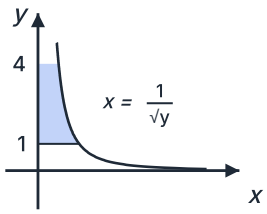
f



g

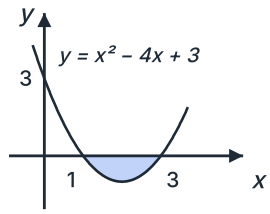


h

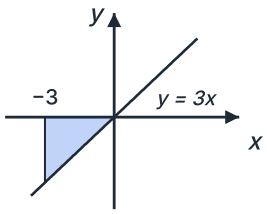


3 Find the area of each shaded region below by evaluating the appropriate integral.

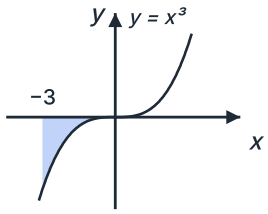
a



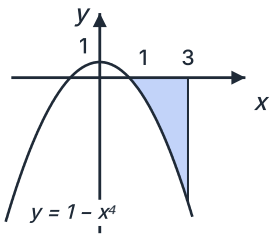
b



c

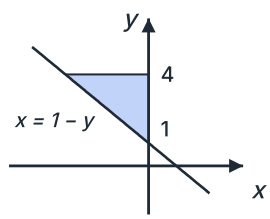


d

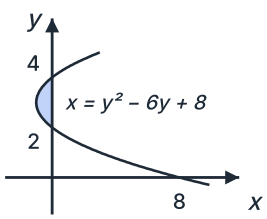


4 Find the area of each shaded region below by evaluating the appropriate integral.

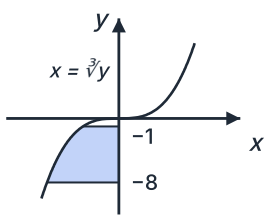
a



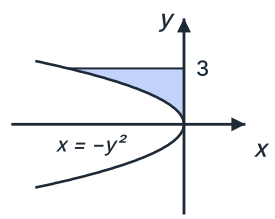
b



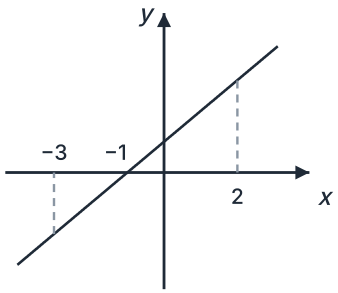
c



d



5 The line $y = x + 1$ is graphed on the right.



a Copy the diagram, and shade the region between the line $y = x + 1$ and the x-axis from $x = -3$ to $x = 2$.

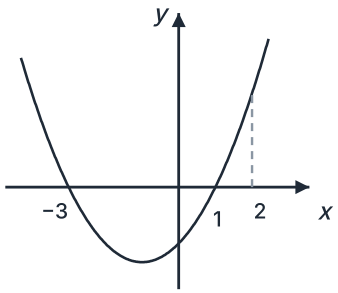
b By evaluating $\int_{-1}^{2} (x + 1)\, dx$, find the area of the shaded region above the x-axis.

c By evaluating $\int_{-3}^{-1} (x + 1)\, dx$, find the area of the shaded region below the x-axis.

d Hence find the area of the entire shaded region.

e Find $\int_{-3}^{2} (x + 1)\, dx$, and explain why this integral does not give the area of the shaded region.

6 The curve $y = (x-1)(x+3) = x^2 + 2x - 3$ is graphed.



a Copy the diagram, and shade the region between the curve $y = (x-1)(x+3)$ and the x-axis from $x = -3$ to $x = 2$.

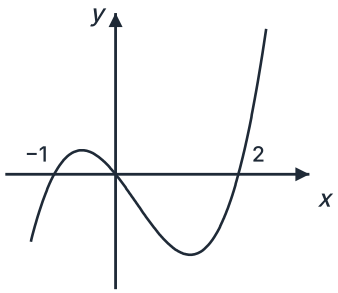
b By evaluating $\int_{-3}^{1} (x^2 + 2x - 3)\,dx$, find the area of the shaded region below the x-axis.

c By evaluating $\int_{1}^{2} (x^2 + 2x - 3)\,dx$, find the area of the shaded region above the x-axis.

d Hence find the area of the entire shaded region.

e Find $\int_{-3}^{2} (x^2 + 2x - 3)\,dx$, and explain why this integral does not give the area of the shaded region.

7 The curve $y = x(x+1)(x-2) = x^3 - x^2 - 2x$ is graphed.



a Copy the diagram, and shade the region bounded by the curve and the x-axis.

b By evaluating $\int_0^2 (x^3 - x^2 - 2x)\,dx$, find the area of the shaded region below the x-axis.

c By evaluating $\int_{-1}^0 (x^3 - x^2 - 2x)\,dx$, find the area of the shaded region above the x-axis.

d Hence find the area of the entire region you have shaded.

e Find $\int_{-1}^2 (x^3 - x^2 - 2x)\,dx$, and explain why this integral does not give the area of the shaded region.

DEVELOPMENT

8 In each part below, find the area of the region bounded by the graph of the given function and the x-axis between the specified values. Remember that areas above and below the x-axis must be calculated separately.

a $y = x^2$, between $x = -3$ and $x = 2$,

b $y = 2x^3$, between $x = -4$ and $x = 1$,

c $y = 3x(x - 2)$, between $x = 0$ and $x = 2$,

d $y = x - 3$, between $x = -1$ and $x = 4$,

e $y = (x - 1)(x + 3)(x - 2)$, between $x = -3$, and $x = 2$

f $y = -2x(x + 1)$, between $x = -2$ and $x = 2$.

9 In each part below, find the area of the region bounded by the graph of the given function and the y-axis between the specified values. Remember that areas to the right and to the left of the y-axis must be calculated separately.

a $x = y - 5$, between $y = 0$ and $y = 6$,

b $x = 3 - y$, between $y = 2$ and $y = 5$,

c $x = y^2$, between $y = -1$ and $y = 3$,

d $x = (y - 1)(y + 1)$, between $y = 3$ and $y = 0$.

10 In each part below you should sketch the curve and look carefully for any symmetries that will simplify the calculation.

a Find the area of the region bounded by the given curve and the x-axis.

i $y = x^7$, for $-2 \leq x \leq 2$,

ii $y = x^3 - 16x = x(x-4)(x+4)$, for $-4 \leq x \leq 4$,

iii $y = x^4 - 9x^2 = x^2(x-3)(x+3)$, for $-3 \leq x \leq 3$.

b Find the area of the region bounded by the given curve and the y-axis.

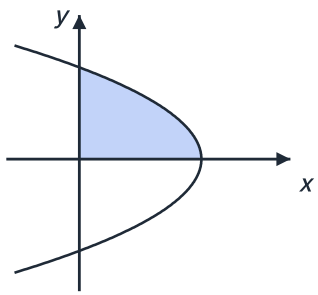
i $x = 2y$, for $-5 \leq y \leq 5$,

ii $x = y^2$, for $-3 \leq y \leq 3$,

iii $x = 4 - y^2 = (2-y)(2+y)$, for $-2 \leq y \leq 2$.

11 Find the area of the region bounded by $y = |x+2|$ and the x-axis, for $-2 \leq x \leq 2$.

12 The diagram shows the parabola $y^2 = 16(2 - x)$.



a Find the x-intercept and the y-intercepts.

b Find the exact area of the shaded region:

i by integrating $y = 4\sqrt{2 - x}$ with respect to x,

ii by integrating with respect to y. (You will need to make x the subject of the equation.)

13 The gradient of a curve is $y' = x^2 - 4x + 3$, and the curve passes through the origin.

a Find the equation of the curve.

b Show that the curve has turning points at $\left(1, 1\frac{1}{3}\right)$ and $(3, 0)$, and sketch its graph.

c Find the area of the region bounded by the curve and the x-axis between the two turning points.

14 Sketch $y = x^2$ and mark the points $A(a, a^2)$, $B(-a, a^2)$, $P(a, 0)$ and $Q(-a, 0)$.

a Show that $\int_0^a x^2\,dx = \frac{2}{3}\,(\text{area } \triangle OAP)$.

b Show that $\int_{-a}^a x^2\,dx = \frac{1}{3}\,(\text{area of rectangle } ABQP)$.

15 Given positive real numbers a and n, let A, P and Q be the points (a, a^n), $(a, 0)$ and $(0, a^n)$ respectively. Find the ratios:

a $\int_0^a x^n\,dx : (\text{area of } \triangle AOP)$

b $\int_0^a x^n\,dx : (\text{area of rectangle } OPAQ)$

16 a Show that $x^4 - 2x^3 + x = x(x-1)(x^2 - x - 1)$. Then sketch a graph of the function $y = x^4 - 2x^3 + x$ and shade the three regions bounded by the graph and the x-axis.

b If $a = \frac{1}{2}\left(1 + \sqrt{5}\right)$, evaluate a^2, a^4 and a^5.

c Show that the area of one shaded region equals the sum of the areas of the other two.

ENRICHMENT

17 Consider the function $G(x) = \int_0^x g(u)\,du$, where

$$g(u) = \begin{cases} 4 - \frac{4}{3}u, & \text{for } 0 \leq u < 6, \\ u - 10, & \text{for } 6 \leq u \leq 12. \end{cases}$$

a Sketch a graph of $g(u)$.

b Find the stationary points of the function $y = G(x)$ and determine their nature.

c Find the values of x for which $G(x) = 0$.

d Sketch the curve $y = G(x)$, indicating all important features.

e Find the area bounded by the curve $y = G(x)$ and the x-axis for $0 \leq x \leq 6$.

18 **a** Show that for $n < -1$, $\int_1^N x^n \, dx$ converges as $N \to \infty$, and find the limit.

b Show that for $n > -1$, $\int_\varepsilon^1 x^n \, dx$ converges as $\varepsilon \to 0^+$, and find the limit.

c Interpret these two results as areas.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 5 minutes **1 mark** Calculator allowed · show working · hints are below each question if you get stuck

This is the only school-paper item that fits this lesson. For more timed practice, do Questions 4(b), 6, 8(e), 9(d), 10(a)(iii) and 12 above in 30 minutes.

PP 1 **Normanhurst Boys High School, 2022, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q10** `1 mark`

What is the area enclosed between the curve $y = x^3 - 1$, the y-axis and the lines $y = 1$ and $y = 2$?

(A) $\int_1^2 (x^3 - 1)\,dy$

(B) $\int_1^2 \sqrt[3]{y+1}\,dy$

(C) $\int_1^2 \left(\sqrt{3}y + 1\right)dy$

(D) $\int_1^2 (y+1)\,dy$

Hint: what are the boundaries of the region?

LESSON 5G

Areas of compound regions

When a region is formed by two or more different curves, it usually has to be cut into pieces before integrals can find its area. This lesson finds where the curves meet, then either adds the areas of the pieces under different curves, or integrates the top curve minus the bottom curve. A careful sketch comes first every time.

CONCEPT 1 · regions under a combination of curves

COMPOUND REGIONS

When a region is formed by two or more different curves, some dissection process is usually needed before integrals can be used to calculate its area. Thus a preliminary sketch of the region becomes all the more important.

AREAS OF REGIONS UNDER A COMBINATION OF CURVES

Some regions are bounded by different curves in different parts of the x-axis.

CUTTING THE REGION INTO PIECES

1. Sketch the curves and find where they meet, by solving their equations simultaneously.
2. Cut the region with a vertical line through the meeting point. Each piece now lies under just one curve.
3. Integrate each curve over its own piece, then add the areas.

THE PRIMITIVE OF $(x-2)^2$

From Lesson 4J, a primitive of $(ax+b)^n$ is $\frac{(ax+b)^{n+1}}{a(n+1)}$. With $a=1$, $b=-2$ and $n=2$, a primitive of $(x-2)^2$ is $\frac{1}{3}(x-2)^3$. Check by differentiating:

$$\frac{d}{dx}\,\frac{(x-2)^3}{3} = \frac{3(x-2)^2}{3} = (x-2)^2.$$

TUTOR DEMO **Two parabolas and the x-axis**

a Sketch the curves $y = x^2$ and $y = (x-4)^2$ on one set of axes.
b Shade the region bounded by $y = x^2$, $y = (x-4)^2$ and the x-axis.
c Find the area of this shaded region.

YOUR TURN **Same method**

a Sketch the curves $y = x^2$ and $y = (x-6)^2$ on one set of axes.
b Shade the region bounded by $y = x^2$, $y = (x-6)^2$ and the x-axis.
c Find the area of this shaded region.

WORKED EXAMPLE 1 **From the text: keep for homework**

a Sketch the curves $y = x^2$ and $y = (x-2)^2$ on one set of axes.
b Shade the region bounded by $y = x^2$, $y = (x-2)^2$ and the x-axis.
c Find the area of this shaded region.

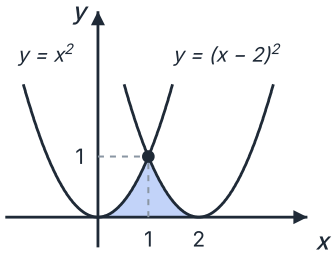
SOLUTION

a The two curves intersect at $(1, 1)$, because it is easily checked by substitution that this point lies on both curves.

b The whole region is above the x-axis, but it will be necessary to find separately the areas of the regions to the left and right of $x = 1$.

c First,

$$\int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}.$$



Secondly,

$$\int_1^2 (x-2)^2\,dx = \left[\frac{(x-2)^3}{3}\right]_1^2 = 0 - \left(-\frac{1}{3}\right) = \frac{1}{3}.$$

Combining these,

$$\text{area} = \frac{1}{3} + \frac{1}{3} = \frac{2}{3} \text{ square units.}$$

NOTE

Note: In this worked example, the second parabola is the first shifted right 2, and a parabola is symmetric about its axis of symmetry. This is why the two pieces have the same area.

WORKED EXAMPLE 1, LINE BY LINE

a Checking $(1, 1)$: $1^2 = 1$ and $(1-2)^2 = (-1)^2 = 1$. To find the point without guessing, solve $x^2 = (x-2)^2$. Expanding, $x^2 = x^2 - 4x + 4$, so $4x = 4$ and $x = 1$. Then $y = 1^2 = 1$.

c First piece: $\frac{1^3}{3} - \frac{0^3}{3} = \frac{1}{3} - 0 = \frac{1}{3}$. Second piece: at $x = 2$, $\frac{(2-2)^3}{3} = \frac{0}{3} = 0$; at $x = 1$, $\frac{(1-2)^3}{3} = \frac{(-1)^3}{3} = -\frac{1}{3}$. So the integral is $0 - \left(-\frac{1}{3}\right) = \frac{1}{3}$.

CONCEPT 2 · regions between curves

AREAS OF REGIONS BETWEEN CURVES

Suppose that one curve $y = f(x)$ is always below another curve $y = g(x)$ in an interval $a \leq x \leq b$. Then the area of the region between the curves from $x = a$ to $x = b$ can be found by subtraction.

17 AREA BETWEEN CURVES

If $f(x) \leq g(x)$ in the interval $a \leq x \leq b$, then

$$\text{area between the curves} = \int_a^b \big(g(x) - f(x)\big)\,dx.$$

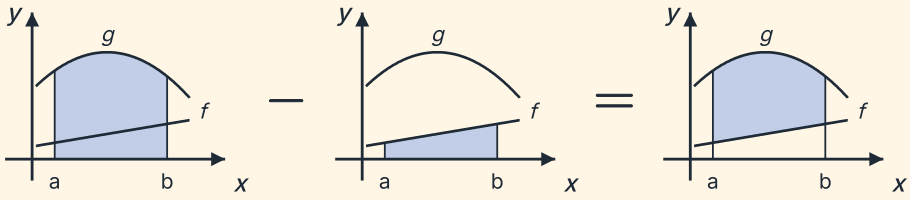
That is, take the integral of the top curve minus the bottom curve.

THE ASSUMPTION $f(x) \leq g(x)$ IS IMPORTANT

The assumption that $f(x) \leq g(x)$ is important. If the curves cross each other, then separate integrals will need to be taken or else the areas of regions where different curves are on top will begin to cancel each other out.

WHY SUBTRACTION WORKS

Suppose first that both curves are above the x-axis. The area under the top curve g, minus the area under the bottom curve f, leaves the area between them:



In symbols, using the properties of the definite integral from Lesson 5C,

$$\int_a^b g(x)\,dx - \int_a^b f(x)\,dx = \int_a^b \big(g(x) - f(x)\big)\,dx.$$

FINDING THE LIMITS AND THE TOP CURVE

1. The limits a and b are usually the x-coordinates of the points where the curves meet. Put the two expressions for y equal, collect every term on one side, and solve the equation.
2. To see which curve is on top, substitute a value of x between a and b into both equations. The larger y-value belongs to the top curve.
3. Simplify (top) — (bottom) before integrating. Put the bottom curve in brackets, because the minus sign changes the sign of every one of its terms.

TUTOR DEMO **A line above a parabola**

a Find the two points where the curve $y = (x-3)^2$ meets the line $y = x + 3$.
b Draw a sketch and shade the area of the region between these two graphs.
c Find the shaded area.

YOUR TURN **Same method**

a Find the two points where the curve $y = (x-1)^2$ meets the line $y = x + 1$.
b Draw a sketch and shade the area of the region between these two graphs.
c Find the shaded area.

WORKED EXAMPLE 2 **From the text: keep for homework**

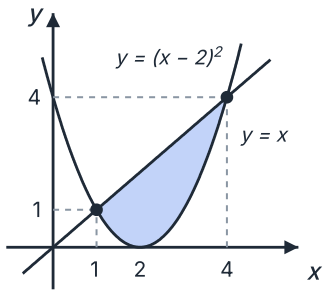
a Find the two points where the curve $y = (x-2)^2$ meets the line $y = x$.
b Draw a sketch and shade the area of the region between these two graphs.
c Find the shaded area.

SOLUTION

a Substituting $y = x$ into $y = (x-2)^2$ gives

$$\begin{aligned}(x-2)^2 &= x\\ x^2 - 4x + 4 &= x\\ x^2 - 5x + 4 &= 0\\ (x-1)(x-4) &= 0,\\ x &= 1 \text{ or } 4,\end{aligned}$$

so the two graphs intersect at $(1, 1)$ and $(4, 4)$.



b The sketch is drawn to the right.

c In the shaded region, the line is above the parabola.

$$\begin{aligned}\text{Hence area} &= \int_1^4 \left(x - (x-2)^2\right) dx\\ &= \int_1^4 \left(x - (x^2 - 4x + 4)\right) dx\\ &= \int_1^4 (-x^2 + 5x - 4)\, dx\\ &= \left[-\frac{x^3}{3} + \frac{5x^2}{2} - 4x\right]_1^4\\ &= \left(-21\tfrac{1}{3} + 40 - 16\right) - \left(-\tfrac{1}{3} + 2\tfrac{1}{2} - 4\right)\\ &= 2\tfrac{2}{3} + 1\tfrac{5}{6}\\ &= 4\tfrac{1}{2} \text{ square units.}\end{aligned}$$

WORKED EXAMPLE 2, LINE BY LINE

At $x = 4$: $-\frac{4^3}{3} = -\frac{64}{3} = -21\frac{1}{3}$, $\frac{5\times 4^2}{2} = \frac{80}{2} = 40$ and $4 \times 4 = 16$. So the first bracket is $-21\frac{1}{3} + 40 - 16 = 2\frac{2}{3}$.

At $x = 1$: $-\frac{1^3}{3} = -\frac{1}{3}$, $\frac{5\times 1^2}{2} = 2\frac{1}{2}$ and $4 \times 1 = 4$. So the second bracket is $-\frac{2}{6} + \frac{15}{6} - \frac{24}{6} = -\frac{11}{6} = -1\frac{5}{6}$.

Subtracting, $2\frac{2}{3} - \left(-1\frac{5}{6}\right) = 2\frac{2}{3} + 1\frac{5}{6} = \frac{16}{6} + \frac{11}{6} = \frac{27}{6} = 4\frac{1}{2}$.

CONCEPT 3 · a region between curves that crosses the x-axis

NOTE

Note: The formula given in Box 17 for the area of the region between two curves holds even if the region crosses the x-axis.

To illustrate this point, the next example is the previous example shifted down 2 units so that the region between the line and the parabola crosses the x-axis. The area of course remains the same — and notice how the formula still gives the correct answer.

WHY THE FORMULA STILL WORKS

Shift both curves up by the same number c, far enough that the whole region is above the x-axis. The area does not change, and Box 17 works for the shifted curves. But the height of each strip is unchanged too:

$$\big(g(x)+c\big)-\big(f(x)+c\big)=g(x)-f(x).$$

So the integral of the top curve minus the bottom curve gives the area wherever the region lies. There is no need to split the region at the x-intercepts, as Lesson 5F did, because $g(x)-f(x)$ is never negative between a and b.

TUTOR DEMO **The region crosses the x-axis**

a Find the two points where the curves $y=x^2-6x+3$ and $y=x-3$ meet.
b Draw a sketch and find the area of the region between these two curves.

YOUR TURN **Same method**

a Find the two points where the curves $y = x^2 - 2x - 2$ and $y = x - 2$ meet.
b Draw a sketch and find the area of the region between these two curves.

WORKED EXAMPLE 3 **From the text: keep for homework**

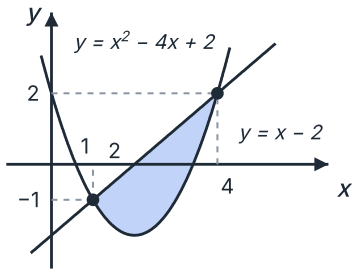
a Find the two points where the curves $y = x^2 - 4x + 2$ and $y = x - 2$ meet.
b Draw a sketch and find the area of the region between these two curves.

SOLUTION

a Substituting $y = x - 2$ into $y = x^2 - 4x + 2$ gives

$$\begin{aligned} x^2 - 4x + 2 &= x - 2 \\ x^2 - 5x + 4 &= 0 \\ (x - 1)(x - 4) &= 0 \\ x &= 1 \text{ or } 4, \end{aligned}$$

so the two graphs intersect at $(1, -1)$ and $(4, 2)$.



b Again, the line is above the parabola,

$$\begin{aligned} \text{so area} &= \int_1^4 \big((x - 2) - (x^2 - 4x + 2)\big)\, dx \\ &= \int_1^4 (-x^2 + 5x - 4)\, dx \\ &= \Big[-\frac{x^3}{3} + \frac{5x^2}{2} - 4x \Big]_1^4 \\ &= \big(-21\tfrac{1}{3} + 40 - 16 \big) - \big(-\tfrac{1}{3} + 2\tfrac{1}{2} - 4 \big) \\ &= 4\tfrac{1}{2} \text{ square units.} \end{aligned}$$

WORKED EXAMPLE 3, LINE BY LINE

The integrand: $(x-2)-(x^2-4x+2)=x-2-x^2+4x-2=-x^2+5x-4$, the same integrand as in Worked example 2.

The brackets are the same as in Worked example 2: $-21\frac{1}{3}+40-16=2\frac{2}{3}$ and $-\frac{1}{3}+2\frac{1}{2}-4=-1\frac{5}{6}$. The step the text leaves out is $2\frac{2}{3}-\left(-1\frac{5}{6}\right)=2\frac{2}{3}+1\frac{5}{6}=4\frac{1}{2}$.

CONCEPT 4 · regions between curves that cross

AREAS OF REGIONS BETWEEN CURVES THAT CROSS

Now suppose that one curve $y=f(x)$ is sometimes above and sometimes below another curve $y=g(x)$ in the relevant interval. In this case, the conditions of Box 17 no longer hold, and separate integrals will need to be calculated.

ONE INTEGRAL FOR EACH PIECE

1. Find every point where the curves meet inside the interval. These points cut the region into pieces.
2. In each piece, decide which curve is on top, and integrate (top) — (bottom) over that piece.
3. Add the areas of the pieces.

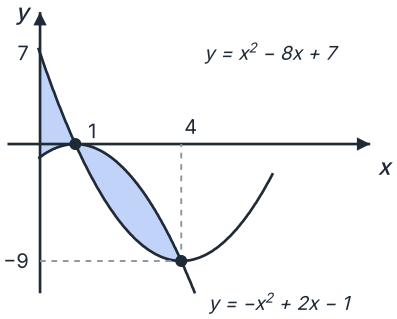
A single integral of (one curve) — (the other) across a crossing point would subtract one piece from the other, which is the cancelling that the box above warns about.

TUTOR DEMO **The curves change places**

The diagram shows the curves

$$y = -x^2 + 2x - 1 \quad \text{and} \quad y = x^2 - 8x + 7$$

meeting at the points $(1, 0)$ and $(4, -9)$. Find the area of the shaded region.

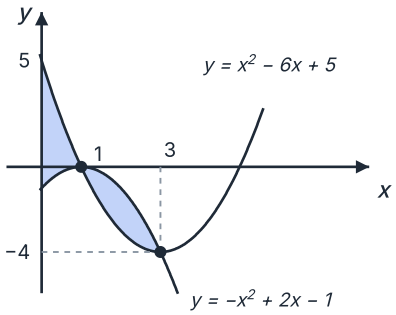


YOUR TURN **Same method**

The diagram shows the curves

$$y = -x^2 + 2x - 1 \quad \text{and} \quad y = x^2 - 6x + 5$$

meeting at the points $(1, 0)$ and $(3, -4)$. Find the area of the shaded region.

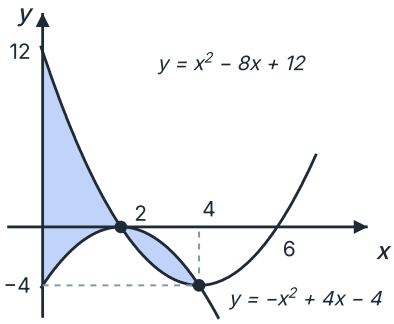


WORKED EXAMPLE 4 **From the text: keep for homework**

The diagram below shows the curves

$$y = -x^2 + 4x - 4 \quad \text{and} \quad y = x^2 - 8x + 12$$

meeting at the points $(2, 0)$ and $(4, -4)$. Find the area of the shaded region.



SOLUTION

In the left-hand region, the second curve is above the first,

$$\begin{aligned}
\text{So} \quad \text{area} &= \int_0^2 \left((x^2 - 8x + 12) - (-x^2 + 4x - 4)\right) dx \\
&= \int_0^2 (2x^2 - 12x + 16)\, dx \\
&= \left[\frac{2x^3}{3} - 6x^2 + 16x\right]_0^2 \\
&= 5\tfrac{1}{3} - 24 + 32 \\
&= 13\tfrac{1}{3} \text{ u}^2.
\end{aligned}$$

In the right-hand region, the first curve is above the second,

$$\begin{aligned}
\text{so} \quad \text{area} &= \int_2^4 \left((-x^2 + 4x - 4) - (x^2 - 8x + 12)\right) dx \\
&= \int_2^4 (-2x^2 + 12x - 16)\, dx \\
&= \left[-\frac{2x^3}{3} + 6x^2 - 16x\right]_2^4 \\
&= \left(-42\tfrac{2}{3} + 96 - 64\right) - \left(-5\tfrac{1}{3} + 24 - 32\right) \\
&= -10\tfrac{2}{3} + 13\tfrac{1}{3} \\
&= 2\tfrac{2}{3} \text{ u}^2.
\end{aligned}$$

$$\begin{aligned}
\text{Hence total area} &= 13\tfrac{1}{3} + 2\tfrac{2}{3} \\
&= 16 \text{ u}^2.
\end{aligned}$$

WORKED EXAMPLE 4, LINE BY LINE

Left integrand: take away each term of the bottom curve.

$$(x^2 - 8x + 12) - (-x^2 + 4x - 4) = x^2 - 8x + 12 + x^2 - 4x + 4$$
$$= 2x^2 - 12x + 16.$$

At $x = 2$: $\frac{2\times 2^3}{3} = \frac{16}{3} = 5\frac{1}{3}$, $6 \times 2^2 = 24$ and $16 \times 2 = 32$. At $x = 0$ every term is 0, so the area is $5\frac{1}{3} - 24 + 32 = 13\frac{1}{3}$.

Right integrand:

$$(-x^2 + 4x - 4) - (x^2 - 8x + 12) = -x^2 + 4x - 4 - x^2 + 8x - 12$$
$$= -2x^2 + 12x - 16.$$

At $x = 4$: $-\frac{2\times 4^3}{3} = -\frac{128}{3} = -42\frac{2}{3}$, $6 \times 4^2 = 96$ and $16 \times 4 = 64$, so the bracket is $-42\frac{2}{3} + 32 = -10\frac{2}{3}$. At $x = 2$: $-5\frac{1}{3} + 24 - 32 = -13\frac{1}{3}$. Then $-10\frac{2}{3} - \left(-13\frac{1}{3}\right) = -10\frac{2}{3} + 13\frac{1}{3} = 2\frac{2}{3}$.

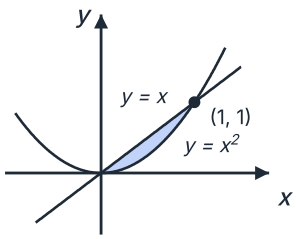
TEXTBOOK HOMEWORK · Exercise 5G

FOUNDATION

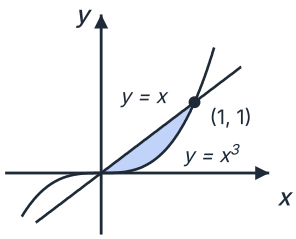
Technology: Graphing programs are particularly useful with compound regions because they allow the separate parts of the region to be identified clearly.

1 Find the area of the shaded region in each diagram below.

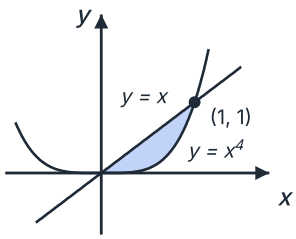
a



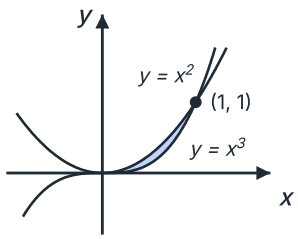
b



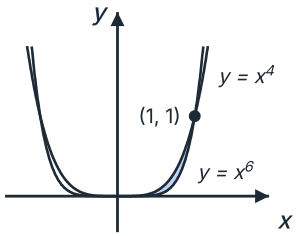
c



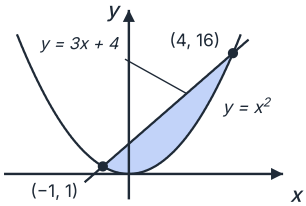
d



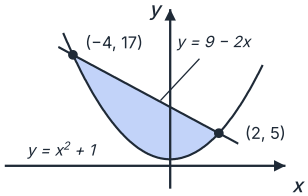
e



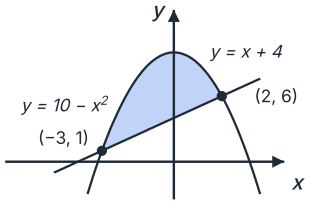
f



g

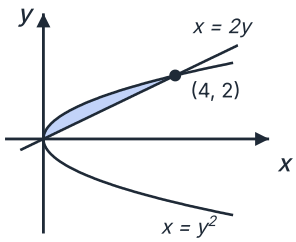


h

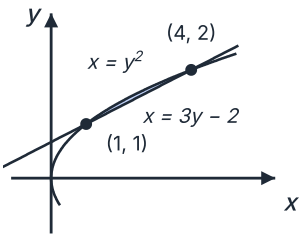


2 By considering regions between the curves and the y-axis, find the area of the shaded region in each diagram below.

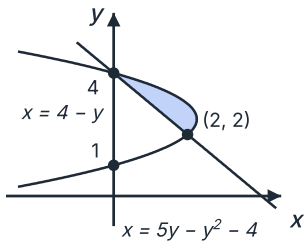
a



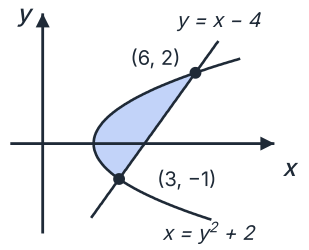
b



c

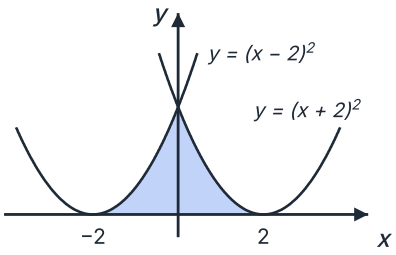


d

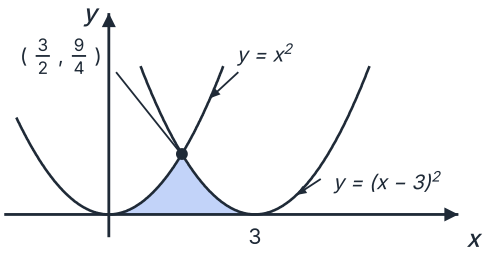


3 Find the areas of the shaded regions in the diagrams below. In each case you will need to find two areas and add them.

a

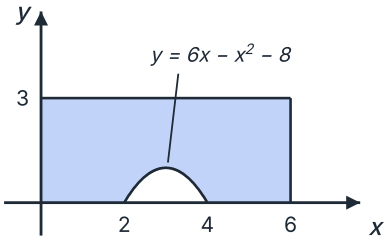


b

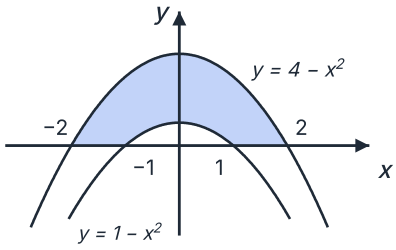


4 Find the areas of the shaded regions in the diagrams below. In each case you will need to find two areas and subtract one from the other.

a



b



DEVELOPMENT

5 a By solving the equations simultaneously, show that the parabola $y = x^2 + 4$ and the line $y = x + 6$ intersect at the points $(-1, 5)$ and $(2, 8)$.

b Sketch the parabola and the line on the same diagram, and shade the region enclosed between them.

c Show that this region has area

$$\int_{-1}^{2} \left((x+6) - (x^2+4)\right) dx = \int_{-1}^{2} (x - x^2 + 2)\, dx,$$

and evaluate the integral.

6 a By solving the equations simultaneously, show that the parabola $y = 3x - x^2 = x(3 - x)$ and the line $y = x$ intersect at the points $(0, 0)$ and $(2, 2)$.

b Sketch the parabola and the line on the same diagram, and shade the region enclosed between them.

c Show that this region has area

$$\int_{0}^{2} (3x - x^2 - x)\, dx = \int_{0}^{2} (2x - x^2)\, dx,$$

and evaluate the integral.

7 a By solving the equations simultaneously, show that the parabola $y = (x-3)^2$ and the line $y = 14 - 2x$ intersect at the points $(-1, 16)$ and $(5, 4)$.

b Sketch the parabola and the line on the same diagram, and shade the region enclosed between them.

c Show that this region has area

$$\int_{-1}^{5} \left((14-2x) - (x-3)^2\right) dx = \int_{-1}^{5} (4x + 5 - x^2)\, dx,$$

and evaluate the integral.

8 Solve simultaneously to find the points of intersection of each pair of graphs. Then sketch the graphs on the same diagram and shade the region enclosed between them. By evaluating the appropriate definite integral, find the area of the shaded region in each case.

a $y = x + 3$ and $y = x^2 + 1$

b $y = 9 - x^2$ and $y = 3 - x$

c $y = x^2 - x + 4$ and $y = -x^2 + 3x + 4$

9 a By solving the equations simultaneously, show that the parabola $y = x^2 + 2x - 8$ and the line $y = 2x + 1$ intersect at the points $(3, 7)$ and $(-3, -5)$.

b Sketch both graphs on the same diagram, and shade the region enclosed between them.

c Despite the fact that it crosses the x-axis, the region has area given by

$$\int_{-3}^{3} \left((2x + 1) - (x^2 + 2x - 8)\right) dx = \int_{-3}^{3} (9 - x^2)\, dx.$$

Evaluate the integral and hence find the area of the region enclosed between the curves.

10 a By solving the equations simultaneously, show that the parabola $y = x^2 - x - 2$ and the line $y = x - 2$ intersect at the points $(0, -2)$ and $(2, 0)$.

b Sketch both graphs on the same diagram, and shade the region enclosed between them.

c Despite the fact that it is below the x-axis, the region has area given by

$$\int_{0}^{2} \left((x - 2) - (x^2 - x - 2)\right) dx = \int_{0}^{2} (2x - x^2)\, dx.$$

Evaluate this integral and hence find the area of the region between the curves.

11 Solve simultaneously to find the points of intersection of each pair of graphs. Then sketch the graphs on the same diagram, and shade the region enclosed between them. By evaluating the appropriate definite integral, find the area of the shaded region in each case.

a $y = x^2 - 6x + 5$ and $y = x - 5$

b $y = -3x$ and $y = 4 - x^2$

c $y = x^2 - 1$ and $y = 7 - x^2$

12 **a** On the same number plane, sketch the graphs of the parabolas $y = x^2$ and $x = y^2$, clearly indicating their points of intersection. Shade the region enclosed between them.

b Explain why the area of this region is given by $\int_0^1 \left(\sqrt{x} - x^2\right) dx$.

c Find the area of the region bounded by the two curves.

13 Tangents are drawn to the parabola $x^2 = 8y$ at the points $A(4, 2)$ and $B(-4, 2)$.

a Draw a diagram of the situation and note the symmetry about the y-axis.

b Find the equation of the tangent at the point A.

c Find the area of the region bounded by the curve and the tangents.

14 **a** Show that the tangent to the curve $y = x^3$ at the point where $x = 2$ has equation $y = 12x - 16$.

b Show by substitution that the tangent and the curve intersect again at the point $(-4, -64)$.

c Find the area of the region enclosed between the curve and the tangent.

15 Consider the curves $y = x^3 - 3$ and $y = -x^2 + 10x - 11$.

a Show by substitution that the curves intersect at three points whose x-values are -4, 1 and 2.

b Sketch the curves showing clearly their intersection points.

c Find the area of the region enclosed by the two curves.

16 **a** Find the points of intersection of the curves $y = x^2(1 - x)$ and $y = x(1 - x)^2$.

b Hence find the area bounded by the two curves.

17 **a** Given the two functions $f(x) = (x+1)(x-1)(x-3)$ and $g(x) = (x+1)(x-1)$, for what values of x is $f(x) > g(x)$?

b Sketch a graph of the two functions on the same number plane, and find the area enclosed between them.

18 **a** Sketch a graph of the function $y = 12x - 32 - x^2$, clearly indicating the x-intercepts.

b Find the equation of the tangent to the curve at the point A where $x = 5$.

c If the tangent meets the x-axis at B, and C is the x-intercept of the parabola closer to the origin, find the area of the region bounded by AB, BC and the arc CA.

ENRICHMENT

19 Find the value of k for which the line $y = kx$ bisects the area enclosed by the curve $4y = 4x - x^2$ and the x-axis.

20 The *average value* of a continuous function $f(x)$ over an interval $a \leq x \leq b$ is defined to be $\dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$ if $a \neq b$, or $f(a)$ if $a = b$. If k is the average value of $f(x)$ on the interval $a \leq x \leq b$, show that the area of the region bounded by $f(x)$ above the line $y = k$ is equal to the area of the region bounded by $f(x)$ below the line $y = k$.

PAST-PAPER HOMEWORK · timed practice

The school trial papers used for this pack (2020–2024) have no question that first needs this lesson's methods: each integration question in the catalogue for these papers can be done with the methods of an earlier lesson, or needs later work. For timed practice, do Questions 8(b), 11(c), 13, 15 and 17 above in 30 minutes.

LESSON 5H

The trapezoidal rule

Some definite integrals cannot be found exactly with a primitive, and some functions are known only from a table of measured values. This lesson approximates such integrals by replacing the curve by chords, so that the region becomes a set of trapezia. It covers the rule with one subinterval, splitting the interval into several equal subintervals, how the concavity of the curve tells you whether the rule overestimates or underestimates, the single formula for n subintervals, and doing the calculation on a spreadsheet.

CONCEPT 1 · the trapezoidal rule, and subdividing the interval

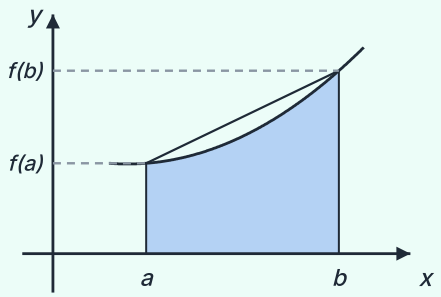
WHY APPROXIMATE?

Methods of approximating definite integrals become necessary when exact calculations using primitives are not possible. This can happen for two reasons.

- The primitives of many important functions cannot be written down in a formula suitable for calculation. This is the case for the important normal distribution in Chapter 16.
- Some values of a function may be known only from experiments, and the function formula may be unknown.

THE TRAPEZOIDAL RULE

Besides taking upper and lower rectangles, the most obvious way to approximate an integral is to replace the curve by a straight line, that is, by a chord joining $\big(a, f(a)\big)$ and $\big(b, f(b)\big)$. The resulting region is a trapezium, so this approximation method is called the *trapezoidal rule*.



Consider the trapezium in the diagram to the right.

$$\text{Here} \quad \text{width} = b - a,$$

$$\text{and} \quad \text{average of parallel sides} = \frac{f(a) + f(b)}{2}.$$

$$\begin{aligned}\text{Hence} \quad \text{area of trapezium} &= \text{width} \times \text{average of parallel sides} \\ &= \frac{b-a}{2}\Big(f(a) + f(b)\Big).\end{aligned}$$

The area of this trapezium is taken as an approximation of the integral.

18 THE TRAPEZOIDAL RULE USING ONE SUBINTERVAL

Let $f(x)$ be a function that is continuous in the closed interval $[a, b]$.

- Approximating the curve from $x = a$ to $x = b$ by a chord allows the region under the curve to be approximated by a trapezium, giving
$$\int_a^b f(x)\,dx \doteqdot \frac{b-a}{2}\Big(f(a)+f(b)\Big).$$
- If the function is linear, then the chord coincides with the curve and the formula is exact.
- Always start a trapezoidal-rule calculation by constructing a table of values.

WHERE THE FORMULA COMES FROM

- The area of a trapezium is $\frac{1}{2}(a+b)h$, where a and b are the lengths of the two parallel sides and h is the distance between them. Here the parallel sides are the two vertical lines at $x = a$ and $x = b$, with lengths $f(a)$ and $f(b)$, and the distance between them is the width $b - a$ along the x-axis.
- So the area is $\frac{1}{2}\big(f(a)+f(b)\big)(b-a)$, which is the same as $\dfrac{b-a}{2}\big(f(a)+f(b)\big)$.
- The sign $\doteqdot$ means ‘is approximately equal to’. The only error comes from the gap between the curve and the chord.

SUBDIVIDING THE INTERVAL

Given an integral over an interval $[a, b]$, we can split that interval $[a, b]$ up into a number of subintervals and apply the trapezoidal rule to each subinterval in turn. This will usually improve the accuracy of the approximation.

Here is the method applied to the reciprocal function $y = \dfrac{1}{x}$, whose primitive we will establish in Chapter 6.

HOW THE PIECES FIT TOGETHER

Splitting at the points between a and b uses the dissection of the interval (Box 7 of Lesson 5C). Each smaller integral is then replaced by the area of one trapezium, and the trapezium areas are added. The chords of the smaller trapezia hug the curve more closely, which is why the accuracy usually improves.

TUTOR DEMO **One subinterval, then three**

Find approximations of $\int_1^4 \frac{12}{x}\,dx$ using the trapezoidal rule with:

a one subinterval, **b** three subintervals.

YOUR TURN **Same method**

Find approximations of $\int_2^5 \frac{60}{x}\,dx$ using the trapezoidal rule with:

a one subinterval, **b** three subintervals.

WORKED EXAMPLE 32 **From the text: keep for homework**

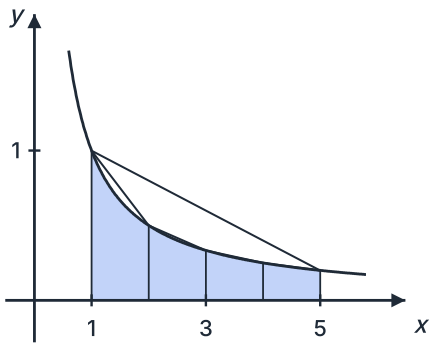
Find approximations of $\int_1^5 \frac{1}{x}\,dx$ using the trapezoidal rule with:

a one subinterval, **b** four subintervals.

SOLUTION

Always begin with a table of values of the function.

x	1	2	3	4	5
$\frac{1}{x}$	1	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$



a One application of the trapezoidal rule, using the whole interval as the subinterval, requires just two values of the function.

$$\begin{aligned}\int_1^5 \frac{1}{x}\,dx &\doteqdot \frac{5-1}{2} \times \Big(f(1)+f(5)\Big)\\ &\doteqdot 2 \times \left(1+\tfrac{1}{5}\right)\\ &\doteqdot 2\tfrac{2}{5}\end{aligned}$$

b Four applications of the trapezoidal rule require five values of the function. Dividing the interval $1 \leq x \leq 5$ into four equal subintervals,

$$\int_1^5 \frac{1}{x}\,dx = \int_1^2 \frac{1}{x}\,dx + \int_2^3 \frac{1}{x}\,dx + \int_3^4 \frac{1}{x}\,dx + \int_4^5 \frac{1}{x}\,dx.$$

Each subinterval has width 1, so applying the trapezoidal rule to each integral,

$$\begin{aligned}\int_1^5 \frac{1}{x}\,dx &\doteqdot \tfrac{1}{2}\Big(f(1)+f(2)\Big) + \tfrac{1}{2}\Big(f(2)+f(3)\Big)\\ &\qquad + \tfrac{1}{2}\Big(f(3)+f(4)\Big) + \tfrac{1}{2}\Big(f(4)+f(5)\Big)\\ &\doteqdot \tfrac{1}{2}\left(\tfrac{1}{1}+\tfrac{1}{2}\right) + \tfrac{1}{2}\left(\tfrac{1}{2}+\tfrac{1}{3}\right) + \tfrac{1}{2}\left(\tfrac{1}{3}+\tfrac{1}{4}\right) + \tfrac{1}{2}\left(\tfrac{1}{4}+\tfrac{1}{5}\right)\\ &\doteqdot 1\tfrac{41}{60}.\end{aligned}$$

STEPS THE BOOK SKIPS: WORKED EXAMPLE 32

a The width is $5-1=4$, so $\dfrac{5-1}{2}=2$. Then $1+\frac{1}{5}=\frac{6}{5}$, and $2\times\frac{6}{5}=\frac{12}{5}=2\frac{2}{5}$.

b Work out each trapezium, then add over the common denominator 120:

$$\begin{aligned}&\tfrac{1}{2}\left(\tfrac{1}{1}+\tfrac{1}{2}\right)+\tfrac{1}{2}\left(\tfrac{1}{2}+\tfrac{1}{3}\right)+\tfrac{1}{2}\left(\tfrac{1}{3}+\tfrac{1}{4}\right)+\tfrac{1}{2}\left(\tfrac{1}{4}+\tfrac{1}{5}\right)\\&=\tfrac{1}{2}\times\tfrac{3}{2}+\tfrac{1}{2}\times\tfrac{5}{6}+\tfrac{1}{2}\times\tfrac{7}{12}+\tfrac{1}{2}\times\tfrac{9}{20}\\&=\tfrac{3}{4}+\tfrac{5}{12}+\tfrac{7}{24}+\tfrac{9}{40}\\&=\tfrac{90}{120}+\tfrac{50}{120}+\tfrac{35}{120}+\tfrac{27}{120}\\&=\tfrac{202}{120}\\&=1\tfrac{41}{60}.\end{aligned}$$

The second answer is smaller than the first. The four chords lie closer to the curve than the single chord does, as the diagram shows.

EQUAL SUBINTERVALS

Note: Always take subintervals of equal width unless otherwise indicated.

CONCEPT 2 · concavity and the trapezoidal rule

CONCAVITY AND THE TRAPEZOIDAL RULE

The curve in the example above is concave up, so every chord is above the curve, and every approximation found using the trapezoidal rule is therefore greater than the integral.

Similarly, if a curve is concave down, then every chord is below the curve, and every trapezoidal-rule approximation is less than the integral. The second derivative can be used to test concavity.

19 CONCAVITY AND THE TRAPEZOIDAL RULE

- If the curve is concave up, the trapezoidal rule overestimates the integral.
- If the curve is concave down, the trapezoidal rule underestimates the integral.
- If the curve is linear, the trapezoidal rule gives the exact value of the integral.

The second derivative $\dfrac{d^2y}{dx^2}$ can be used to test the concavity.

RECALL: CONCAVITY (YEAR 11)

- If $f''(x) > 0$ throughout an interval, the curve is concave up there: it bends upwards, and each chord lies above the curve. If $f''(x) < 0$ throughout the interval, the curve is concave down, and each chord lies below the curve.
- A chord above the curve gives a trapezium that holds the whole region plus a thin sliver, so its area is too big. A chord below the curve leaves a sliver out, so its area is too small.
- Box 19 needs one concavity across the whole interval. If the concavity changes inside the interval, the errors on the two sides can partly cancel, and the box gives no conclusion.

TUTOR DEMO **Over or under?**

a Use the trapezoidal rule with one subinterval (that is, two function values) to approximate $\int_1^3 (50x - x^4)\,dx$.

b Use the second derivative to explain why the approximation underestimates the integral.

c Find the exact value of the integral, and check part **b**.

YOUR TURN **Same method**

a Use the trapezoidal rule with one subinterval (that is, two function values) to approximate $\int_1^3 (x^3+1)\,dx$.

b Use the second derivative to explain why the approximation overestimates the integral.

c Find the exact value of the integral, and check part **b**.

WORKED EXAMPLE 33 **From the text: keep for homework**

a Use the trapezoidal rule with one subinterval (that is, two function values) to approximate $\int_1^5 (200x - x^4)\,dx$.

b Use the second derivative to explain why the approximation underestimates the integral.

SOLUTION

a Construct a table of values for $y = 200x - x^4$.

x	1	5
y	199	375

$$\int_1^5 (200x - x^4)\,dx \doteqdot \frac{5-1}{2} \times \Big(f(1) + f(5)\Big)$$
$$\doteqdot 2 \times (199 + 375)$$
$$\doteqdot 1148$$

b The function is $y = 200x - x^4$.

Differentiating, $y' = 200 - 4x^3$
and $y'' = -12x^2$.

Because $y'' = -12x^2$ is negative throughout the interval $1 \leq x \leq 5$, the curve is concave down throughout this interval.
Hence the trapezoidal rule underestimates the integral.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 33

- The table: $f(1) = 200 \times 1 - 1^4 = 200 - 1 = 199$ and $f(5) = 200 \times 5 - 5^4 = 1000 - 625 = 375$.
- The arithmetic: $\dfrac{5-1}{2} = 2$, $199 + 375 = 574$ and $2 \times 574 = 1148$.
- Why y'' is negative: $x^2 > 0$ for $1 \leq x \leq 5$, so $-12x^2 < 0$.
- A check with the exact value:

$$\begin{aligned}\int_1^5 (200x - x^4)\,dx &= \left[100x^2 - \tfrac{1}{5}x^5\right]_1^5 \\ &= \left(100 \times 25 - \tfrac{1}{5} \times 3125\right) - \left(100 \times 1 - \tfrac{1}{5} \times 1\right) \\ &= (2500 - 625) - (100 - 0.2) \\ &= 1875 - 99.8 \\ &= 1775.2,\end{aligned}$$

  which is greater than 1148, as part **b** says.

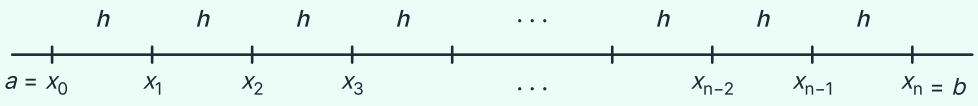
CONCEPT 3 · a formula for n subintervals

A FORMULA FOR MULTIPLE APPLICATIONS OF THE TRAPEZOIDAL RULE

When the trapezoidal rule is being applied two or three times, it is easier to perform the two or three calculations required. These separate calculations also reinforce the meaning of the approximation, and help to gain an intuitive understanding of the accuracy of the estimates.

But increasing accuracy with the trapezoidal rule requires larger numbers of applications of the rule, and this can quickly become tedious. Let us then develop a single formula that splits an integral into n subintervals of equal width and applies the trapezoidal rule to each. Anyone writing a program or using a spreadsheet to estimate integrals would want to do this.

The first step is to divide the interval $[a, b]$ into n equal subintervals, each of width h, like this:



DERIVING THE FORMULA

There are $n+1$ points altogether, and they divide the interval into n equal subintervals. The endpoints are $a = x_0$ and $b = x_n$, and the $n-1$ division points in between are x_1, x_2, ..., x_{n-1}.

There are n subintervals, so $nh = b - a$, and the width h of each subinterval is

$$h = \frac{b-a}{n}.$$

Thus starting with $a = x_0$, the successive values of the division points are

$$\begin{aligned} x_0 &= a \\ x_1 &= a + h \\ x_2 &= a + 2h \\ &\vdots \end{aligned} \qquad \begin{aligned} x_{n-2} &= a + (n-2)h \\ x_{n-1} &= a + (n-1)h \\ x_n &= a + nh = a + (b-a) = b \end{aligned}$$

That is, $x_r = a + rh$, for $r = 0, 1, 2, \ldots, n-1, n$.

Now we can apply the trapezoidal rule to each subinterval in turn,

$$\begin{aligned} \int_a^b f(x)\,dx &= \int_a^{x_1} f(x)\,dx + \int_{x_1}^{x_2} f(x)\,dx + \cdots \\ &\qquad + \int_{x_{n-2}}^{x_{n-1}} f(x)\,dx + \int_{x_{n-1}}^{b} f(x)\,dx \\ &\doteqdot \frac{h}{2}\Big(f(a) + f(x_1)\Big) + \frac{h}{2}\Big(f(x_1) + f(x_2)\Big) + \cdots \\ &\qquad + \frac{h}{2}\Big(f(x_{n-2}) + f(x_{n-1})\Big) + \frac{h}{2}\Big(f(x_{n-1}) + f(b)\Big) \\ &\doteqdot \frac{h}{2}\Big(f(a) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(b)\Big). \end{aligned}$$

20 TRAPEZOIDAL-RULE FORMULA USING n SUBINTERVALS

Let $f(x)$ be a function that is continuous in the closed interval $[a, b]$. Then

$$\int_a^b f(x)\,dx \doteqdot \frac{h}{2}\Big(f(a) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(b)\Big)$$

where $h = \dfrac{b-a}{n}$ and $x_r = a + rh$, for $r = 1, 2, \ldots, n-1$.

A COMMON REARRANGEMENT

A common rearrangement of this formula, using three sets of nested brackets, is

$$\int_a^b f(x)\,dx \doteqdot \frac{b-a}{2n}\left(f(a) + f(b) + 2\Big(f(x_1) + f(x_2) + \cdots + f(x_{n-1})\Big)\right).$$

THE STEPS OF THE DERIVATION, FILLED IN

- Each subinterval has width h, so each trapezium has area $\frac{h}{2} \times (\text{left height} + \text{right height})$.
- Each division point $x_1, \ldots, x_{n-1}$ is the right-hand end of one subinterval and the left-hand end of the next. So each of $f(x_1), \ldots, f(x_{n-1})$ appears in two trapezia, and the terms collect to $2f(x_1), \ldots, 2f(x_{n-1})$. The end values $f(a)$ and $f(b)$ each appear in only one trapezium.
- For the rearrangement, $\frac{h}{2} = \frac{b-a}{n} \div 2 = \frac{b-a}{2n}$. Then the two end values are written first, and the factor 2 is taken out of the middle terms.
- Check the count: n subintervals use $n+1$ function values. For example, four subintervals use five function values.

USING THE FORMULA FOR THE TRAPEZOIDAL RULE

The formula may look complicated at first sight, but it is actually quite straightforward to use, provided that:

- We begin with a sensible value of the width h of each subinterval.
- We construct a clear table of values to work from.

Here is an example where there is no equation of the function, but simply a set of experimental results gathered by recording equipment.

TUTOR DEMO **A table of measurements**

The flow rate into a dam after a storm was measured every 3 hours, from $t = 0$ to $t = 18$ hours. The flow rates, in cubic metres per second, are shown in the table.

t	0	3	6	9	12	15	18
Flow rate	50	400	900	700	500	300	200

a Convert each flow rate to R, the flow rate in millions of cubic metres per hour.

b Use the trapezoidal rule with 3-hour subintervals to estimate the total amount of water that flowed into the dam in these 18 hours. Use both forms of the formula.

YOUR TURN **Same method**

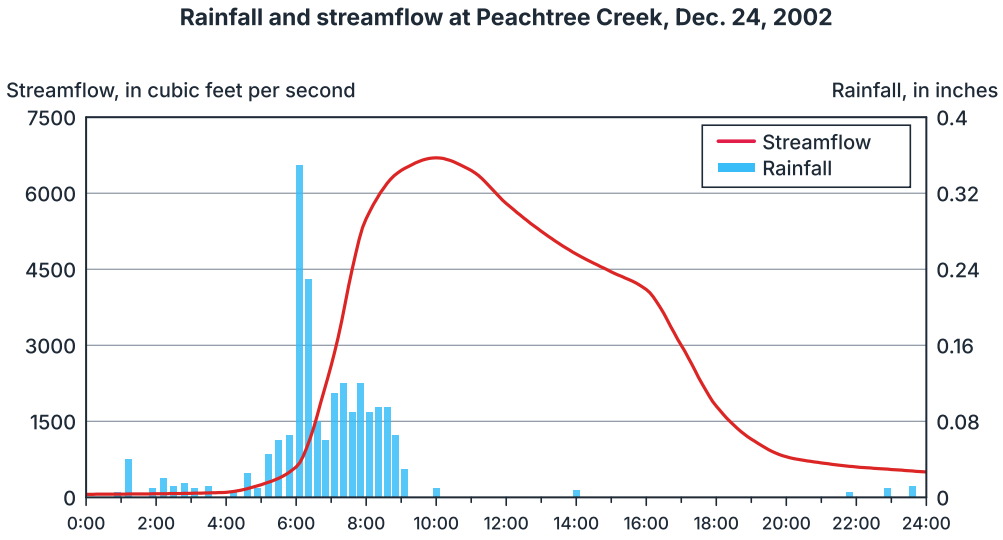
The rate at which water flows from a pipe into a tank was measured every 10 minutes for 40 minutes. The rates, in litres per minute, are shown in the table.

t	0	10	20	30	40
Rate	20	35	40	30	15

Use the trapezoidal rule with 10-minute subintervals to estimate the total amount of water that flowed into the tank. Use both forms of the formula.

WORKED EXAMPLE 34 **From the text: keep for homework**

The flow at Peachtree Creek on 24th December 2002 after a storm is shown in the graph below. The flow rate in cubic feet per second is sketched as a function of the time t in hours.



We can estimate the total amount of water that flowed down the creek after the storm that day by integrating from $t = 4$ to $t = 24$. Use the trapezoidal rule with two-hour subintervals to approximate the total amount of water.

SOLUTION

The graph is very inaccurate, like so much internet data, but here is a rough table of values of the flow rate in cubic feet per second as a function of time t in hours.

t	4	6	8	10	12	14	16	18	20	22	24
Flow rate	100	600	5500	6700	5800	4800	4100	1800	800	600	500

The units need attention. The time is in hours, so the flow rates must be converted to cubic feet per hour by multiplying by $60 \times 60 = 3600$.

To avoid zeroes, let R be the flow rate in millions of cubic feet per hour.

t	4	6	8	10	12	14	16	18	20	22	24
R	0.36	2.16	19.8	24.12	20.88	17.28	14.76	6.48	2.88	2.16	1.8

Here $h = 2$ and $n = 10$. Also $a = x_0 = 4$, $x_1 = 6$, ..., $x_{n-1} = 22$, $x_n = b = 24$.

$$\text{Hence} \quad \int_4^{24} R\,dt \doteqdot \frac{2}{2}\Big(f(4) + 2f(6) + 2f(8) + \cdots + 2f(22) + f(24)\Big)$$
$$\doteqdot 0.36 + 4.32 + 39.6 + 48.24 + 41.76 + 34.56$$
$$+29.52 + 12.96 + 5.76 + 4.32 + 1.8$$
$$\doteqdot 223.2.$$

Alternatively, using the second formula,

$$\begin{aligned}\int_4^{24} R\,dt &\doteqdot \frac{24-4}{20}\Big(f(4)+f(24)+2\big(f(6)+f(8)+\cdots+f(22)\big)\Big)\\ &\doteqdot 0.36+1.8+2(2.16+19.8+24.12+20.88+17.28\\ &\qquad +14.76+6.48+2.88+2.16)\\ &= 2.16+2\times 110.52\\ &= 223.2.\end{aligned}$$

Thus about 223 million cubic feet of water flowed down the creek from 4:00 am to midnight.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 34

- Each flow rate F in cubic feet per second becomes $3600F$ cubic feet per hour, which is $3600F \div 1\,000\,000$ million cubic feet per hour. For example, at $t=4$, $100\times 3600 = 360\,000$, which is 0.36 million. At $t=10$, $6700\times 3600 = 24\,120\,000$, which is 24.12 million.
- The first formula: $\dfrac{h}{2}=\dfrac{2}{2}=1$. The end values are $f(4)=0.36$ and $f(24)=1.8$. The nine middle values are doubled: $2\times 2.16=4.32$, $2\times 19.8=39.6$, $2\times 24.12=48.24$, $2\times 20.88=41.76$, $2\times 17.28=34.56$, $2\times 14.76=29.52$, $2\times 6.48=12.96$, $2\times 2.88=5.76$ and $2\times 2.16=4.32$.
- The second formula: $n=10$, so $\dfrac{b-a}{2n}=\dfrac{24-4}{2\times 10}=\dfrac{20}{20}=1$. The end values add to $0.36+1.8=2.16$, and the nine middle values add to 110.52, so the total is $2.16+221.04=223.2$.
- The units of the answer: R is in millions of cubic feet per hour and t is in hours, so the integral is in millions of cubic feet. The interval from $t=4$ to $t=24$ is 4:00 am to midnight.

CONCEPT 4 · using a spreadsheet for calculations

USING A SPREADSHEET FOR CALCULATIONS

The authors used a spreadsheet for all the calculations above. The trapezoidal-rule formula is well suited for machine computation. The next worked example shows how to use an Excel spreadsheet to carry out such a calculation, but any spreadsheet can be used. Note that:

- Excel commands and procedures have been changing over successive versions.
- Mac users will need some adjustments, particularly when implementing the 'fill down' and 'fill right' commands.

The calculation involves the integration of $\dfrac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}$. We will see in Chapter 16 that this function is the probability density function of the normal distribution, and is the most important function in statistics. There is no simple equation for its primitive, so approximations are always necessary.

READING THE SPREADSHEET

- Row 2 holds the x-values $x_0, x_1, \ldots, x_{10}$, row 3 holds the function values, and row 4 holds each function value times its weight in Box 20: weight 1 for the two end values and weight 2 for the nine middle values.
- The sum of row 4 is the big bracket in Box 20, and the last step multiplies it by $\frac{1}{2}h$.
- No spreadsheet? A calculator gives the same table one value at a time. The spreadsheet just does all the values at once.

TUTOR DEMO **Spreadsheet, 10 subintervals**

Let $\phi(x) = \dfrac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}$. Approximate the integral $\displaystyle\int_0^{0.5} \phi(x)\,dx$ using the trapezoidal rule with 10 subintervals. You may use the fact that $\phi''(x) = (x^2 - 1)\,\phi(x)$ to say whether your answer is an overestimate or an underestimate.

YOUR TURN **Same method**

With the same function $\phi(x)$, approximate the integral $\displaystyle\int_1^2 \phi(x)\,dx$ using the trapezoidal rule with 10 subintervals. Use $\phi''(x) = (x^2 - 1)\,\phi(x)$ to say whether your answer is an overestimate or an underestimate.

WORKED EXAMPLE 35 **From the text: keep for homework**

Let $\phi(x) = \dfrac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}$. Approximate the integral $\displaystyle\int_0^1 \phi(x)\,dx$ using the trapezoidal rule with 10 subintervals.
(The symbol ϕ is the lower case Greek letter 'phi', corresponding to Latin f.)

SOLUTION

In these instructions, we enter a formula into a cell by typing the = sign as the first character. The cell in column D and row 3 is labelled D3, and in formulae, this refers to its contents. Normally leave the top row clear for later titles.

1. On a new sheet in Excel, enter 0 into Cell A2 and press `Enter`.
   Select Cell B2 and type `=0.1+A2` and press `Enter`.
   Cell B2 should now show 0.1.
2. Select Cells `B2:K2` and press `Ctrl+R` to 'fill right'.
   Cells `C2:K2` should now show 0.2, 0.3, ..., 1.
3. Type `=EXP(-A2*A2/2)/SQRT(2*PI())` into Cell A3.
   Cell A3 should now show $\phi(0) = 0.398942$.
4. Select Cells `A3:K3` and press `Ctrl+R` to 'fill right'.
   Cells `B3:K3` should now show $\phi(0.1) = 0.396953$, ..., $\phi(1) = 0.241971$.

We now have the table of values for the function $\phi(x)$, and we need to add

$$\phi(0) + 2\phi(0.1) + 2\phi(0.2) + \cdots + 2\phi(0.9) + \phi(1)$$

5. Select Cell A4 and enter `=A3`. This should duplicate the value in A3.
   Select Cell K4 and enter `=K3`. Again, this duplicates the value in K3.
   Select Cell B4 and enter `=2*B3`. This should double the value in B3.
   Select Cells `B4:J4` and press `Ctrl+R` to 'fill right'.
   Add the row by selecting Cell L4 and typing `=SUM(A4:K4)`.

In this case, $h = 0.1$ so we multiply by $\frac{1}{2}h = 0.05$.

6. Select Cell L5 and type `=L4*0.05`. This shows the final answer.

Hence $\displaystyle\int_0^1 \phi(x)\,dx \doteqdot 0.341$. We shall find in Chapter 16 that this is approximately the probability that a score in a normal distribution lies between the mean 0 and one standard deviation above the mean. The correct value to four decimal places is 0.3413. We will see in Chapter 16 that the curve is concave down in the interval $[0, 1]$, which explains why our estimate is a little smaller than it should be.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 35

The sheet after step 6 (values to six decimal places):

column	row 2	row 3	row 4
A	0	0.398942	0.398942
B	0.1	0.396953	0.793905
C	0.2	0.391043	0.782085
D	0.3	0.381388	0.762776
E	0.4	0.368270	0.736540
F	0.5	0.352065	0.704131
G	0.6	0.333225	0.666449
H	0.7	0.312254	0.624508
I	0.8	0.289692	0.579383
J	0.9	0.266085	0.532170
K	1	0.241971	0.241971
L4	6.822861		
L5	0.341143		

- Row 2 holds the x-values, and row 3 holds $\phi(x)$.
- Row 4 holds $\phi(0)$, then $2\phi(0.1), \ldots, 2\phi(0.9)$, then $\phi(1)$. Their sum is $L4 \doteqdot 6.822861$. (The sheet works with more decimal places than it shows.)
- Then $\frac{1}{2}h = \frac{1}{2} \times 0.1 = 0.05$, and $0.05 \times 6.822861 \doteqdot 0.341143$, which is 0.341 to three decimal places.
- The estimate 0.3411 is a little smaller than the correct value 0.3413, as the concavity predicts.

MORE SUBINTERVALS

Readers may like to repeat the calculations above using 100 subintervals and see how close the approximation is then.

TEXTBOOK HOMEWORK · Exercise 5H

FOUNDATION

Technology: It is not difficult to write (or download) a program that will allow the calculations of the trapezoidal rule to be automated. It can then be applied to many examples from this exercise. The number of subintervals used can be steadily increased, and the approximations may then converge to the exact value of the integral. An accompanying screen sketch showing the curve and the chords would be helpful in giving a visual impression of the size and the sign of the error.

1 Approximate $\int_2^6 f(x)\,dx$ using the formula $\frac{1}{2}(a+b)h$ for the area of a trapezium.

a

x	2	6
$f(x)$	8	12

b

x	2	6
$f(x)$	6.2	4.8

c

x	2	6
$f(x)$	-4	-9

2 Three function values are given in the table to the right.

x	2	6	10
$f(x)$	12	20	30

a Approximate $\int_2^{10} f(x)\,dx$ by calculating the areas of two trapezia and then adding.

b Check your answer to part **a** by using the formula for the trapezoidal rule.

3 Use the trapezoidal rule with the three given function values to approximate $\int_{-5}^{5} f(x)\,dx$.

x	-5	0	5
$f(x)$	2.4	2.6	4.4

4 Show, by means of a diagram, that the trapezoidal rule will:

a overestimate $\int_a^b f(x)\,dx$, if $f''(x) > 0$ for $a \leq x \leq b$,

b underestimate $\int_a^b f(x)\,dx$, if $f''(x) < 0$ for $a \leq x \leq b$.

5 **a** Complete this table for the function $y = x(4 - x)$.

x	0	1	2	3	4
y					

b Hence use the trapezoidal rule with five function values to approximate $\int_0^4 x(4 - x)\,dx$.

c What is the exact value of $\int_0^4 x(4 - x)\,dx$, and why does it exceed the approximation? Sketch the curve and the four chords involved.

d Calculate the percentage error in the approximation (that is, divide the error by the exact answer and convert to a percentage).

6 a Complete this table for the function $y = \dfrac{6}{x}$.

x	1	2	3	4	5
y					

b Use the trapezoidal rule with the five function values above, that is with four subintervals, to approximate $\displaystyle\int_1^5 \frac{6}{x}\,dx$.

c Show that the second derivative of $y = \dfrac{6}{x}$ is $y'' = 12x^{-3}$, and use this result to explain why the approximation will exceed the exact value of the integral.

DEVELOPMENT

8 Use the trapezoidal rule with three function values to approximate each definite integral, writing your answer correct to two significant figures where necessary.

a $\displaystyle\int_0^1 2^{-x}\,dx$ b $\displaystyle\int_{-2}^0 2^{-x}\,dx$ c $\displaystyle\int_1^3 \sqrt[3]{9-2x}\,dx$ d $\displaystyle\int_{-13}^{-1} \sqrt{3-x}\,dx$

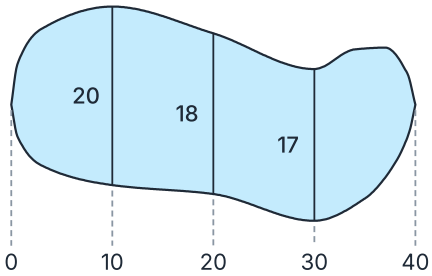
9 Use the trapezoidal rule with five function values to approximate each definite integral, writing your answer correct to three significant figures where necessary.

a $\displaystyle\int_2^6 \frac{1}{x}\,dx$ b $\displaystyle\int_0^2 \frac{1}{2+\sqrt{x}}\,dx$ c $\displaystyle\int_4^8 \sqrt{x^2-3}\,dx$ d $\displaystyle\int_1^2 \log_{10} x\,dx$

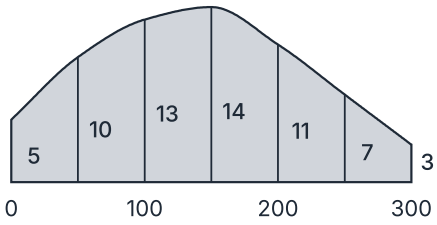
10 An object is moving along the x-axis with values of the velocity v in m/s at various times t given in the table to the right. Given that the distance travelled may be found by calculating the area under the velocity/time graph, use the trapezoidal rule to estimate the distance travelled by the particle in the first 5 seconds.

t	0	1	2	3	4	5
v	1.5	1.3	1.4	2.0	2.4	2.7

11 The diagram to the right shows the width of a lake at 10-metre intervals. Use the trapezoidal rule to estimate the surface area of the water.



12 The diagram to the right shows a vertical rock cutting of length 300 metres alongside a straight horizontal section of highway. The heights of the cutting are measured at 50-metre intervals. Use the trapezoidal rule to estimate the area of the vertical rock cutting.



13

a Use the trapezoidal rule with five function values to approximate $\int_0^1 \sqrt{1-x^2}\,dx$, giving your answer correct to four decimal places.

b Use part **a** and the fact that $y = \sqrt{1-x^2}$ is a semi-circle to approximate π. Give your answer correct to one decimal place, and explain why your approximation is less than π.

14 Use the trapezoidal rule with five function values, together with appropriate log laws, to show that $\int_1^5 \ln x\,dx \doteqdot \ln 54$.

ENRICHMENT

15 a Show that the function $y = \sqrt{x}$ is increasing for all $x > 0$.

b By dividing the area under the curve $y = \sqrt{x}$ into n equal subintervals, show that

$$\sqrt{1} + \sqrt{2} + \sqrt{3} + \cdots + \sqrt{n} \geq \int_0^n \sqrt{x}\, dx = \frac{2n\sqrt{n}}{3}.$$

c Use the trapezoidal rule to show that for all integers $n \geq 1$,

$$\sqrt{1} + \sqrt{2} + \sqrt{3} + \cdots + \sqrt{n} \leq \frac{\sqrt{n}\,(4n+3)}{6}.$$

d Hence estimate $\sqrt{1} + \sqrt{2} + \sqrt{3} + \cdots + \sqrt{12\,000}$, correct to the nearest hundred.

An investigation using a spreadsheet for trapezoidal-rule calculations

16 Work through the spreadsheet example just above this exercise. Then use a spreadsheet to estimate these integrals using the trapezoidal rule with 5, 10, 20 and perhaps more subintervals.

a $\displaystyle\int_1^{11} \frac{1}{x}\, dx$ b $\displaystyle\int_1^{11} \log_e x\, dx$ c $\displaystyle\int_0^{10} e^{-x^2}\, dx$

You will need to look at the results and perhaps vary the number of decimal places that you are using in the calculations and recording in your answers.

Possible spreadsheet projects

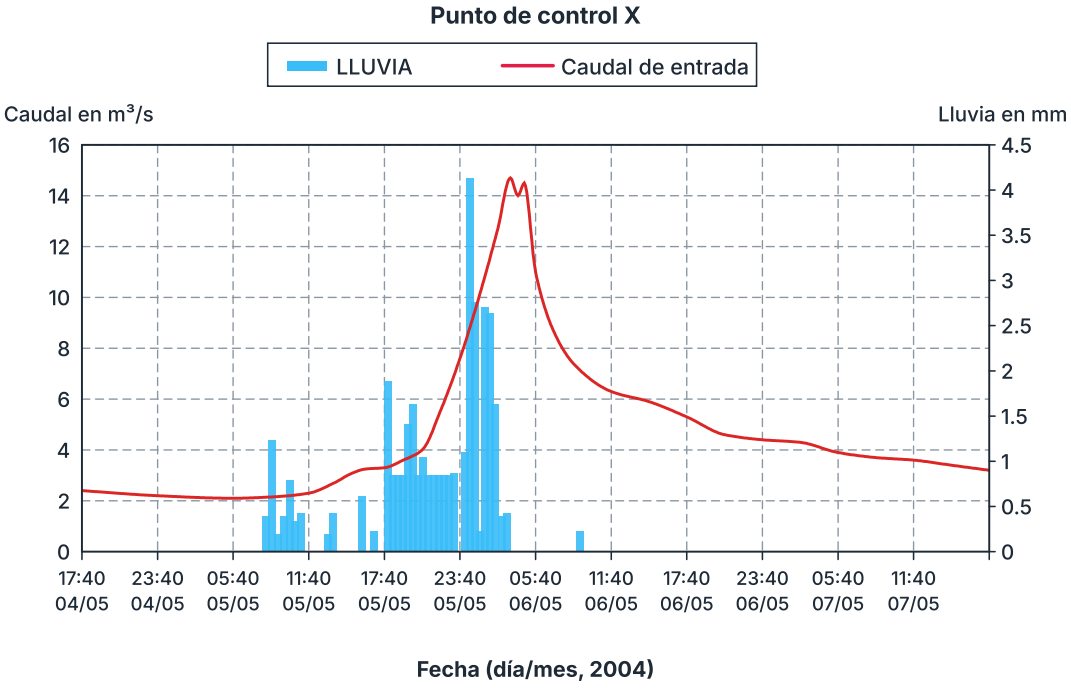
It is possible to program a spreadsheet so that the number of subintervals can be entered as a single variable. The construction of such a program and similar programs could be incorporated into a longer project examining the usefulness and accuracy of the trapezoidal rule, or examining some physical phenomena.

In the next diagram, it is clear that in parts of the graph where there is a lot of activity, the subintervals should be quite narrow, whereas in other calmer parts they can be far wider. Such variability could also be incorporated into the spreadsheet and its formulae.

An investigation (and possible project) integrating a graph from the web by the trapezoidal rule

There is a great deal of data available on the web for a sustained investigation of river flow. The following question suggests some interesting questions about one such situation, but there are many more situations and questions. Integrating graphs of all kinds from the web using the trapezoidal rule could be the basis of various different projects.

17



The hydrograph above shows the rate of flow through Control Point X on the Turia River in Spain over a three-day period in May 2004. The rate of flow ('Caudal') is given as a function of the date–times ('Fecha'). Notice that the successive date–times on the horizontal axis are separated by exactly 6 hours. The rainfall ('Lluvia') is given by the vertical bars.

The units of time are hours, and the units of the flow rate are 'cubic metres per second'. The flow rate R should be converted to units of 'thousands of cubic metres per hour' so that time is in hours and there are fewer zeroes: multiply by $\dfrac{60 \times 60}{1000} = 3.6$.

a From the graph, copy and complete the table of values of the flow rate R at the first four date–times, 04/05/2004 17:40 to 05/05/2004 11:40. Then use the trapezoidal rule to estimate the total volume of water that flowed through the control point in those 18 hours.

t	17:40	23:40	05:40	11:40
R				

b Draw up a similar table for the 18 hours of heavy flow from 05/05/2004 17:40 to 06/05/2004 11:40, but use 3 hours as the separation between successive times.

t	17:40	20:40	23:40	02:40	05:40	08:40	11:40
R							

Then use the trapezoidal rule to estimate the total volume of water that flowed through the Control Point in those 18 hours. Why are 3 hours suggested here in part **b** for the width of the subintervals, where 6 hours was used in part **a**?

c How many times more water flowed down the river in the second 18-hour period? Look at the rainfall record, and discuss how the river flow responded to the rainfall.

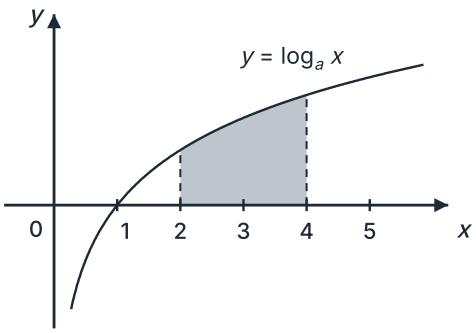
PAST-PAPER HOMEWORK · mini exam

Suggested time: 5 minutes **1 mark** Calculator allowed · show working · hints are below each question if you get stuck

This is the only school-paper item that needs just this lesson's ideas. For more timed practice, do Questions 5, 9(c), 12, 13 and 14 above in 25 minutes.

PP 1 **Sydney Grammar School, 2023, Trial Examination, Form VI Mathematics Extension 1, Q3** `1 mark`

Consider the area between the curve $y = \log_a x$, the x-axis, $x = 2$ and $x = 4$ as shown in the diagram.



What is the area approximated by one application of the trapezoidal rule?

(A) $3\log_a 2$ (B) $\log_a 6$ (C) $\dfrac{\log_a 2 + \log_a 4}{2}$ (D) $\dfrac{3a}{2}$

Hint: what are the boundaries of the shaded region?

LESSON 5I

The reverse chain rule

When the chain rule is used to differentiate a composite function, the result is a product of two terms. This lesson reverses a chain-rule differentiation to find a primitive. A formula for the reverse chain rule is then developed in two notations, using u or using $f(x)$. The vital step is to identify u or $f(x)$ and its derivative in the function that is being integrated. The lesson ends with the Chapter 5 review exercise.

CONCEPT 1 · reversing a chain-rule differentiation

THE REVERSE CHAIN RULE

When we use the chain rule to differentiate a composite function, the result is a product of two terms. For example, in the first worked example below,

$$\frac{d}{dx}(x^3+5)^4 = 4(x^3+5)^3 \times 3x^2.$$

This section deals with the problem of reversing a chain-rule differentiation.

REVERSING A CHAIN-RULE DIFFERENTIATION

Finding primitives is the reverse process of differentiation. Thus once any differentiation has been performed, the process can then be reversed to give a primitive.

THE CHAIN RULE (FROM YEAR 11)

The revision pages at the start of this pack set out the chain rule. To differentiate $y = (x^3+5)^4$, work on the right first: let $u = x^3+5$, so that $y = u^4$. Then $\frac{du}{dx} = 3x^2$ and $\frac{dy}{du} = 4u^3$. On the left, multiply them, and put $u = x^3+5$ back:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = 4u^3 \times 3x^2 = 4(x^3+5)^3 \times 3x^2.$$

The two terms of the product are the derivative of the outer power, $4(x^3+5)^3$, and the derivative of the inside function, $3x^2$.

READING A DERIVATIVE BACKWARDS

If $\frac{d}{dx}F(x) = g(x)$, then $F(x)$ is a primitive of $g(x)$, so $\int g(x)\,dx = F(x) + C$. If the function to be integrated is $g(x)$ divided by a number k, then its primitive is $F(x)$ divided by k:

$$\int \frac{g(x)}{k}\,dx = \frac{F(x)}{k} + C, \qquad \text{because} \qquad \frac{d}{dx}\left(\frac{F(x)}{k}\right) = \frac{g(x)}{k}.$$

TUTOR DEMO **Differentiate, then reverse**

a Differentiate $(x^4 - 3)^5$ with full setting-out of the chain rule.

b Hence find a primitive of $20x^3(x^4 - 3)^4$.

c Hence find the primitive of $x^3(x^4 - 3)^4$.

YOUR TURN **Same method**

a Differentiate $(x^2 + 7)^3$ with full setting-out of the chain rule.

b Hence find a primitive of $6x(x^2 + 7)^2$.

c Hence find the primitive of $x(x^2 + 7)^2$.

WORKED EXAMPLE 36 **From the text: keep for homework**

a Differentiate $(x^3+5)^4$ with full setting-out of the chain rule.

b Hence find a primitive of $12x^2(x^3+5)^3$.

c Hence find the primitive of $x^2(x^3+5)^3$.

SOLUTION

a Let $y=(x^3+5)^4$.

Then $\dfrac{dy}{dx}=\dfrac{dy}{du}\times\dfrac{du}{dx}$

$=4(x^3+5)^3\times 3x^2$

$=12x^2(x^3+5)^3$.

Let $u=x^3+5$.

Then $y=u^4$.

Hence $\dfrac{du}{dx}=3x^2$

and $\dfrac{dy}{du}=4u^3$.

b By part **a**, $\dfrac{d}{dx}(x^3+5)^4=12x^2(x^3+5)^3$.

Reversing this, $\displaystyle\int 12x^2(x^3+5)^3\,dx=(x^3+5)^4$.

c Dividing by 12,

$\displaystyle\int x^2(x^3+5)^3\,dx=\tfrac{1}{12}(x^3+5)^4+C,$ for some constant C.

ADDING THE CONSTANT

Note: It is best not to add the arbitrary constant until the last line, because it would be pointless to divide C by 12 as well.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 36

a On the right, $\dfrac{dy}{du}=4u^3$. Putting $u=x^3+5$ back gives $4u^3=4(x^3+5)^3$, and this is the factor used on the left. Then
$4(x^3+5)^3\times 3x^2=4\times 3\times x^2(x^3+5)^3=12x^2(x^3+5)^3$.

b Reversing the line in part **a** means that $(x^3+5)^4$ is a primitive of $12x^2(x^3+5)^3$. Any constant could be added; the book leaves it until part **c**.

c The function $x^2(x^3+5)^3$ is $12x^2(x^3+5)^3$ divided by 12, so its primitive is $(x^3+5)^4$ divided by 12: $\displaystyle\int x^2(x^3+5)^3\,dx=\tfrac{1}{12}\int 12x^2(x^3+5)^3\,dx=\tfrac{1}{12}(x^3+5)^4+C$.

Check: $\dfrac{d}{dx}\left(\tfrac{1}{12}(x^3+5)^4\right)=\tfrac{1}{12}\times 12x^2(x^3+5)^3=x^2(x^3+5)^3$.

CONCEPT 2 · reversing with a negative index

A FRACTION IS A NEGATIVE POWER

To differentiate a fraction such as $\dfrac{1}{1+x^2}$ by the chain rule, write it as $(1+x^2)^{-1}$. Let $u = 1+x^2$, so that $y = u^{-1}$ and $\dfrac{dy}{du} = -u^{-2}$. The answer is written as a fraction again: $-(1+x^2)^{-2} = -\dfrac{1}{(1+x^2)^2}$.

When the derivative is reversed, the number to divide by may be negative. Dividing by -2 is the same as multiplying by $-\frac{1}{2}$.

TUTOR DEMO **A negative index**

a Differentiate $\dfrac{1}{4+x^3}$ with full setting-out of the chain rule.

b Hence find the primitive of $\dfrac{x^2}{(4+x^3)^2}$.

YOUR TURN **Same method**

a Differentiate $\dfrac{1}{9+x^2}$ with full setting-out of the chain rule.

b Hence find the primitive of $\dfrac{x}{(9+x^2)^2}$.

WORKED EXAMPLE 37 **From the text: keep for homework**

a Differentiate $\dfrac{1}{1+x^2}$ with full setting-out of the chain rule.

b Hence find the primitive of $\dfrac{x}{(1+x^2)^2}$.

SOLUTION

a Let $y = \dfrac{1}{1+x^2}$.

Then $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$

$= -(1+x^2)^{-2} \times 2x$

$= \dfrac{-2x}{(1+x^2)^2}$.

Let $u = 1+x^2$.

Then $y = u^{-1}$.

Hence $\dfrac{du}{dx} = 2x$

and $\dfrac{dy}{du} = -u^{-2}$.

b By part **a**, $\dfrac{d}{dx}\left(\dfrac{1}{1+x^2}\right) = \dfrac{-2x}{(1+x^2)^2}$.

Reversing this, $\displaystyle\int \frac{-2x}{(1+x^2)^2}\,dx = \frac{1}{1+x^2} + C$, for some constant C,

$\boxed{\div(-2)}$ $\displaystyle\int \frac{x}{(1+x^2)^2}\,dx = \frac{-1}{2(1+x^2)} + C$.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 37

a On the right, $\dfrac{dy}{du} = -u^{-2}$. Putting $u = 1 + x^2$ back gives $-u^{-2} = -(1+x^2)^{-2}$.
Then $-(1+x^2)^{-2} \times 2x = -\dfrac{2x}{(1+x^2)^2} = \dfrac{-2x}{(1+x^2)^2}$.

b The function $\dfrac{x}{(1+x^2)^2}$ is $\dfrac{-2x}{(1+x^2)^2}$ divided by -2, so divide the primitive by -2:
$\dfrac{1}{1+x^2} \div (-2) = \dfrac{1}{1+x^2} \times \left(-\dfrac{1}{2}\right) = \dfrac{-1}{2(1+x^2)}$. The constant is added last.

21 REVERSING A CHAIN-RULE DIFFERENTIATION

Once a chain-rule differentiation, or any differentiation, has been performed, the result can be written down in reverse as an indefinite integral.

CONCEPT 3 · a formula for the reverse chain rule

A FORMULA FOR THE REVERSE CHAIN RULE

There is a formula for the reverse chain rule. Start with the formula for differentiating a function using the chain rule — we gave the formula in two forms:

$$\frac{d}{dx}(u^n) = nu^{n-1}\frac{du}{dx} \qquad \frac{d}{dx}\big(f(x)\big)^n = n\big(f(x)\big)^{n-1}f'(x)$$

and we can reverse both forms of the formula,

$$\int u^{n-1}\frac{du}{dx}\,dx = \frac{u^n}{n} \qquad \int \big(f(x)\big)^{n-1}f'(x)\,dx = \frac{\big(f(x)\big)^n}{n}.$$

Then replacing $n-1$ by n and n by $n+1$,

$$\int u^n\frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1} \qquad \text{OR} \qquad \int \big(f(x)\big)^n f'(x)\,dx = \frac{\big(f(x)\big)^{n+1}}{n+1}.$$

Take your pick which formula you prefer to use. The difficult part is recognising what u or $f(x)$ should be in the function that you are integrating.

We shall do Worked example 36c again using the formula.

WHY THE FORMULA WORKS

Differentiate the right-hand side by the chain rule:
$\dfrac{d}{dx}\left(\dfrac{u^{n+1}}{n+1}\right) = \dfrac{(n+1)u^n}{n+1} \times \dfrac{du}{dx} = u^n\dfrac{du}{dx}$, which is the function inside the integral.
The index on the right is one more than the index on the left, and we divide by that new index. So the index $n = -1$ cannot be used, because then $n+1 = 0$ (as in Lesson 5E).

USING THE FORMULA, STEP BY STEP

1. Choose u (or $f(x)$): the function inside the power.
2. Find $\frac{du}{dx}$ (or $f'(x)$) on the right-hand side.
3. If the other factor is a multiple of $\frac{du}{dx}$, multiply inside the integral by a number and outside by its reciprocal, so that $\frac{du}{dx}$ appears exactly.
4. Write the standard form with the particular value of n, then simplify the numbers in front.

TUTOR DEMO **Using the formula**

Use the formula for the reverse chain rule to find $\int x^3(x^4+2)^5\,dx$.

YOUR TURN **Same method**

Use the formula for the reverse chain rule to find $\int x(x^2-3)^4\,dx$.

WORKED EXAMPLE 38 **From the text: keep for homework**

Use the formula for the reverse chain rule to find $\int x^2(x^3+5)^3\,dx$.

SOLUTION

The key to finding this integral is to realise that x^2 is a multiple of the derivative of x^3+5. At this point, we can put $u = x^3+5$ or $f(x) = x^3+5$, and everything works smoothly after that. Here are two strong recommendations:

- Show working identifying u or $f(x)$ and its derivative on the right-hand side.
- Write down the standard form above substituting the particular value of n.

Using the first formula,

$$\begin{aligned}\int x^2(x^3+5)^3\,dx &= \tfrac{1}{3}\int 3x^2(x^3+5)^3\,dx \\ &= \tfrac{1}{3}\times\tfrac{1}{4}(x^3+5)^4 + C, \\ &\quad \text{for some constant } C, \\ &= \tfrac{1}{12}(x^3+5)^4 + C.\end{aligned}$$

$$\begin{aligned}&\text{Let } u = x^3+5. \\ &\text{Then } \frac{du}{dx} = 3x^2. \\ &\text{Here } \int u^3\frac{du}{dx}\,dx \\ &\quad = \tfrac{1}{4}u^4.\end{aligned}$$

Using the second formula,

$$\begin{aligned}\int x^2(x^3+5)^3\,dx &= \tfrac{1}{3}\int 3x^2(x^3+5)^3\,dx \\ &= \tfrac{1}{3}\times\tfrac{1}{4}(x^3+5)^4 + C, \\ &\quad \text{for some constant } C, \\ &= \tfrac{1}{12}(x^3+5)^4 + C.\end{aligned}$$

$$\begin{aligned}&\text{Let } f(x) = x^3+5. \\ &\text{Then } f'(x) = 3x^2. \\ &\text{Here } \int \big(f(x)\big)^3 f'(x)\,dx \\ &\quad = \tfrac{1}{4}\big(f(x)\big)^4.\end{aligned}$$

TWO NOTATIONS

Notice that the two notations differ only in the working in the right-hand column.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 38

First line. The derivative of x^3+5 is $3x^2$, and $x^2 = \frac{1}{3}\times 3x^2$. Multiplying inside the integral by 3 and outside by $\frac{1}{3}$ does not change it, because $\frac{1}{3}\times 3 = 1$:

$$\int x^2(x^3+5)^3\,dx = \tfrac{1}{3}\int 3x^2(x^3+5)^3\,dx.$$

Second line. Here $n = 3$, so the formula gives $\dfrac{u^{3+1}}{3+1} = \frac{1}{4}u^4$, and $u^4 = (x^3+5)^4$.

Third line. $\frac{1}{3}\times\frac{1}{4} = \frac{1}{12}$. This is the answer to Worked example 36c.

CONCEPT 4 · fractional indices, and where the integrand is defined

A SQUARE ROOT IS A POWER

Write $\sqrt{1-x^2} = (1-x^2)^{\frac{1}{2}}$, so that $n = \frac{1}{2}$ and the new index is $n+1 = \frac{3}{2}$. Dividing by $\frac{3}{2}$ is the same as multiplying by $\frac{2}{3}$.

A definite integral needs the integrand to be defined at every point of the interval. Since $\sqrt{1-x^2}$ needs $1-x^2 \geq 0$, it is defined only for $-1 \leq x \leq 1$. For a definite integral, the primitive goes in square brackets, with no constant, and the limits are substituted (Lesson 5B).

TUTOR DEMO **A square root**

Use the formula for the reverse chain rule to find:

a $\int x\sqrt{9-x^2}\,dx$ b $\int_0^3 x\sqrt{9-x^2}\,dx$ c $\int_0^4 x\sqrt{9-x^2}\,dx$

YOUR TURN **Same method**

Use the formula for the reverse chain rule to find:

a $\int x\sqrt{4-x^2}\,dx$ b $\int_0^2 x\sqrt{4-x^2}\,dx$ c $\int_0^3 x\sqrt{4-x^2}\,dx$

WORKED EXAMPLE 39 **From the text: keep for homework**

Use the formula for the reverse chain rule to find:

a $\int x\sqrt{1-x^2}\,dx$ b $\int_0^2 x\sqrt{1-x^2}\,dx$

SOLUTION

a This integral is based on the recognition that $\dfrac{d}{dx}(1-x^2) = -2x$.

Using the first formula,

$$\begin{aligned}\int x\sqrt{1-x^2}\,dx &= -\tfrac{1}{2}\int (-2x)\times(1-x^2)^{\frac{1}{2}}\,dx\\ &= -\tfrac{1}{2}\times\tfrac{2}{3}(1-x^2)^{\frac{3}{2}}+C,\\ &\quad \text{for some constant } C,\\ &= -\tfrac{1}{3}(1-x^2)^{\frac{3}{2}}+C.\end{aligned}$$

$$\begin{aligned}&\text{Let } u = 1-x^2.\\ &\text{Then } \frac{du}{dx} = -2x.\\ &\text{Here } \int u^{\frac{1}{2}}\frac{du}{dx}\,dx\\ &= \tfrac{2}{3}u^{\frac{3}{2}}.\end{aligned}$$

Using the second formula,

$$\begin{aligned}\int x\sqrt{1-x^2}\,dx &= -\tfrac{1}{2}\int (-2x)\times(1-x^2)^{\frac{1}{2}}\,dx\\ &= -\tfrac{1}{2}\times\tfrac{2}{3}(1-x^2)^{\frac{3}{2}}+C,\\ &\quad \text{for some constant } C,\\ &= -\tfrac{1}{3}(1-x^2)^{\frac{3}{2}}+C.\end{aligned}$$

$$\begin{aligned}&\text{Let } f(x) = 1-x^2.\\ &\text{Then } f'(x) = -2x.\\ &\text{Here } \int (f(x))^{\frac{1}{2}} f'(x)\,dx\\ &= \tfrac{2}{3}(f(x))^{\frac{3}{2}}.\end{aligned}$$

b The definite integral is meaningless because $\sqrt{1-x^2}$ is undefined for $x > 1$.

STEPS THE BOOK SKIPS: WORKED EXAMPLE 39

a The derivative of $1 - x^2$ is $-2x$, and $x = -\frac{1}{2} \times (-2x)$, so multiply inside by -2 and outside by $-\frac{1}{2}$. Here $n = \frac{1}{2}$, so the formula gives $\frac{u^{\frac{3}{2}}}{\frac{3}{2}} = \frac{2}{3}u^{\frac{3}{2}}$. Then $-\frac{1}{2} \times \frac{2}{3} = -\frac{1}{3}$.

b The interval $0 \leq x \leq 2$ includes the values $1 < x \leq 2$, where $1 - x^2 < 0$. For example, at $x = 2$, $1 - x^2 = 1 - 4 = -3$, which has no square root.

22 A FORMULA FOR THE REVERSE CHAIN RULE

- The reversing of the chain rule can be written as a formula in two ways:

$$\int u^n \frac{du}{dx}\, dx = \frac{u^{n+1}}{n+1} + C \quad \text{OR} \quad \int \big(f(x)\big)^n f'(x)\, dx = \frac{\big(f(x)\big)^{n+1}}{n+1} + C.$$

- The vital step in using this formula is to identify u or $f(x)$ and its derivative.

WRITING THE INTEGRAND AS A PRODUCT

To use the formula, we have to write the integrand as a product. One factor is a power of a function $f(x)$ or u. The other factor is the derivative of that function.

TEXTBOOK HOMEWORK · Exercise 5I

FOUNDATION

1 a Find $\frac{d}{dx}(2x+3)^4$.

b Hence find:

i $\int 8(2x+3)^3\, dx$

ii $\int 16(2x+3)^3\, dx$

2 a Find $\frac{d}{dx}(3x-5)^3$.

b Hence find:

i $\int 9(3x-5)^2\,dx$

ii $\int 27(3x-5)^2\,dx$

3 a Find $\frac{d}{dx}(1+4x)^5$.

b Hence find:

i $\int 20(1+4x)^4\,dx$

ii $\int 10(1+4x)^4\,dx$

4 a Find $\frac{d}{dx}(1-2x)^4$.

b Hence find:

i $\int -8(1-2x)^3\,dx$

ii $\int -2(1-2x)^3\,dx$

5 a Find $\frac{d}{dx}(4x+3)^{-1}$.

b Hence find:

i $\int -4(4x+3)^{-2}\,dx$

ii $\int (4x+3)^{-2}\,dx$

6 a Find $\dfrac{d}{dx}(2x-5)^{\frac{1}{2}}$.

b Hence find:

i $\displaystyle\int (2x-5)^{-\frac{1}{2}}\,dx$

ii $\displaystyle\int \tfrac{1}{3}(2x-5)^{-\frac{1}{2}}\,dx$

7 a Find $\dfrac{d}{dx}(x^2+3)^4$.

b Hence find:

i $\displaystyle\int 8x(x^2+3)^3\,dx$

ii $\displaystyle\int 40x(x^2+3)^3\,dx$

8 a Find $\dfrac{d}{dx}(x^3-1)^5$.

b Hence find:

i $\displaystyle\int 15x^2(x^3-1)^4\,dx$

ii $\displaystyle\int 3x^2(x^3-1)^4\,dx$

9 a Find $\dfrac{d}{dx}\sqrt{2x^2+3}$.

b Hence find:

i $\displaystyle\int \frac{2x}{\sqrt{2x^2+3}}\,dx$

ii $\displaystyle\int \frac{x}{\sqrt{2x^2+3}}\,dx$

10 a Find $\dfrac{d}{dx}\left(\sqrt{x}+1\right)^3$.

b Hence find:

i $\displaystyle\int \frac{3\left(\sqrt{x}+1\right)^2}{2\sqrt{x}}\,dx$

ii $\displaystyle\int \frac{\left(\sqrt{x}+1\right)^2}{\sqrt{x}}\,dx$

11 a Find $\dfrac{d}{dx}\left(x^3+3x^2+5\right)^4$.

b Hence find:

i $\displaystyle\int 12(x^2+2x)(x^3+3x^2+5)^3\,dx$

ii $\displaystyle\int (x^2+2x)(x^3+3x^2+5)^3\,dx$

12 a Find $\dfrac{d}{dx}\left(5-x^2-x\right)^7$.

b Hence find:

i $\displaystyle\int (-14x-7)(5-x^2-x)^6\,dx$

ii $\displaystyle\int (2x+1)(5-x^2-x)^6\,dx$

DEVELOPMENT

13 Find these indefinite integrals using the reverse chain rule in either form

$$\int f'(x)\big(f(x)\big)^n\,dx = \frac{\big(f(x)\big)^{n+1}}{n+1} + C \qquad \text{OR} \qquad \int u^n \frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1} + C.$$

a $\displaystyle\int 5(5x+4)^3\,dx$ (Let $f(x) = 5x+4$ or $u = 5x+4$.)

b $\displaystyle\int -3(1-3x)^5\,dx$ (Let $f(x) = 1-3x$ or $u = 1-3x$.)

c $\displaystyle\int 2x(x^2-5)^7\,dx$ (Let $f(x) = x^2-5$ or $u = x^2-5$.)

d $\displaystyle\int 3x^2(x^3+7)^4\,dx$ (Let $f(x) = x^3+7$ or $u = x^3+7$.)

e $\displaystyle\int \frac{6x}{(3x^2+2)^2}\,dx$ (Let $f(x) = 3x^2+2$ or $u = 3x^2+2$.)

f $\displaystyle\int \frac{-6x^2}{\sqrt{9-2x^3}}\,dx$ (Let $f(x) = 9-2x^3$ or $u = 9-2x^3$.)

14 Find these indefinite integrals using the reverse chain rule.

a $\int 10x(5x^2+3)^2\,dx$

b $\int 2x(x^2+1)^3\,dx$

c $\int 12x^2(1+4x^3)^5\,dx$

d $\int x(1+3x^2)^4\,dx$

e $\int x^3(1-x^4)^7\,dx$

f $\int 3x^2\sqrt{x^3-1}\,dx$

g $\int x\sqrt{5x^2+1}\,dx$

h $\int \dfrac{2x}{\sqrt{x^2+3}}\,dx$

i $\int \dfrac{x+1}{\sqrt{4x^2+8x+1}}\,dx$

j $\int \dfrac{x}{(x^2+5)^3}\,dx$

15 Evaluate these definite integrals using the reverse chain rule.

a $\int_{-1}^{1} x^2(x^3+1)^4\,dx$

b $\int_{0}^{1} \dfrac{x}{(5x^2+1)^3}\,dx$

c $\int_{0}^{\frac{1}{2}} x\sqrt{1-4x^2}\,dx$

d $\int_{-3}^{-1} (x+5)(x^2+10x+3)^2\,dx$

ENRICHMENT

16 Use the reverse chain rule to find:

a $\int \dfrac{\left(1-\frac{1}{x}\right)^5}{x^2}\,dx$

b $\int_{1}^{4} \dfrac{1}{\sqrt{x}\left(1+\sqrt{x}\right)^2}\,dx$

17 **a** What is the domain of the function $f(x) = x\sqrt{x^2 - 1}$?

b Find $f'(x)$, and hence show that the function has no stationary points in its domain.

c Show that the function is odd, and hence sketch its graph.

d By evaluating the appropriate definite integral, find the area of the region bounded by the curve and the x-axis from $x = 1$ to $x = 3$.

18 a Sketch $y = x(7 - x^2)^3$, indicating all stationary points and intercepts with the axes.

b Find the area of the region enclosed between the curve and the x-axis.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 5 minutes **2 marks** Calculator allowed · show working · hints are below each question if you get stuck

These are the only school-paper items that need just this lesson's ideas. For more timed practice, do Questions 14(g), 14(i), 14(j), 15(b), 15(d), 16(b) and 17(d) above in 25 minutes.

PP 1 **Ascham School, 2020, Mathematics Extension 1 Trial HSC Examination, Q5** 1 mark

$$\int \frac{x}{\sqrt{1-x^2}}\,dx =$$

A. $\sin^{-1} x + c$

B. $-x\cos^{-1} x + c$

C. $-\frac{1}{2}\sqrt{1-x^2} + c$

D. $-\sqrt{1-x^2} + c$

Hint: what is being integrated here?

PP 2 **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q10** 1 mark

Which of the following is an expression for $\int \frac{x}{\sqrt{9-x^2}}\,dx$?

A. $-\sqrt{9-x^2} + C$

B. $-2\sqrt{9-x^2} + C$

C. $\sqrt{9-x^2} + C$

D. $2\sqrt{9-x^2} + C$

Hint: what is under the square root, and what is in the numerator?

CHAPTER 5 REVIEW

Chapter review exercise

Mixed practice from the whole chapter, Lessons 5A to 5I.

1 Evaluate these definite integrals, using the fundamental theorem.

a $\int_0^1 3x^2\,dx$ b $\int_1^2 x\,dx$ c $\int_2^5 4x^3\,dx$

d $\int_{-1}^1 x^4\,dx$ e $\int_{-4}^{-2} 2x\,dx$ f $\int_{-3}^{-1} x^2\,dx$

g $\int_0^2 (x+3)\,dx$ h $\int_{-1}^4 (2x-5)\,dx$ i $\int_{-3}^1 (x^2-2x+1)\,dx$

2 By expanding the brackets where necessary, evaluate these definite integrals.

a $\int_1^3 x(x-1)\,dx$ b $\int_{-1}^0 (x+1)(x-3)\,dx$ c $\int_0^1 (2x-1)^2\,dx$

3 Write each integrand as separate fractions, then evaluate the integral.

a $\displaystyle\int_1^2 \frac{x^2 - 3x}{x}\,dx$ b $\displaystyle\int_2^3 \frac{3x^4 - 4x^2}{x^2}\,dx$ c $\displaystyle\int_{-2}^{-1} \frac{x^3 - 2x^4}{x^2}\,dx$

4

a i Show that $\displaystyle\int_4^k 5\,dx = 5k - 20$.

ii Hence find the value of k if $\displaystyle\int_4^k 5\,dx = 10$.

b i Show that $\displaystyle\int_0^k (2x - 1)\,dx = k^2 - k$.

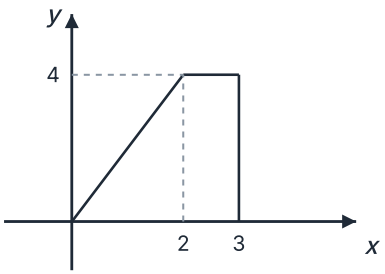
ii Hence find the positive value of k for which $\displaystyle\int_0^k (2x - 1)\,dx = 6$.

5 Without finding a primitive, use the properties of the definite integral to evaluate these integrals, stating reasons.

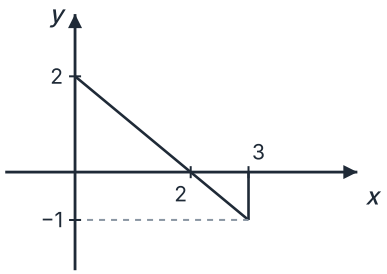
a $\displaystyle\int_3^3 (x^3 - 5x + 4)\,dx$ b $\displaystyle\int_{-2}^2 x^3\,dx$ c $\displaystyle\int_{-3}^3 (x^3 - 9x)\,dx$

6 Use area formulae to find $\int_0^3 f(x)\,dx$, given the following sketches of $f(x)$.

a



b



7 a Find each signed area function.

i $A(x) = \int_{-2}^{x} (4 - t)\,dt$

ii $A(x) = \int_{2}^{x} t^{-2}\,dt$

b Differentiate the results in part **a** to find:

i $\frac{d}{dx}\int_{-2}^{x} (4 - t)\,dt$

ii $\frac{d}{dx}\int_{2}^{x} t^{-2}\,dt$

c Without first performing the integration, use the fundamental theorem of calculus to find these functions.

i $\frac{d}{dx}\int_{7}^{x} (t^5 - 5t^3 + 1)\,dt$

ii $\frac{d}{dx}\int_{3}^{x} \frac{t^2 + 4}{t^2 - 1}\,dt$

8 Find these indefinite integrals.

a $\int (x+2)\,dx$

b $\int (x^3+3x^2-5x+1)\,dx$

c $\int x(x-1)\,dx$

d $\int (x-3)(2-x)\,dx$

e $\int x^{-2}\,dx$

f $\int \frac{1}{x^7}\,dx$

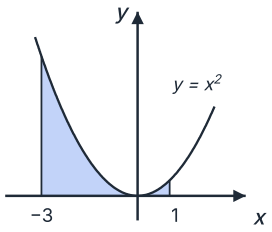
g $\int \sqrt{x}\,dx$

h $\int (x+1)^4\,dx$

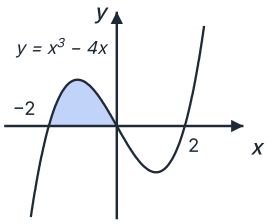
i $\int (2x-3)^5\,dx$

9 Find the area of each shaded region below by evaluating a definite integral.

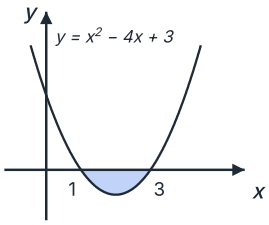
a



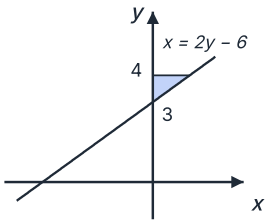
b



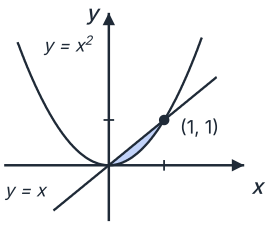
c



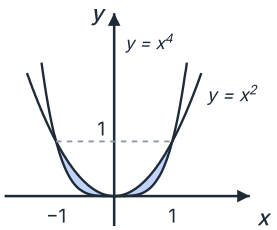
d



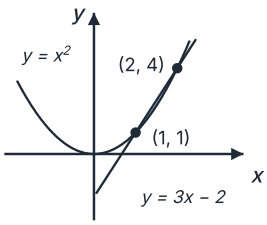
e



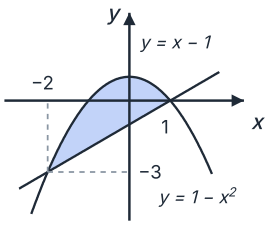
f



g



h



10 a By solving the equations simultaneously, show that the curves $y = x^2 - 3x + 5$ and $y = x + 2$ intersect at the points $(1, 3)$ and $(3, 5)$.

b Sketch both curves on the same diagram and find the area of the region enclosed between them.

11 a Use the trapezoidal rule with three function values to approximate $\int_1^3 2^x \, dx$.

b Use the trapezoidal rule with five function values to approximate $\int_1^3 \log_{10} x \, dx$. Give your answer correct to two significant figures.

12 a Find $\dfrac{d}{dx}(3x + 4)^6$.

b Hence find:

i $\displaystyle\int 18(3x + 4)^5 \, dx$

ii $\displaystyle\int 9(3x + 4)^5 \, dx$

13 a Find $\dfrac{d}{dx}(x^2-1)^3$.

b Hence find:

i $\displaystyle\int 6x(x^2-1)^2\,dx$

ii $\displaystyle\int x(x^2-1)^2\,dx$

14 Find these indefinite integrals using the reverse chain rule.

a $\displaystyle\int 3x^2(x^3+1)^4\,dx$

b $\displaystyle\int \frac{2x}{(x^2-5)^3}\,dx$

15 Use the reverse chain rule to show that $\displaystyle\int_0^1 \frac{x}{\sqrt{x^2+3}}\,dx = 2-\sqrt{3}$.